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ALLEGRETTI et al.(10) **Pub. No.: US 2025/0282831 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **NOVEL PROTEIN AND NUCLEIC ACID
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(2013.01)

(57)

ABSTRACT

The present invention relates to a mutated SARS-COV-2 spike protein, a variant or fragment thereof or an mRNA or DNA encoding them for use in the prevention of COVID-19.

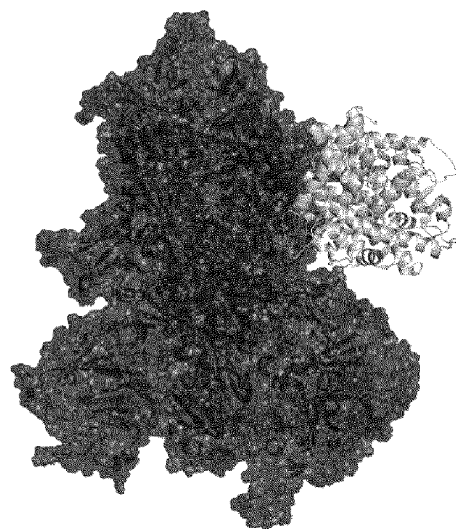
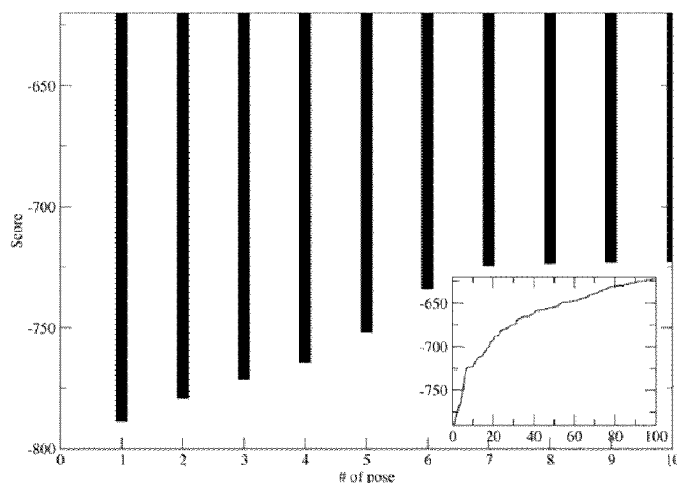
Specification includes a Sequence Listing.

Figure 1

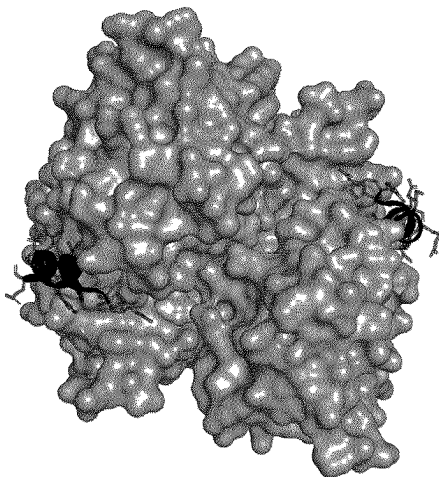


Figure 2

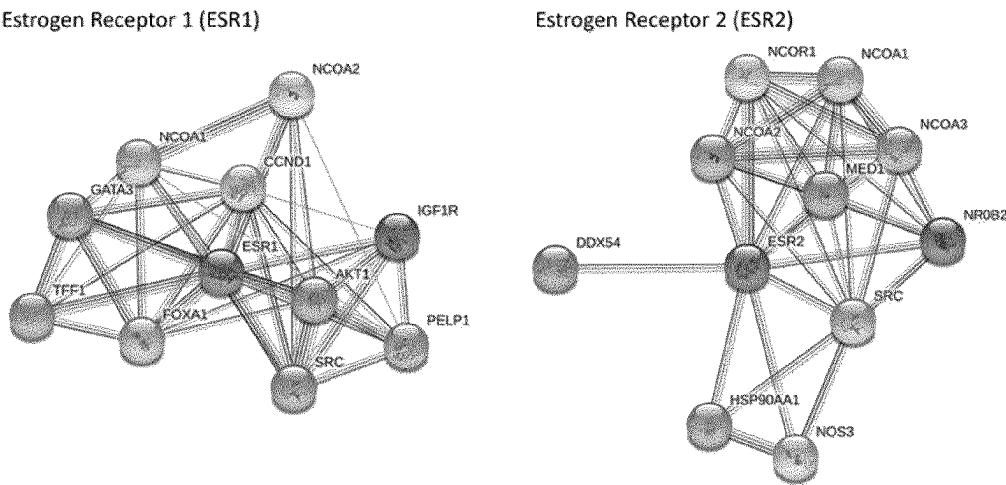


Figure 3

SPIKELMX1	PSKRSFIEDLLFNKVTL	aa positions 812-828
NCOA1LXD1	SQTSHKLVQLTTTAEQ	aa positions 627-643
NCOA2LXD1	SKGQTKLLQLLTKSDQ	aa positions 635-651
SPIKELMX1	PSKRSFIEDLLFNKVTL	aa positions 812-828
NCOA1LXD2	TERHKILHRLQEGSPS	aa positions 684-700
NCOA2LXD2	KECHKILHRLQDSSSP	aa positions 684-700
SPIKELMX1	PSKRSFIEDLLFNKVTL	aa positions 812-828
NCOA1LXD3	SKDHQLLRYLLDKDEKD	aa positions 743-759
NCOA2LXD3	KKENALLRYLLDKDDTK	aa positions 739-755
SPIKELMX1	PSKRSFIEDLLFNKVTL	aa positions 812-828
NCOA1LXD4	QCISSQLDELLCPPTTV	aa positions 907-923
NCOA2LXD4	PSDEGALLDQLYLALRN	aa positions 1073-1089

Figure 4

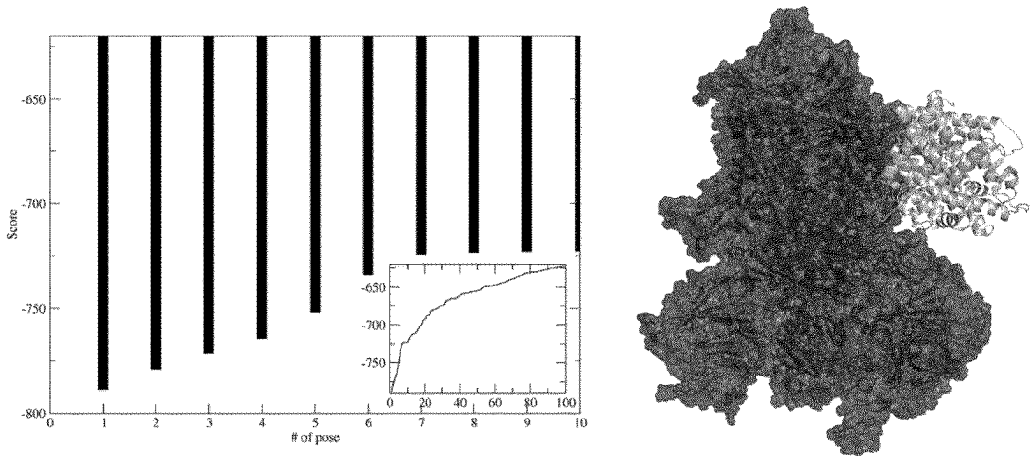


Figure 5

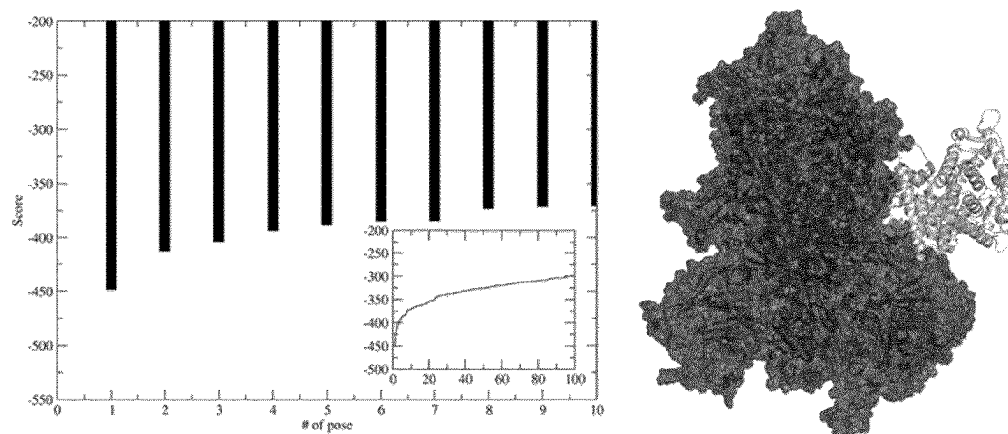


Figure 6

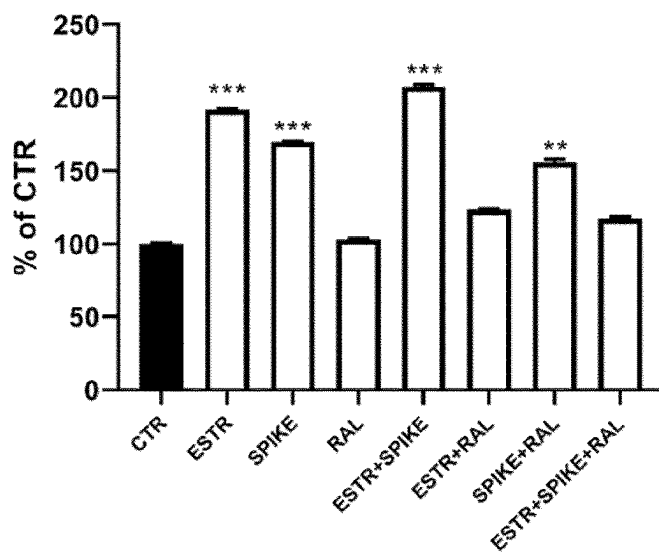


Figure 7

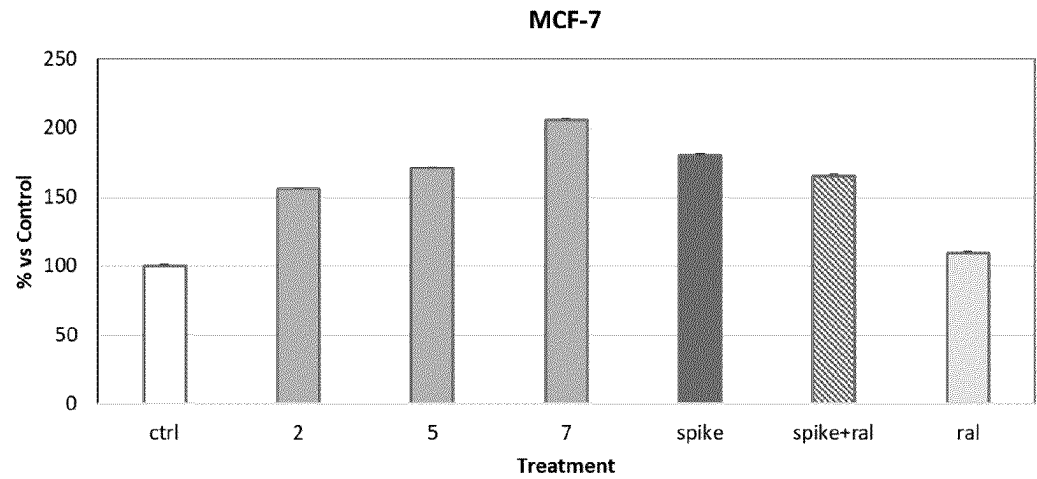


Figure 8

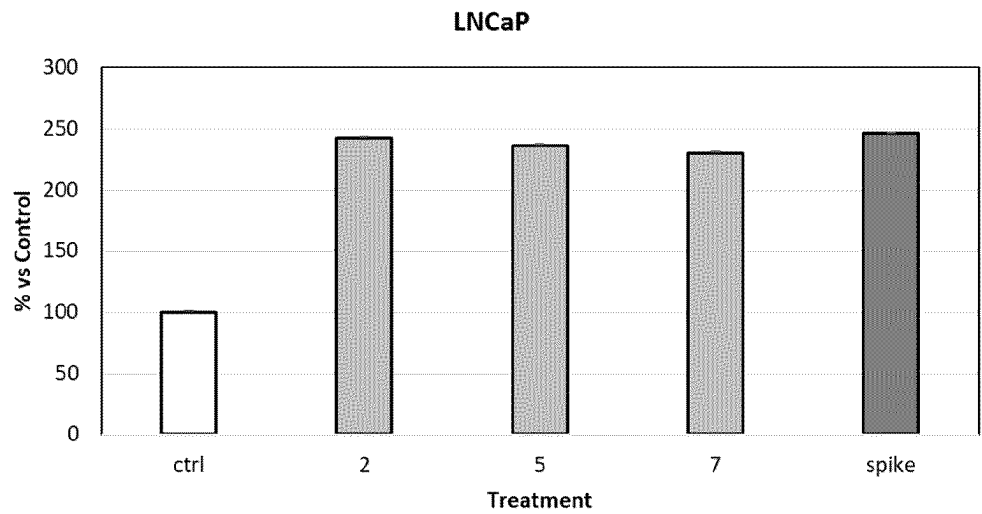


Figure 9

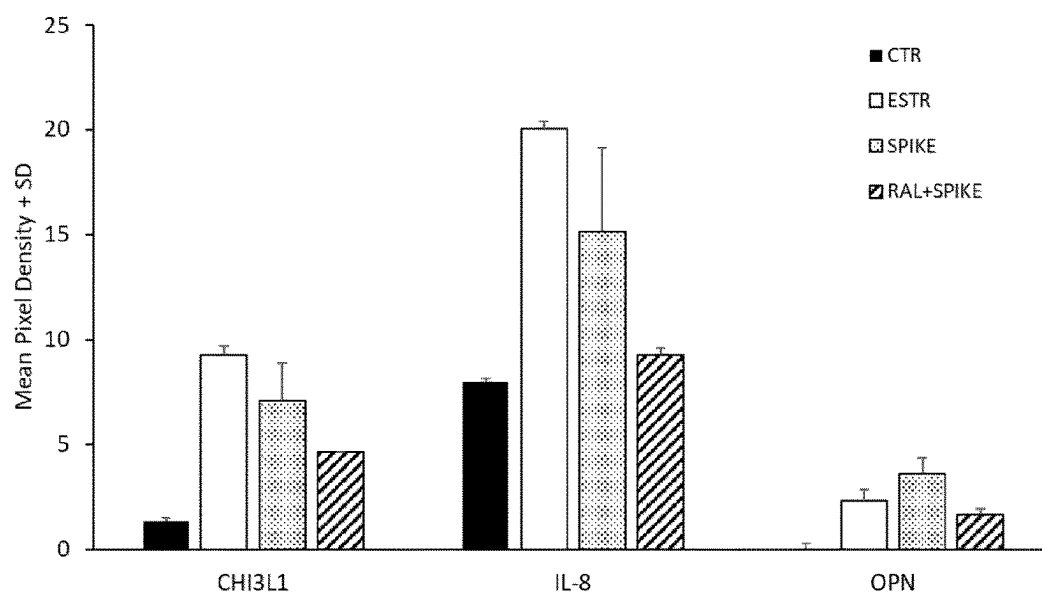
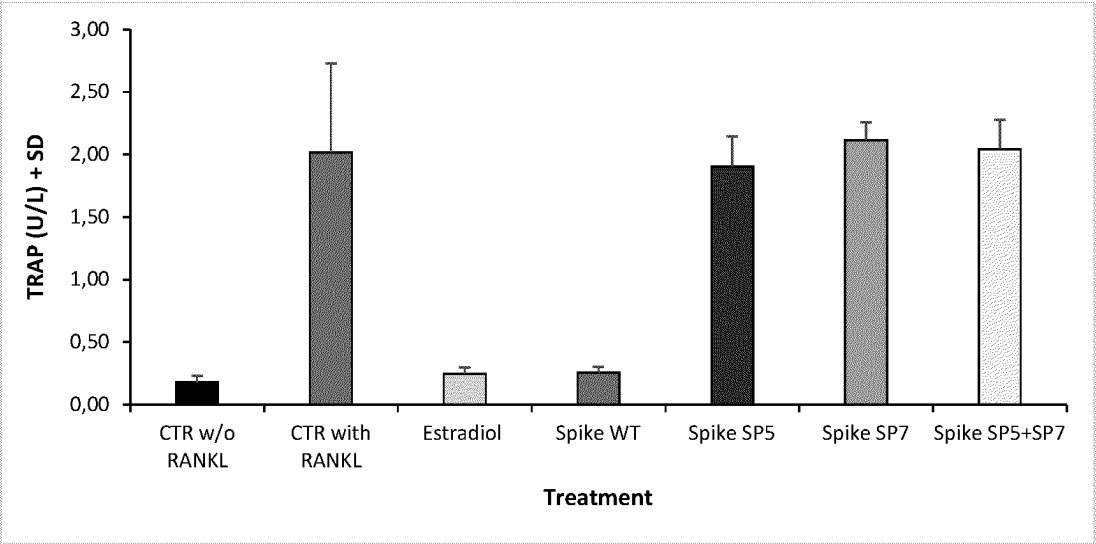


Figure 10



NOVEL PROTEIN AND NUCLEIC ACID SEQUENCES FOR COVID-19 VACCINES

SEQUENCE LISTING

[0001] The instant application contains a Sequence Listing which has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. Said ASCII copy, created on Jul. 5, 2024, is named 50600-023001_Sequence_Listing_7_5_24_ST25 and is 480,152 bytes in size.

FIELD OF THE INVENTION

[0002] The present invention relates to proteins and nucleic acid sequences suitable for use in COVID-19 vaccines.

STATE OF THE ART

[0003] Coronaviruses (Covs) are a large family of single-stranded, enveloped RNA viruses that belong to the Coronaviridae family. The limited number of coronaviruses known to be able to infect humans were considered in the past as relatively harmless respiratory human pathogens, causing mild infections. However, two coronavirus subtypes have emerged, Severe Acute Respiratory Syndrome Coronavirus (SARS-COV) and the Middle East Respiratory Syndrome Coronavirus (MERS-COV), which cause severe and sometimes fatal respiratory tract infections in humans (Pereira, H et al., 1989 Coronaviridae. Andrewes 'Viruses of Vertebrates, 5th ed. pp. 42-57, Holmes, K. V. et al., Virology 1996, 1:1075-1093). In December 2019, atypical pneumonia cases occurred in China and the cause was later identified as being a novel coronavirus. The World Health Organization (WHO) named the virus as SARS-COV-2 and the related disease as COVID-19.

[0004] The virus spread rapidly worldwide, and on 11 Mar. 2020 the WHO declared SARS-CoV-2 infection as a pandemic. Most people infected with COVID-19 experience mild to moderate respiratory illness (fever, fatigue, dry cough and dyspnoea) and recover without requiring special treatments. Older people, and those with underlying medical problems like cardiovascular diseases, diabetes, chronic respiratory disease, and cancer are more likely to develop serious illness.

[0005] Analysis of epidemiological and clinical characteristics and outcomes of patients infected by SARS-COV-2 have shown that 15% of people with symptomatic COVID-19 have significant disease including severe pneumonia, and 5% experience critical disease with life-threatening complications. Critical disease includes acute respiratory distress syndrome (ARDS), sepsis, septic shock, cardiac disease, thromboembolic events, such as pulmonary embolism and multi-organ failure.

[0006] There is also growing evidence that in those who develop critical COVID-19 disease, long-term consequences such as rare neurological and psychiatric complications are reasonably expected. These may include stroke, delirium, anxiety, depression, damage or inflammation of the brain and sleep disturbances.

[0007] Due to the dramatic rise in cases and deaths worldwide and the social and economic consequences thereof, efforts have focused on the development of safe and effective vaccines to help control and bring to an end the COVID-19 pandemic.

[0008] Different vaccines are under development and some have already been approved for prevention of COVID-19.

[0009] All COVID-19 vaccines and vaccine candidates act by inducing an immune response against virus protein antigens, making use of different strategies and technologies: traditional inactivated or live-attenuated virus vaccines, based on the introduction in the host of a number of different SARS-COV-2 antigens, vaccines based on recombinantly produced SARS-COV-2 proteins in association with adjuvants and gene-based vaccines, such as mRNA, DNA or vector vaccines, which deliver genes encoding the viral protein antigens for in vivo production in the cells of the host.

[0010] SARS-COV-2 is characterised by four structural proteins, namely spike(S) protein, envelope (E) protein, membrane (M) protein and nucleocapsid (N) protein (Chen Y et al, J Med Virol 2020, 92:418-423).

[0011] Currently, most COVID-19 vaccines already approved or under development use the spike protein as target antigen. This protein is in fact considered an important antigenic determinant capable of inducing a protective immune response (Ou et al, Nat Commun 2020; 11:1620). Furthermore, it is also an essential molecule for entry of the virus into cells through the cell entry receptor Angiotensin-converting enzyme II (ACE2). Therefore, antibodies targeting the S protein prevent the virus from entering and replicating inside the host cells.

[0012] The SARS-COV-2 S protein has the 1273 amino-acid sequence of SEQ.ID. No: 1 and consists of a signal peptide located at the N-terminus (amino acids 1-13), the S1 subunit (14-685 residues) and the S2 subunit (686-1273 residues). The S1 subunit mediates receptor binding to ACE2 through the receptor binding domain (RBD) and the S2 subunit is responsible for the viral and host cell membrane fusion (Letko et al, Nat Microbiol 2020, 5:562-569, Huang et al, Acta Pharmacologica Sinica 2020, 41:1141-1149). During the fusion process the S2 subunit changes from a prefusion to a postfusion (hairpin) conformation. Subsequently, surface proteases cleave the S2 subunit (Silverira et al, Life Sci 2021, 267:1128919).

[0013] The vaccines approved so far by EMA are based on nucleic acids that induce the production in the host of the spike protein as protein antigen: the mRNA based Comirnaty vaccine developed by Pfizer and BioNTech, the mRNA based Moderna COVID-19 vaccine developed by Moderna and the DNA, viral vector delivered vaccines Vaxzevria developed by Oxford University and AstraZeneca and COVID-19 Vaccine Janssen developed by Janssen-Cilag International.

[0014] Other vaccines based on these approaches are under development as well as vaccines using recombinantly produced SARS-COV-2 spike protein as the active ingredient, for example the vaccine NVX-CoV2373 by Novavax and Co-VLP by Medicago and GSK (Forni et al, Cell Death & Differentiation 2021, 28:626-639).

[0015] Recent evidence has suggested a potential activity of the SARS-COV-2 spike protein on cell signalling that is independent from facilitation of viral infection.

[0016] In particular, it has been shown that the spike protein can induce pro-inflammatory responses in vitro in murine and human macrophages (Shirato et al, Heliyon 2021, 7 (2) e06187) and can activate cell signaling events in

cultured human vascular cells (Suzuki et al, Vascular Pharmacology 2021, 137:106823).

[0017] This aspect is very relevant since most of the COVID-19 vaccines introduce the spike protein in the organism.

[0018] It has been reported that after administration of the mRNA and DNA vaccines approved so far by intramuscular injection, the spike protein is expressed mainly in myocytes, fibroblasts, dendritic cells and lymphocytes but levels of expression have been detected with some of the vaccines also in other organs (information available at <https://www.ema.europa.eu/>).

[0019] In view of the above, it therefore needs to be investigated whether the presence in the body of the spike protein produced by vaccination may interfere with biological processes, thereby causing side effects in the short and long term.

[0020] Nuclear hormone receptors are a family of ligand-regulated transcription factors that are activated by thyroid and steroid hormones and various other lipid-soluble signals, including retinoic acid and vitamin D.

[0021] Steroid hormones, in particular androgens and estrogens, have been shown to regulate a wide range of physiological functions such as development and maintenance of secondary sexual characteristics, metabolism, blood salt balance, immune and inflammatory responses, response to stress and/or neuronal function.

[0022] Furthermore, an unbalance in the level or activity of these hormones has been implicated in many pathological processes.

[0023] In particular, estrogens have been implicated in the etiology of breast, endometrial, renal, and uterine cancer (Okamoto et al, Toxicology Letters 2020, 318:99-103), while a pivotal role of androgens has been demonstrated in the pathogenesis of prostate cancer (Cold Spring Harb Perspect Med 2017; 7: a030452).

[0024] Furthermore, administration of exogenous androgens and estrogens has been associated to a disturbed regulation of numerous aspects of the hemostatic and fibrinolytic pathways that contribute to the generation of a pro-thrombotic milieu and to a higher risk of thrombotic events (Rqsca et al, J. Clin. Med. 2021, 10:147; Abou-Ismaïl et al, Thrombosis Research 2020, 192:40-51; Walker et al, JAMA Intern Med. 2020, 180 (2): 190-197).

[0025] The activity of these molecules is mediated by nuclear receptors, the estrogen and the androgen receptor, that function as hormone-activated transcription factors. These receptors are involved in diverse activities but share a remarkable structural and functional similarity. In the absence of the ligand they exist within target cells in a transcriptionally inactive form. Upon ligand activation, they bind to the promoters of target genes and form a transcriptional complex with a number of interacting proteins, collectively known as co-regulators, that can either activate (coactivators (CoA)) or inactivate (corepressors (CoR)) transcriptional activity, thereby triggering the activation or repression of target gene expression (Patel et al, Pharm Ther 2018, 186:1-24; Zella et al, Arch Biochem Biophys 2007, 460 (2): 206-212). The interaction between nuclear receptors and coactivators is mediated by LxxLL motifs (where L is leucine and x is any amino acid) present and shared within the coactivator proteins of the different nuclear hormone receptor. These motifs are necessary and sufficient for the binding of these proteins to the receptor and for enhancing

its transcriptional activity (Patel et al, Pharm Ther 2018, 186:1-24; Zella et al, Arch Biochem Biophys 2007, 460 (2): 206-212; Chang et al, Trends Pharm Sci 2005, 26 (5): 225-228).

SUMMARY OF THE INVENTION

[0026] The present inventors have now surprisingly found that the SARS-COV-2 spike protein contains a number of LxxLL-like motifs homolog to those of the nuclear coactivator 1 (NCOA1), that are able to bind to the NCOA1 binding domain and activate the transcription activity of the estrogen and androgen receptors. Furthermore, the sequence homology of the NCOA1 binding domain throughout the different nuclear receptors supports a potential ability of the spike protein to also activate transcription induced by other nuclear receptors, such as the Vitamin D and thyroid hormone receptor.

[0027] From this finding derives that the spike protein produced or contained in vaccines can interfere with the activity of nuclear receptors, in particular the estrogen and androgen receptors, and create an unbalance between physiological and pathological activities mediated by these receptors, thus generating unwanted, serious side effects.

[0028] The present inventors have identified mutations in the LxxLL motifs of SARS-COV-2 spike protein that abolish the binding activity of the protein to the NCOA1 binding domain of the estrogen and androgen receptors.

[0029] These sequences are therefore useful for the development of safer vaccines against COVID-19.

[0030] Accordingly, an object of the invention is a mutated SARS-COV-2 spike protein or a variant thereof, wherein at least one of the aminoacids isoleucine or leucine at a position corresponding to aminoacid 818, 821, 822, 841, 844, 861, 864 and 865 of the wild type SARS-Cov-2 protein sequence (SEQ ID NO: 1) are replaced by alanine.

[0031] A further object of the invention is a mutated SARS-COV-2 spike protein according to the invention having the aminoacid sequence of SEQ ID NO: 2, SEQ ID NO: 28, SEQ ID NO: 29, SEQ ID NO: 30 or SEQ ID NO: 31.

[0032] A further object of the invention is an immunogenic fragment of a mutated SARS-COV-2 spike protein or of a variant according to the invention.

[0033] A further object of the invention is an mRNA encoding for a mutated SARS-COV-2 spike protein, a variant or an immunogenic fragment according to the invention.

[0034] A further object of the invention is said mutated SARS-COV-2 spike protein, variant, fragment, mRNA or DNA for use in the prevention of COVID-19 in a subject.

[0035] A further object of the invention is a vaccine comprising said mutated SARS-COV-2 spike protein, variant or fragment, mRNA or DNA.

[0036] A further object of the invention is said vaccine for use in the prevention of COVID-19 in a subject.

DESCRIPTION OF THE FIGURES

[0037] FIG. 1 shows the Estrogen receptor (ER) complexed with nuclear receptor coactivator. The whole complex is reported in the cartoon; the ER monomers are in grey surface, while the NCOA segments are reported in black cartoon and lines.

[0038] FIG. 2 shows the network of the most significant interactions of ER1 and ER2. Among the most reliable

interacting proteins, the Nuclear receptor COActivators (NCOAs) directly bind the nuclear receptors and stimulate transcriptional activities.

[0039] FIG. 3 shows the sequence alignment among LDX region in spike and other LDX cofactor-receptor domains.

[0040] FIG. 4 shows the 3D best docking hypothesis for spike-ER blind docking. The proteins are reported in cartoon. The ER dimer is in white cartoon while the spike protein is reported in black surface.

[0041] FIG. 5 shows the 3D best docking hypothesis for spike-ER motif-oriented docking. The proteins are reported in cartoon. The ER dimer is in white cartoon while the spike protein is reported in black surface.

[0042] FIG. 6 shows cell proliferation, expressed as percentage of control, induced by treatment of MCF-7 breast cancer cells with vehicle (CTR), estradiol (ESTR), wild type spike protein (SPIKE) or combinations of the two, either alone or in the presence of Raloxifene, measured as described in Example 2.

[0043] FIG. 7 shows cell proliferation, expressed as percentage of control, induced by treatment of MCF-7 breast cancer cells with vehicle (CTRL), peptide 2, 5 or 7, wild type spike protein, wild type spike protein plus raloxifene or raloxifene alone, measured as described in Example 2.

[0044] FIG. 8 shows cell proliferation, expressed as percentage of control, induced by treatment of LNCap prostate cancer cells with peptide 2, 5 or 7 or wild type spike protein, as described in Example 2.

[0045] FIG. 9 shows results of proteomic analysis carried with Proteome Profiler Human XL Cytokine Array Kit (THP1 cell line differentiated with TPA/LPS and treated for 24 h), after treatment with control (CTR), estrogen (ESTR), wild type spike protein (SPIKE) or raloxifene and wild type spike protein (RAL+SPIKE), as described in Example 3.

[0046] FIG. 10 shows the Tartrate Resistant Acid Phosphatase (TRAP) activity determination by Enzyme-linked immunosorbent assay (UL) in Osteoclasts differentiated from RAW264.7 macrophages and treated with 1 nM estradiol and 10 ng/ml of spike wt, or mutant SP5, SP7 or SP5+SP7, as described in Example 4b.

DETAILED DESCRIPTION OF THE INVENTION

[0047] As will be described in detail in the Experimental section, the present invention is based on the surprising finding that the SARS-COV-2 spike protein exposes three LxxLL-like motifs that are able to mediate the binding and activation of nuclear receptors, in particular of estrogen and androgen receptor. As will be shown in the Examples, each of these motifs is able to activate proliferation of cancer cells induced by the estrogen or androgen receptor.

[0048] This finding raises concerns as regards potential side effects of COVID-19 vaccines that are based on the administration or the production in the host of the SARS-COV-2 spike protein or of variants or fragments thereof that contain one or more of the above motifs. The inventors have found that the introduction of specific aminoacid mutations in correspondence with one or more of the three motifs identified abolishes the ability of the spike protein to interact with these receptors while not modifying the efficiency of the antigen-specific immune response against SARS-COV-2.

[0049] Therefore, the use in new or commercially available spike based COVID-19 vaccines of a SARS-COV-2

spike protein having a sequence carrying these mutations or an mRNA or DNA sequences encoding it, allows to maintain an equally effective immune response with a higher safety profile compared to the use of wild type SARS-COV-2 spike protein. The present invention is directed to a mutated SARS-COV-2 spike protein, variants and fragments thereof as well as mRNA and DNA encoding for the same, suitable to be used as active ingredient in vaccines for the prevention of COVID-19.

[0050] Accordingly, a first object of the invention is a mutated SARS-COV-2 spike protein or a variant thereof, wherein at least one of the aminoacids isoleucine or leucine at a position corresponding to aminoacid 818, 821, 822, 841, 844, 861, 864 and 865 of the wild type SARS-Cov-2 protein sequence (UniProtKB—PODTC2, SEQ ID NO: 1) is replaced by alanine.

[0051] Preferably, in said mutated SARS-COV-2 spike protein or variant thereof according to the invention, at least one of the aminoacids isoleucine or leucine at a position corresponding to aminoacid 818, 821, 822, 841, 861, 864 and 865 of the wild type SARS-Cov-2 protein sequence (SEQ ID NO: 1) is replaced by alanine.

[0052] Preferably, in said mutated SARS-COV-2 spike protein or variant thereof according to the invention, at least the aminoacids 818, 821 and 822, or the aminoacid 841 or the aminoacids 861, 864 and 865 of SEQ ID NO: 1 are replaced by alanine.

[0053] Preferably, in said mutated SARS-COV-2 spike protein or variant thereof according to the invention, at least the aminoacids 818, 821 and 822 or the aminoacid 841 and 844 or the aminoacids 861, 864 and 865 of SEQ ID NO: 1 are replaced by alanine.

[0054] According to one embodiment, in said mutated SARS-COV-2 spike protein or variant thereof according to the invention, the isoleucine or leucine at positions corresponding to aminoacids 818, 821, 822, 841, 861, 864 and 865 of the wild type SARS-Cov-2 protein sequence (SEQ ID NO: 1) are replaced by alanine.

[0055] According to this embodiment, the mutated SARS-COV-2 spike protein has the aminoacid sequence of SEQ ID NO: 2.

[0056] According to another preferred embodiment, in said mutated SARS-COV-2 spike protein or variant thereof according to the invention, the aminoacids isoleucine or leucine at positions corresponding to aminoacids 841 and 844 of the wild type SARS-COV-2 protein sequence (SEQ ID NO: 1) are replaced by alanine. According to this embodiment, the mutated SARS-COV-2 spike protein has the aminoacid sequence of SEQ ID NO: 28.

[0057] According to another preferred embodiment, in said mutated SARS-COV-2 spike protein or variant thereof according to the invention, the aminoacids isoleucine or leucine at positions corresponding to aminoacids 861, 864 and 865 of the wild type SARS-COV-2 protein sequence (SEQ ID NO: 1) are replaced by alanine. According to this embodiment, the mutated SARS-COV-2 spike protein has the aminoacid sequence of SEQ ID NO: 29. According to another preferred embodiment, in said mutated SARS-COV-2 spike protein or variant thereof according to the invention, the aminoacids isoleucine or leucine at positions corresponding to aminoacids 841, 844, 861, 864 and 865 of the wild type SARS-Cov-2 protein sequence (SEQ ID NO: 1) are replaced by alanine. According to this embodiment,

the mutated SARS-COV-2 spike protein has the aminoacid sequence of SEQ ID NO: 30.

[0058] According to another preferred embodiment, in said mutated SARS-COV-2 spike protein or variant thereof according to the invention, the aminoacids isoleucine or leucine at positions corresponding to aminoacids 818, 821, 822, 841, 844, 861, 864 and 865 of the wild type SARS-Cov-2 protein sequence (SEQ ID NO: 1) are replaced by alanine.

[0059] According to this embodiment, the mutated SARS-COV-2 spike protein has the aminoacid sequence of SEQ ID NO: 31.

[0060] As discussed above and demonstrated in the experimental section, the aminoacid replacements introduced in the above described mutated SARS-COV-2 spike protein are able to inhibit the activity of the spike protein on the nuclear hormone receptors and, at the same time, do not modify the conformation and antigenic properties of the SARS-CoV-2 spike protein.

[0061] The mutated SARS-COV-2 spike protein according to the invention can be modified by introducing further substitutions, or deletions or insertions that improve the stability of the mutated protein or improve the immunogenic activity against variants of concern of the SARS-COV-2 virus characterised by modifications in the aminoacid sequence of the spike protein.

[0062] Accordingly, a further object of the invention is a variant of any of the mutated SARS-CoV-2 spike protein, as described above.

[0063] In the variant according to the invention, at least one of the aminoacids isoleucine or leucine at a position corresponding to aminoacid 818, 821, 822, 841, 844, 861, 864 and 865 of the wild type SARS-Cov-2 protein sequence (SEQ ID NO: 1) is replaced by alanine. Preferably, said variant has the same aminoacids of the corresponding mutated SARS-COV-2 protein at positions corresponding to aminoacid 818, 821, 822, 841, 844, 861, 864 and 865 of the wild type SARS-Cov-2 protein sequence (SEQ ID NO: 1).

[0064] More preferably, said variant has the same aminoacid sequences of the corresponding mutated SARS-COV-2 spike protein or variant in the following positions:

[0065] aminoacid sequence from a position corresponding to aminoacid 818 of SEQ ID NO: 1 to a position corresponding to aminoacid 822 of SEQ ID NO: 1,

[0066] aminoacid sequence from a position corresponding to aminoacid 841 of SEQ ID NO: 1 to a position corresponding to aminoacid 844 of SEQ ID NO: 1, and

[0067] aminoacid sequence from a position corresponding to aminoacid 861 of SEQ ID NO: 1 to position 865 of SEQ ID NO: 1.

[0068] Preferably, the variant presents the aminoacid replacements at positions corresponding to aminoacid 818, 821, 822, 841, 844, 861, 864 and 865 of SEQ ID NO: 1 described above.

[0069] According to the present invention, by “variant” of the mutated SARS-COV-2 spike protein it is meant a protein having an aminoacid sequence which differs from the sequence of the mutated SARS-COV-2 spike protein as described above, in that it contains one or more substitutions, deletions or insertions of aminoacids. Said substitutions, deletions or insertions of aminoacids may be in any position except a position corresponding to aminoacid 818, 821, 822, 841, 844, 861, 864 and 865 of SEQ ID NO: 1. Preferably, said substitutions, deletions or insertions do not

interfere with the conformation and the antigenic properties of the mutated SARS-COV-2 spike protein according to the present invention. More preferably, said substitutions, deletions or insertions improve the antigenic properties and/or stability of the mutated SARS-COV-2 spike protein. Preferably, the variant has a sequence with an aminoacid identity with the corresponding mutated SARS-COV-2 spike protein as described above of at least 80%, 85%, 90%, 95%, 98%, 99%.

[0070] In one preferred embodiment, said variant has a sequence with aminoacid replacements that stabilize the protein in the prefusion conformation. The use as an antigen of the spike protein in this conformation induces the production of antibodies that are able to respond and inactivate the virus in the early stages of the infection and before this enters into the cells, thus significantly increasing vaccine efficacy.

[0071] A number of spike variants with aminoacid replacements that stabilize the spike protein in the prefusion state have already been identified. Among these, for example, is the mutant SARS-2-S spike protein (S-2P) having the aminoacid replacements K986P and V987P, which is encoded by the mRNA vaccine from both Pfizer/BioNTech (BNT162b) and Moderna (mRNA-1273). Furthermore, alternative variants of SARS-COV-2 spike protein with a stable prefusion conformation and a higher yield of expression compared to S-2P have also been described in the literature. These are characterized by the replacements A892P and A942P or F817P and A899P. Finally, the most promising variant as regards stability and expression described in the literature is the HexaPro variant that includes all 6 aminoacid replacements described above.

[0072] According to one preferred embodiment, in the variant according to the invention at least two of the following aminoacids of said mutated SARS-COV-2 spike protein are replaced by proline:

[0073] alanine at positions corresponding to aminoacids 892, 899 and 942 of SEQ ID NO: 1,

[0074] phenylalanine at a position corresponding to aminoacid 817 of SEQ ID NO: 1,

[0075] lysine at a position corresponding to aminoacid 986 of SEQ ID NO: 1, and

[0076] valine at a position corresponding to aminoacid 987 of SEQ ID NO: 1.

[0077] Particularly preferred variants of the mutated SARS-COV-2 spike protein according to this embodiment of the invention, that stabilize the prefusion conformation of the mutated SARS-COV-2 spike protein, are provided hereinbelow.

[0078] A first variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 2, wherein the aminoacid lysine at position 986 of SEQ ID NO: 2 and the valine at position 987 of SEQ ID NO: 2 are replaced by proline (SEQ ID NO: 3).

[0079] A second variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 2, wherein the alanines at positions 892 and 942 of SEQ ID NO: 2 are replaced by proline (SEQ ID NO: 4).

[0080] A third variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 2, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 2 and alanine at position 899 of SEQ ID NO: 2 are replaced by proline (SEQ ID NO: 5).

[0081] A fourth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 2, wherein the aminoacid lysine at position 986 of SEQ ID NO: 2, valine at position 987 of SEQ ID NO: 2 and alanine at positions 892 and 942 of SEQ ID NO: 2 are replaced by proline (SEQ ID NO: 6).

[0082] A fifth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 2, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 2, alanine at positions 892, 899 and 942 of SEQ ID NO: 2, lysine at position 986 of SEQ ID NO: 2 and valine at position 987 of SEQ ID NO: 2 are replaced by proline (SEQ ID NO: 7).

[0083] A sixth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 2, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 2, alanine at positions 892, 899 and 942 of SEQ ID NO: 2 are replaced by proline (SEQ ID NO: 8).

[0084] A seventh variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 28, wherein the aminoacid lysine at position 986 of SEQ ID NO: 28 and the valine at position 987 of SEQ ID NO: 28 are replaced by proline (SEQ ID NO: 32).

[0085] An eighth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 28, wherein the alanines at positions 892 and 942 of SEQ ID NO: 28 are replaced by proline (SEQ ID NO: 33).

[0086] A ninth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 28, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 28 and alanine at position 899 of SEQ ID NO: 28 are replaced by proline (SEQ ID NO: 34).

[0087] A tenth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 28, wherein the aminoacid lysine at position 986 of SEQ ID NO: 28, valine at position 987 of SEQ ID NO: 28 and alanine at positions 892 and 942 of SEQ ID NO: 28 are replaced by proline (SEQ ID NO: 35).

[0088] An eleventh variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 28, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 28, alanine at positions 892, 899 and 942 of SEQ ID NO: 28, lysine at position 986 of SEQ ID NO: 28 and valine at position 987 of SEQ ID NO: 28 are replaced by proline (SEQ ID NO: 36).

[0089] A twelfth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 28, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 28, alanine at positions 892, 899 and 942 of SEQ ID NO: 28 are replaced by proline (SEQ ID NO: 37).

[0090] A thirteenth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 29, wherein the aminoacid lysine at position 986 of SEQ ID NO: 29 and the valine at position 987 of SEQ ID NO: 29 are replaced by proline (SEQ ID NO: 38).

[0091] A fourteenth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 29, wherein the alanines at positions 892 and 942 of SEQ ID NO: 29 are replaced by proline (SEQ ID NO: 39).

[0092] A fifteenth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 29, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 29 and alanine at position 899 of SEQ ID NO: 29 are replaced by proline (SEQ ID NO: 40).

[0093] A sixteenth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 29, wherein the aminoacid lysine at position 986 of SEQ ID NO: 29, valine

at position 987 of SEQ ID NO: 29 and alanine at positions 892 and 942 of SEQ ID NO: 29 are replaced by proline (SEQ ID NO: 41).

[0094] A seventeenth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 29, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 29, alanine at positions 892, 899 and 942 of SEQ ID NO: 29, lysine at position 986 of SEQ ID NO: 29 and valine at position 987 of SEQ ID NO: 29 are replaced by proline (SEQ ID NO: 42).

[0095] An eighteenth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 29, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 29, alanine at positions 892, 899 and 942 of SEQ ID NO: 29 are replaced by proline (SEQ ID NO: 43).

[0096] A nineteenth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 30, wherein the aminoacid lysine at position 986 of SEQ ID NO: 30 and the valine at position 987 of SEQ ID NO: 30 are replaced by proline (SEQ ID NO: 44).

[0097] A twentieth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 30, wherein the alanines at positions 892 and 942 of SEQ ID NO: 30 are replaced by proline (SEQ ID NO: 45).

[0098] A twenty-first variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 30, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 30 and alanine at position 899 of SEQ ID NO: 30 are replaced by proline (SEQ ID NO: 46).

[0099] A twenty-second variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 30, wherein the aminoacid lysine at position 986 of SEQ ID NO: 30, valine at position 987 of SEQ ID NO: 30 and alanine at positions 892 and 942 of SEQ ID NO: 30 are replaced by proline (SEQ ID NO: 47).

[0100] A twenty-third variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 30, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 30, alanine at positions 892, 899 and 942 of SEQ ID NO: 30, lysine at position 986 of SEQ ID NO: 30 and valine at position 987 of SEQ ID NO: 30 are replaced by proline (SEQ ID NO: 48).

[0101] A twenty-fourth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 30, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 30, alanine at positions 892, 899 and 942 of SEQ ID NO: 30 are replaced by proline (SEQ ID NO: 49).

[0102] A twenty-fifth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 31, wherein the aminoacid lysine at position 986 of SEQ ID NO: 31 and the valine at position 987 of SEQ ID NO: 31 are replaced by proline (SEQ ID NO: 50).

[0103] A twenty-sixth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 31, wherein the alanines at positions 892 and 942 of SEQ ID NO: 31 are replaced by proline (SEQ ID NO: 51).

[0104] A twenty-seventh variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 31, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 31 and alanine at position 899 of SEQ ID NO: 31 are replaced by proline (SEQ ID NO: 52).

[0105] A twenty-eighth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 31, wherein the aminoacid lysine at position 986 of SEQ ID NO: 31, valine

at position 987 of SEQ ID NO: 31 and alanine at positions 892 and 942 of SEQ ID NO: 31 are replaced by proline (SEQ ID NO: 53).

[0106] A twenty-ninth variant is a variant of the mutated SARS-COV-2 protein of SEQ ID NO: 31, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 31, alanine at positions 892, 899 and 942 of SEQ ID NO: 31, lysine at position 986 of SEQ ID NO: 31 and valine at position 987 of SEQ ID NO: 31 are replaced by proline (SEQ ID NO: 54).

[0107] A thirtieth variant is a variant of the mutated SARS-Cov-2 protein of SEQ ID NO: 31, wherein the aminoacids phenylalanine at position 817 of SEQ ID NO: 31, alanine at positions 892, 899 and 942 of SEQ ID NO: 31 are replaced by proline (SEQ ID NO: 55).

[0108] In order to stabilize the SARS-COV-2 spike protein used as active ingredient in the COVID-19 vaccine, it has also been described as advantageous to use a variant of the protein that makes the protein resistant to proteases. For example, the vaccine candidate NVX-CoV2373 contains as active ingredient a variant of the SARS-COV-2 spike protein having the aminoacid replacements R682Q, R683Q, R685Q that does not interfere with antigenic properties of the protein and confers a significantly increased protease resistance. A further aminoacid replacement that prevents cleavage of the protein by the protease furine is the aminoacid replacement R685A.

[0109] According to a further preferred embodiment, also in combination with any of the above preferred embodiments, in the variants of the mutated SARS-COV-2 spike protein according to the invention, the aminoacids arginine at a position corresponding to aminoacid 682, 683 and 685 of SEQ ID NO: 1 is replaced by glutamine.

[0110] Particularly preferred variants according to this embodiment are any of the variants as described above, further containing the replacement of the aminoacids arginine at a position corresponding to aminoacid 682, 683 and 685 of SEQ ID NO: 1 with glutamine. Particularly preferred variants of the mutated SARS-COV-2 protein according to this embodiment are those having the aminoacid sequence of SEQ. ID. NO: 9 to SEQ. ID. NO: 15 (variants 31 to 37).

[0111] According to a further preferred embodiment, also in combination with any of the above preferred embodiments, in the variants of the mutated SARS-COV-2 spike protein according to the invention, the aminoacid arginine at a position corresponding to aminoacid 685 of SEQ ID NO: 1 is replaced by alanine.

[0112] Particularly preferred variants according to this embodiment are any of the variants as described above, further containing the replacement of the aminoacids arginine at a position corresponding to aminoacid 685 of SEQ ID NO: 1 with alanine.

[0113] In order to increase immune response against variants of concern of the SARS-COV-2 virus bearing modifications in the aminoacid sequence of the spike protein, vaccines are under development based on SARS-COV-2 spike proteins sequences which incorporate such modifications.

[0114] Thus, according to a further preferred embodiment, also in combination with any of the above preferred embodiments, the variants of the mutated SARS-COV-2 spike protein according to the invention, comprise the aminoacid modifications found in the spike protein of Delta (B.1.617.2); and Omicron (BA.1) virus variant.

[0115] According to this embodiment, the variant of the mutated SARS-COV-2 spike protein contains the following aminoacid modification at positions corresponding to the aminoacids of SEQ ID NO.1 listed below:

[0116] T19R, EF156/157del, R158G, L452R, T478K, D614G, P681R and D950N, or

[0117] A67V, H69del, V70del, T95I, G142D, V143del, Y144del, Y145del, N211del, L212I, ins214EPE, G339D, S371L, S373P, S375F, K417N, N440K, G446S, S477N, T478K, E484A, Q493R, G496S, Q498R, N501Y, Y505H, T547K, D614G, H655Y, N679K, P681H, N764K, D796Y, N856K, Q954H, N969K, L981F.

[0118] Particularly preferred variants according to this embodiment are any of the variants as described above, further containing the following aminoacid modification at positions corresponding to the aminoacids of SEQ ID NO.1 listed below:

[0119] T19R, EF156/157del, R158G, L452R, T478K, D614G, P681R and D950N, or

[0120] A67V, H69del, V70del, T95I, G142D, V143del, Y144del, Y145del, N211del, L212I, ins214EPE, G339D, S371L, S373P, S375F, K417N, N440K, G446S, S477N, T478K, E484A, Q493R, G496S, Q498R, N501Y, Y505H, T547K, D614G, H655Y, N679K, P681H, N764K, D796Y, N856K, Q954H, N969K, L981F.

[0121] A number of COVID-19 vaccine candidates are based on immunogenic fragments of the SARS-COV-2 spike protein or mRNA or DNA encoding for them (Forni et al, Cell Death & Differentiation 2021, 28:626-639).

[0122] As demonstrated in the experimental section, the LxxLL motifs identified by the inventors in the wild type spike protein have the ability to activate nuclear receptors.

[0123] Therefore, in the context of this type of vaccines, the use of immunogenic fragments of the mutated SARS-COV-2 spike protein or variants thereof according to the present invention is advantageous for the reasons already discussed above.

[0124] Accordingly, a further object of the present invention is an immunogenic fragment of the mutated SARS-COV-2 spike protein or of a variant thereof as described above, said fragment comprising at least one sequence of said mutated SARS-COV-2 spike protein or variant selected from:

[0125] aminoacid sequence from a position corresponding to aminoacid 818 of SEQ ID NO: 1 to a position corresponding to aminoacid 822 of SEQ ID NO: 1,

[0126] aminoacid sequence from a position corresponding to aminoacid 841 of SEQ ID NO: 1 to a position corresponding to aminoacid 844 of SEQ ID NO: 1, and

[0127] aminoacid sequence from a position corresponding to aminoacid 861 of SEQ ID NO: 1 to position 865 of SEQ ID NO: 1.

[0128] Accordingly, a further object of the present invention is an immunogenic fragment of the mutated SARS-COV-2 spike protein or of a variant thereof as described above, said fragment comprising at least one sequence of said mutated SARS-COV-2 spike protein or variant selected from:

[0129] aminoacid sequence from position 818 to position 822 of SEQ ID NO: 2 or SEQ ID NO: 31.

[0130] aminoacid sequence from position 841 to position 844 of SEQ ID NO: 2, SEQ ID NO: 28, SEQ ID NO: 30 or SEQ ID NO: 31, and

[0131] aminoacid sequence from position 861 to position 865 of SEQ ID NO: 2, SEQ ID NO: 29, SEQ ID NO: 30 or SEQ ID NO: 31.

[0132] Accordingly, a further object of the present invention is an immunogenic fragment of the mutated SARS-COV-2 spike protein or a variant thereof as described above, said fragment comprising a sequence corresponding to the aminoacid sequences from aminoacid 818 to aminoacid 865 of SEQ ID NO: 2, SEQ ID NO: 28, SEQ ID NO: 29, SEQ ID NO: 30 or SEQ ID NO: 31.

[0133] A “fragment” of the mutated SARS-COV-2 spike protein or a variant thereof according to the present invention refers to proteins having an aminoacid sequence which correspond to an N-terminally and/or C-terminally truncated sequence of a mutated SARS-COV-2 spike protein or variant thereof according to the invention.

[0134] The immunogenic fragments according to the invention contain epitopes able to trigger protective immune responses in the host.

[0135] Epitopes of wild type SARS-COV-2 spike protein that have a reduced risk of triggering autoimmune response and stimulate the production of protective antibodies have been identified, corresponding to sequences 232-246, 233-247, 471-503, 604-625, 817-833, 891-907, 897-913, 1164-1191, 1182-1209 of SEQ ID NO: 1. On the contrary it was found that the peptide 597-603 induces antibodies that enhance infection through an epitope sequence-dependent.

[0136] Accordingly, preferably, the immunogenic fragments according to the invention further contain at least two, non-overlapping sequences selected from the following aminoacid sequences:

[0137] aminoacid sequence from position 232 to position 246 of SEQ ID NO: 1,

[0138] aminoacid sequence from position 233 to position 247 of SEQ ID NO: 1,

[0139] aminoacid sequence from position 471 to position 503 of SEQ ID NO: 1,

[0140] aminoacid sequence from position 604 to position 625 of SEQ ID NO: 1,

[0141] aminoacid sequence from position 817 to position 833 of SEQ ID NO: 2 or SEQ ID NO: 31,

[0142] aminoacid sequence from position 891 to position 907 of SEQ ID NO: 1

[0143] aminoacid sequence from position 897 to position 913 of SEQ ID NO: 1,

[0144] aminoacid sequence from position 1164 to position 1191 of SEQ ID NO: 1, and

[0145] aminoacid sequence from position 1182 to position 1209 of SEQ ID NO: 1.

[0146] Preferably, the immunogenic fragment according to the invention comprises the aminoacid sequence from position 865 to position 1209 of SEQ ID NO: 2, SEQ ID NO: 29, SEQ ID NO: 30 or SEQ ID NO: 31.

[0147] Preferably, the immunogenic fragment according to the invention comprises the aminoacid sequence from position 471 to position 817 of SEQ ID NO: 1.

[0148] Preferably, the immunogenic fragment according to the invention comprises the aminoacid sequence from position 232 to position 470 of SEQ ID NO: 1.

[0149] Preferably, the variants or fragments of the invention do not contain the aminoacid sequence from position 597 to position 603 of SEQ ID NO: 1.

[0150] A further object of the present invention are proteins comprising the aminoacid sequence of a mutated SARS-COV-2 spike protein, variant or fragment thereof according to the present invention and one or more additional sequences.

[0151] According to one preferred embodiment, said additional sequences are not immunogenic and are used to confer the protein or fragment specific functional properties, for example as regards tissue distribution.

[0152] According to another preferred embodiment, said additional sequences are able to stimulate the immune response towards epitopes of the mutated SARS-COV-2 spike protein, variant or fragment thereof.

[0153] A further object of the present invention is an mRNA encoding for the above described mutated SARS-COV-2 spike protein, variant, peptide or protein.

[0154] According to a preferred embodiment, in order to increase stability and avoid in vivo degradation, said mRNA is a nucleoside modified mRNA sequence.

[0155] According to a preferred embodiment, in said nucleoside modified mRNA sequence uridine residues are replaced with a modified nucleoside selected from pyridin-4-one ribonucleoside, 5-aza-uridine, 2-thio-5-azauridine, 2-thiouridine, 4-thio-pseudouridine, 2-thio-pseudouridine, 5-hydroxyuridine, 3-methyluridine, 5-carboxymethyl-uridine, 1-carboxymethyl-pseudouridine, 5-propynyl-uridine, 1-propynyl-pseudouridine, 5-taurinomethyluridine, 1-taurinomethyl-pseudouridine, 5-taurinomethyl-2-thio-uridine, 1-taurinomethyl-4-thiouridine, 5-methyl-uridine, 1-methyl-pseudouridine, 4-thio-1-methyl-pseudouridine, 2-thio-1-methylpseudouridine, 1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-1-deaza-pseudouridine, dihydrouridine, dihydropseudouridine, 2-thio-dihydrouridine, 2-thio-dihydropseudouridine, 2-methoxyuridine, 2-methoxy-4-thio-uridine, 4-methoxy-pseudouridine, 4-methoxy-2-thio-pseudouridine, preferably with 1-methyl-pseudouridine.

[0156] According to another embodiment, preferably in order to further stabilize the mRNA, the coding region of the mRNA according to the invention contain an increased G/C compared to the G/C content present in the corresponding coding region for the wild type SARS-COV-2 spike protein. This is obtained by introducing into the sequence synonymous codon substitutions that allows an increase of G/C ratio.

[0157] According to another embodiment, preferably in the mRNA according to the present invention codon usage has been optimized for expression in human cells.

[0158] A further object of the present invention is a DNA sequence encoding for the mutated SARS-COV-2 spike protein, a variant or a peptide thereof as described above.

[0159] According to a preferred embodiment, said DNA is contained in a plasmid. Preferably, said plasmid contains at least a promoter and a stop codon.

[0160] According to an alternative preferred embodiment, said DNA is contained in a viral vector. Said viral vector may be non-replicating or replicating. Said viral vector is preferably selected from adeno or pox viruses. More preferably, said viral vector is selected from an adenovirus type 5 (Ad5) vector, adenovirus type 26 (Ad26) vector, replication-deficient simian adenovirus ChAdOx1 vector.

[0161] The proteins, mRNA or DNA according to the present invention can be prepared by methods well known in the art.

[0162] A further object of the present invention is a mutated SARS-COV-2 spike protein, a variant or peptide thereof or an mRNA or DNA encoding them, as above described, for use in the prevention of COVID-19 in a subject.

[0163] A further object of the present invention is a vaccine composition containing a mutated SARS-COV-2 spike protein, a variant or fragment thereof or an mRNA or DNA encoding them, as above defined.

[0164] The vaccine composition is preferably for use in prevention of COVID-19.

[0165] The vaccine can be formulated in accordance with well-known techniques for the preparation of protein or nucleic acids-based vaccines, as for example described in Batty et al, *AdvDrug Delivery Rev* 2021, 169:168-189.

[0166] Preferably, said vaccine contains an adjuvant agent.

[0167] An “adjuvant” according to the invention is a component that is able to enhance the immunostimulatory properties of the composition.

[0168] Said adjuvant is selected from adjuvants well known to the skilled person and suitable for the specific antigen and formulation used.

[0169] The vaccine according to the invention can be administered by parenteral administration, preferably intradermal, intramuscular or subcutaneous administration, oral administration or nasal administration.

EXPERIMENTAL PART

Example 1

Bioinformatic Analysis

Materials and Methods

1. Interactome Analysis

[0170] The STRING database [Szklarczyk D et al., *Nucleic Acids Res.* 2021, 49: D605-D612, doi: 10.1093/nar/gkaa1074], that integrate all known and predicted associations between proteins, including both physical interactions as well as functional associations was used to analyse functional associations between biomolecules. Each protein-protein interaction is annotated with a ‘score’. This score does not indicate the strength or the specificity of the interaction, but only the confidence. All scores rank from 0 to 1, with 1 being the highest possible confidence.

2. 3D Model Selection

[0171] Spike 3D model was built based on PDB 6VYB returned to its wild-type form and fully glycosylated. An asymmetric glycosylation of the three protomers was derived by glycoanalytic data for the N-glycans and O-glycans as published (Casalini L et al, *ACS Cent. Sci.* 2020, 6:1722-1734, doi.org/10.1021/acscentsci.0c01056). The proteins were modeled using Amber14SB force field (Maier J A et al, *J. Chem. Theory Comput.* 2015, 11:3696-3713, doi.org/10.1021/acs.jctc.5b00255) and the carbohydrate moieties by the GLYCAM06j-1 version of GLYCAM06 force field (Kirschner K N et al, *J. Comput. Chem.* 2008, 29:622-655, doi.org/10.1002/jcc.20820). The so prepared

structure was used as starting point for Molecular Docking simulations. Topology files were generated with the pdb2gmX GROMACS tool, using the amber99sb forcefield (Lindorff-Larsen K et al, *Proteins* 2010,78:1950-1958, dx.doi.org/10.1002/prot.22711). Protein was inserted in a triclinic box, extending up to 15 Å from the solute, and immersed in TIP3P water molecules (Jorgensen W L et al, *J Chem Phys* 1983, 79:926-935, dx.doi.org/10.1063/1.445869). Counter ions were added to neutralize the overall charge with the genion GROMACS tool. After energy minimizations, the system was relaxed for 5 ns by applying positional restraints of 1000 KJ mol⁻¹ nm⁻² to the protein atoms. Following this step, unrestrained MD simulation was carried out for a length of 1 microsecond, with a time step of 2 fs, using GROMACS 2018.3 simulation package (supercomputer Galileo and Marconi-100, CINECA, Bologna, Italy) (Abraham M J et al., *SoftwareX* 2015, 1:19-25, dx.doi.org/10.1016/j.softx.2015.06.001). V-rescale temperature coupling was employed to keep the temperature constant at 300 K (Bussi G et al, *J Chem Phys*, 2007 126: 014101. dx.doi.org/10.1063/1.2408420). The Particle-Mesh Ewald method was used for the treatment of the long-range electrostatic interactions (Darden T et al., *J. Chem. Phys* 1993, 98:10089, doi.org/10.1063/1.464397). The first 5 ns of each trajectory were excluded from the analysis. The trajectory obtained after 1 microsecond MD simulation was clustered in order to obtain representative structures. In particular, the structure used for the docking studies is the first centroid of the first cluster extracted from the MD experiment.

[0172] For the estrogen receptor, the XRAY PDB model with code 3OLL was used, containing Estradiol and Nuclear receptor coactivator 1 (Möcklinghoff S et al, *ChemBioChem* 2010, 11:2251-2254, doi.org/10.1002/cbic.201000532).

3. Protein-Protein Docking Procedure

[0173] The input of two individual proteins, one for the receptor and the other for the ligand, were provided. In particular, the Spike protein and ER were used as receptor and ligand, respectively. Then, the HDock tool performs docking to sample putative binding modes through an FFT-based search method, and then scoring the protein-protein interactions. Finally, the top 100 predicted complex structures were provided, and the best ten hypotheses were visually inspected to confirm the reliability of the calculation. The entire workflow is described (Yan Y et al, *Nat Protoc* 2020, 15:1829-1852, doi.org/10.1038/s41596-020-0312-x).

[0174] Given the lack of structural information on the binding mechanism between the Spike protein and the nuclear estrogen receptor (ER), the first check involved the identification of proteins known to interact with estrogen receptors. After identifying these ER interacting proteins, the second step involved the study of any sequence analogies between the spike protein and the ER effector proteins.

Example 1a

Nuclear Receptor Coactivators and LxxLL Motif

[0175] Many transcription factors and co-factors exhibit a common structural motif that ensures interaction with effector proteins. The motif that participates in these protein-protein interactions is termed LxxLL (LXX), and it is

associated with different aspects of transcriptional regulation (Plevin M J et al, Trends Biochem Sci 2005, 30:66-69, dx.doi.org/10.1016/j.tibs.2004.12.001). The LxxLL sequence was originally identified in proteins that bind the activation function-2 (AF-2) region of nuclear receptor ligand-binding domains (LBDs). These motifs are fundamentals in nuclear-receptor regulation with many nuclear-receptor-binding proteins, including co-activators (NCOA-1, 2 and 3) (Heery D M et al, Nature 1997, 387:733-736, doi.org/10.1038/42750).

[0176] The experimental and structural confirmations which show the interaction between ER and NCOAs, made it possible to focus on the LxxLL motif and starting from this, a mapping of the spike protein sequence was carried out searching for structural motifs and homologous portions of spike able to mimic the interaction between the ER and its nuclear co-activators (FIG. 1).

Example 1b

Estrogen Receptor Binding Protein Network Analysis

[0177] The network of most significant interactions of ER1 and ER2 (FIG. 2) highlights that, among the most reliable interacting proteins with ER1 and ER2, the Nuclear receptor COActivators (NCOAs) directly bind the nuclear receptors and stimulate transcriptional activities. Combining the sequence alignment between NACOs and spike with a three-dimensional analysis of the viral protein, it emerged that the spike protein contains a LDX motif (LPPLL, aa 861-865 in spike wild type) and two motifs homologous to LxxLL (IEDLL, aa 818-822 and LGDIA, aa 841-846, in spike wild type), also valid from a structural point of view (in fact, these regions assume alpha-helix conformations), in an outer zone which could, in principle, act as interacting region site with ER. Different LDX domain are associated to specific cofactor-receptor interaction and the spike LDX1 resemble NCOA1 LDX4 and belong to Class III domain (FIG. 3) (Savkur R S et al, J Pept Res 2004, 63:207-212, dx.doi.org/10.1111/j.1399-3011.2004.00126.x

Example 1c

Spike Protein-ER Blind Docking

[0178] It has been experimentally verified that ER does not bind the Receptor Binding Domain (RBD) of the viral protein (data not shown). To confirm and validate the in silico prediction, the ability of ER to interact in a region other than the viral RBD was evaluated by means of a blind-docking between the two proteins, using the HDock server [hdock.phys.hust.edu.cn]. The blind-docking approach does not consider any structural bias and is fully unguided. The best binding hypothesis found puts in evidence a high affinity of ER towards the lateral region of the spike protein, belonging to the so called "fusion peptide portion" (FIG. 4).

Example 1d

Spike Protein-ER Motif-Oriented Docking

[0179] The structural information that ER residues are recognized by NCOAs was used to guide the docking study of ER on spike by optimizing protein-protein interactions. With the protein structures and residue restraints as input, a

suitable model was generated by the HDock server. Given that the structural information of interaction between ER and NCOA are known, a second docking study was carried out, that considered the ER residues that guarantee interaction with the NCOAs, thus guiding the molecular docking procedure. The best binding hypothesis obtained from the guided docking study is shown in FIG. 5 that highlights the binding of the ER to the spike region containing the motifs homologous to the LxxLL pattern.

Example 2

In Vitro Evaluation of Proliferation Activity in Cancer Cell Lines

Materials and Methods

Peptides 2, 5 and 7

[0180] Starting from homologous and LxxLL-like cores, three peptides, named 2, 5 and 7, having the following aminoacid sequences were synthesized:

Peptide 2:	(SEQ ID NO: 16)
SKRSFIEDLLFNKVTL	
Peptide 5:	(SEQ ID NO: 17)
IKQYGDCLGDIAARDLI	
Peptide 7:	(SEQ ID NO: 18)
NGLTVLPPLLTDEMI	

[0181] Each of the peptides contains one of the LxxLL motifs found in the wild type spike protein. The peptides were synthesized by CASLO ApS (Technical University of Denmark, DTU-Science Park Diplomvej 381, DK-2800 Kongens Lyngby, Denmark), by using a Fmoc-based solid-phase peptide synthesis protocol. Each peptide was acetylated at N-terminus and amidated at C-terminus; they were stored as lyophilized hydrochloride salts. The purity of the peptides was assessed by analytical RP-HPLC (C18-250 mm×4.6 mm I.D., flow rate of 1 mL/min; absorbance detected at 220 nm) and verified also by MALDI-TOF mass spectrometry.

Testing Concentrations

[0182] Spike protein was used at the final concentration of 10 ng/ml in medium. Raloxifene was used at 2 µM final concentration in medium. Estradiol stock solution was prepared in ethanol and used at the final concentration of 1 nM diluted in medium. Peptides were diluted in PBS at 4 mg/ml concentration and then 10 µg/ml was the concentration used for the assay (final concentration in medium).

Proliferation Assay

[0183] To evaluate cell proliferation, BrdU assay was performed (Abcam, UK). BrdU is incorporated into newly synthesized DNA of actively proliferating cells. 20 µL of the diluted 1×BrdU label were added to the appropriate wells and incubated 24 hours. BrdU was incorporated into the DNA of dividing cells. To enable antibody binding to the incorporated BrdU cells must be fixed, permeabilized, and the DNA denatured. This is all done in one step by treatment

with Fixing Solution. Detector anti-BrdU monoclonal anti-body is pipetted into the wells and allowed to incubate for one hour, during which time it binds to any incorporated BrdU. Different washes were performed to wash away the unbound antibody, and horseradish peroxidase-conjugated goat anti-mouse antibody was added, which binds to the Detector Antibody. Then, TMB Peroxidase substrate was added and incubated for 30 minutes at room temperature in the dark. Finally, to stop the reaction, the Stop solution was added. The plate was read using a spectrophotometric micro-titer plate reader set at 450 nm

Breast Cancer Cell Line

[0184] MCF-7 cell line is an invasive ductal/breast carcinoma cell hormone-dependent (both estrogen and progesterone receptor-positive).

[0185] MCF-7 was obtained from ATCC and grown in DMEM (Dulbecco's Modified Eagle Medium), supplemented with 10% fetal bovine serum (FBS, Corning, USA), penicillin/streptomycin and glutamine (Sigma, USA) at 37° C. in a 5% CO2 and 95% humidified incubator (Thermo, USA). Before treatments, to reduce estrogen levels in FBS the cells were cultured for 24 h in phenol red-free medium containing 5% dextran-coated charcoal treated serum, followed by incubation in phenol-red free, serum-free medium, supplemented with 0.2% BSA for at least 24 h.

[0186] In a first set of experiments, cells were treated with estradiol, wild type spike protein, 5 raloxifene, estradiol and raloxifene, wild type spike protein and raloxifene, estradiol, spike protein and raloxifene for 24 hours and proliferation was measured.

[0187] The results are shown in FIG. 6. As can be seen both estradiol and spike were able to substantially increase the proliferation of cells and this activity was counteracted by co-treatment with raloxifene. These data demonstrate that wild type spike is able to induce proliferation of breast cancer cells with an ER mediated mechanism.

[0188] In a second set of experiments, cells treated with peptides 2, 5 and 7, wild type spike, spike plus raloxifene or raloxifene alone for 24 hours and proliferation was measured. The results are shown in FIG. 7. The mean and SD are shown in Table 1 below:

TABLE 1							
MCF-7							
	ctrl	2	5	7	spike	spike + ral	ral
Mean	100.0	153.7	170.6	205.5	179.9	165.3	109.5
SD	0.53	1.31	1.54	0.97	0.57	2.13	0.68

[0189] As can be seen, the peptides are all able to induce proliferation similar to that induced by spike. The data obtained demonstrated that each of the LxxL motifs in wild type spike protein are able to induce activation of the estrogen receptor.

Prostate Cancer Cell Line

[0190] Established androgen-dependent cell line LNCap was used as a model for this study. Cells were cultured in DMEM, supplemented with 10% fetal bovine serum (FBS, Corning, USA), penicillin/streptomycin and glutamine (Sigma, USA) at 37° C. in a 5% CO2 and 95% humidified

incubator (Thermo, USA). Before treatments, to reduce estrogen levels in FBS, the cells were cultured for 24 h in phenol red-free medium containing 5% dextran-coated charcoal treated serum, followed by incubation in phenol-red free, serum-free medium, supplemented with 0.2% BSA for at least 24 h. Then cells were treated with peptides 2, 5 and 7 or wild type spike for 24 h and proliferation was measured.

[0191] The results are shown in FIG. 8. The mean and SD are shown in Table 2 below:

TABLE 2					
LNCaP					
	ctrl	2	5	7	spike
Mean	100.0	242.2	236.2	230.2	245.9
SD	0.21	0.72	0.24	0.7	0.29

[0192] The results obtained demonstrate that both the peptides and spike protein are able to induce proliferation, by activation of the androgen receptor.

Example 3

Proteomic Analysis

[0193] Human monocytes THP-1 were purchased from ATCC and cultured following manufacturer's instructions. Then, THP-1 were differentiated using 20 ng/ml phorbol 12-myristate 13-acetate (PMA) for 72 hours, followed by LPS stimulation (100 ng/ml) for 24 hours and treated with estradiol, raloxifene and wild type spike, wild type spike+ raloxifene, estradiol+wild type spike, estradiol+raloxifene+ wild type spike, at the concentrations used in Example 2, for 24 h. The media were then collected and the proteome profile assay was carried out.

[0194] In details, conditioned media from THP-1 treated cells were centrifuged to remove particulates and assayed immediately. The sample amount was adjusted as suggested (500 µl). The reagents were prepared following manufacturer's protocols (R&D, USA). Briefly, the membranes were incubated with Array Buffer 6 for 1 hour on a rocking platform shaker. The samples were prepared by adding 1 ml of Array Buffer 4 in two separated tubes and then 15 µl of reconstituted mouse cytokine detection antibody cocktail (>40 mouse cytokines) to each prepared sample and incubated 1 hour. Then, the Array Buffer 6 was aspirated and replaced with the sample/antibody mixtures and incubated at 4° C. overnight. The following day, membranes were washed 3 times, and Streptavidin-HRP (1:2000) was incubated for 30 minutes at room temperature on a rocking platform shaker. The membranes were washed again, and 1 ml of prepared Chemi Reagent mix was placed onto each membrane. Multiple exposure times were acquired using UVITEC digital analyzer (Alliance, Cambridge, UK). The positive signals seen in the developed membranes can be identified by placing the transparency overlay template on the array image and aligning it with pairs of reference spots in the three corners of each array. Reference spots are included to demonstrate that the array has been incubated with Streptavidin-HRP during the assay procedure. Pixel densities (average signals of pair of duplicates which represent each cytokine) were analyzed by Fiji software, and the average background was subtracted from each spot.

[0195] The results are reported in FIG. 9. As can be seen, IL-8 appears the most abundant, being up-regulated by both estradiol and wild type spike. Raloxifene was effective in counteracting the effect of spike. The non-enzymatic chitinase-3 like-protein-1 (CHI3L1) is up-regulated by both estradiol and wild type spike but raloxifene was able to counteract only the effects of spike alone.

[0196] Osteopontin (OPN) was upregulated by both estradiol and wild type spike and raloxifene was able to counteract the effects of spike alone

Example 4

Production and Purification of Wild Type Spike Protein and Mutants

[0197] The nucelotide sequences encoding for aminoacids from 16 to 1213 of the wild type spike protein (extracellular domain) and for the corresponing SP5, SP7 and SP5+SP7 mutants described below were generated and inserted into a plasmid DNA vector having the Human cytomegalovirus (CMV) immediate early enhancer and promoter, the murine Ig kappa chain leader sequence for protein secretion, the bovine growth hormone polyadenylation (bGH-PolyA) signal, a specialized termination sequence for protein expression in eukaryotic cells and the kanamycin resistance gene from *Staphylococcus aureus* for plasmid amplification in bacteria.

[0198] In details the mutated sequences were as follows:

[0199] SP5: mutations L841A, 1844A, with reference to SEQ. ID NO: 1,

[0200] SP7: mutations L861A, L864A, L865A with reference to SEQ. ID NO: 1,

[0201] SP5+SP7: mutations L841A, 1844A, L861A, L864A, L865A with reference to SEQ. ID NO: 1,

each also containing the stabilising aminoacid replacements R685A, K986P and V987P, with reference to SEQ. ID NO: 1

[0202] The spike protein and SP5, SP7 and SP5+SP7 were produced by transient transfection of Expi293F high-density cells with the ExpiFectamine 293 (Thermo Fisher) lipid cationic transfection reagent according to the manufacturer's instructions. The supernatant containing the proteins was collected after 5 days of incubation from start of transfection and subjected to clarification by centrifugation and filtration for the subsequent purification steps. The proteins were batch purified using IMAC (Immobilized Metal Chelate Affinity Chromatography) using PureCube Ni-NTA Agarose resin slurry (Cube-Biotech). Briefly, the resin slurry was centrifuged and incubated with equilibration buffer (50 mM NaH2PO4, 500 mM NaCl, pH 7.4, 10 mM imidazole). The equilibrated resin was combined with the culture supernatant containing the recombinant proteins and incubated o/n, 4° C., on a rotating platform. The resin was subsequently collected by centrifugation, washed, and the protein was eluted by an elution buffer containing 300 mM Imidazole and subjected to dialysis in phosphate buffer (PBS) using slide-A-lyzers (Thermo Fisher) as indicated in the product datasheet. Once recovered from dialysis, the proteins were quantified on a spectrophotometer measuring the absorbance at 280 nm. The purity of the proteins was assessed by SDS-PAGE and Western Blot analysis, conducted under both reduced and non-reducing conditions. Further characterization of their binding capacity to the

hACE2 receptor and the related affinity was evaluated by BioLayer interferometry BLI as described in example 4a.

Example 4a

hACE-2 Receptor Binding Assay

[0203] Binding studies were performed in order to verify the ability of produced spike proteins (wt and SP5, SP7 and SP5+SP7) to bind the hACE2 receptor by using the Octet Red (Forte Bio) system. All steps were performed at 25° C. with stirring at 600 rpm in a 96-well plate (96-well microplate, bottom F, black, 655209, from Greiner bio-one) containing 200 µl of solution in each well. Kinetic buffer 1× (cat. No. 18-1105, Forte Bio) was used in this study for the dilution of all proteins and analytes and for washing of the sensors. Kinetic assays were performed by first capturing the hACE2-hFc protein using anti-human Fc Octet biosensors (anti-human IgG Fc Capture Biosensors, cat. No. 18-5060, FORTEBIO). The biosensors were immersed for 10 minutes in 1× buffer followed by a measurement of the basal signal for 60 s, at this point the hACE2-hFc was loaded for 300 s (until the biosensor was completely saturated). After a wash step in 1× kinetic buffer for 120 s, the biosensor tips conjugated with hACE2-hFc were immersed for 300 s in wells containing different concentrations of antigen (spike wt protein and its mutants SP5, SP7 and SP5+SP7) to evaluate the association curves, followed by 900 s of dissociation time in kinetic buffer. The binding curve data was collected and then analyzed using the data analysis software v11.1 (FORTEBIO). The binding sensorgrams were aligned at the start of the antigen binding cycle and after the single reference subtraction. Kd values were calculated using a 1:1 global Langmuir binding model.

[0204] The biosensor tips with the various antigens were also immersed in wells containing kinetic buffer to allow for single reference subtractions to compensate for the natural dissociations of the captured antigens. The tips of the biosensor were used without regeneration.

[0205] The results, shown in Table 3 below, demonstrates that the wt spike protein and its mutants bind the hACE2 receptor with high affinity.

TABLE 3

	Spike WT	Spike SP5	Spike SP7	Spike SP5 + SP7
KD (M)	6.36×10^{-10}	2.55×10^{-10}	4.15×10^{-10}	2.38×10^{-8}
KD error (M)	5.92×10^{-12}	4.73×10^{-12}	5.13×10^{-12}	5.00×10^{-9}

Example 4b

Estrogen-Like Activity

[0206] To evaluate estrogen-like activity of the spike proteins the Tartrate Resistant Acid Phosphatase (TRAP) activity was evaluated by Enzyme-linked immunosorbent assay (Myobiosource, cat no. MBS1601167).

[0207] RAW264.7 (murine macrophages ATCC, USA) were cultured as manufacturer's protocol. Then 1.5×10^5 cells/cm² in 24-well dishes were seeded and mouse receptor activator of nuclear factor kb ligand (RANKL, Miltenyi Biotec, Germany) was added at the final concentration of 35 ng/ml to initiate osteoclasts (OC) development (day 0) as previously described (Collin-Osdoby and Osdoby 2012,

PMID: 22130930). At day 3, cells were examined under the microscope and refed with fresh medium containing RANKL. At day 6, RAW-OC population was prevalent and ready for treatments and then biochemical studies. Cells were treated with 1 nM Estradiol or with 10 ng/ml spike wt, SP5, SP7 or SP5+SP7 for 24 h. After 24 h of treatment, cells were quickly collected by sterile tubes and resuspended using PBS (pH 7.4), to dilute cell suspension to the concentration of approximately 1 million/ml. Then, cells were subjected to repeated freeze-thaw cycles to let out the inside components. In the meantime, the reagents of the kit were brought to room temperature. The standard curve, reagents and samples were prepared following manufacturer's protocol. Briefly, 50 μ l of standard were added to "standard wells" and 40 μ l of sample to "sample wells". Then 10 μ l of anti-TRAP antibody were added to sample wells and 50 μ l of streptavidin-HRP to sample wells and standard wells. The plate was covered with a sealer, mixed well on a rocking platform, and incubated 1 hour at 37° C. The plate was washed 5 times with wash buffer and 50 μ l of substrate solution A were added to each well plus 50 μ l of substrate

solution B and incubated 10 minutes at 37° C. in the dark. Finally, 50 μ l of stop solution to each well were added and the optical density was immediately determined using a microplate reader set at 450 nm. The results are shown in FIG. 10. The mean and SD are shown in Table 4 below:

TABLE 4

Osteoclasts from RAW264.7 macrophages (TRAP levels U/L)							
	CTR w/o RANKL	CTR with RANKL	Estradiol	Spike WT	Spike SP5	Spike SP7	Spike SP5 + SP7
Mean	0.18	2.02	0.25	0.26	1.90	2.11	2.04
SD	0.05	0.71	0.05	0.05	0.24	0.14	0.24

[0208] The results demonstrate that spike wt exerts the same effect on the cells observed with estradiol. On the contrary, all mutants SP5, SP7 and SP5+SP7 showed the same TRAP levels of the control stimulated with RANKL, indicating that those mutations prevent the ability of spike protein to activate the estrogen receptor.

SEQUENCE LISTING

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<213> ORGANISM: SARS coronavirus
<220> FEATURE:
<223> OTHER INFORMATION: Wild-type Sars CoV-2 spike protein

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Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
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His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
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Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65            70            75            80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85            90            95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100           105           110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
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Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
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Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
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Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser
			245						250					255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
			260					265					270		
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala
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Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys
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Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala
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Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
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Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
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Asp Cys Thr Met Tyr Ile	Cys Gly Asp Ser Thr	Glu Cys Ser Asn Leu
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805	810	815
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835	840	845
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Thr Leu Val Lys Gln Leu	Ser Ser Asn Phe Gly	Ala Ile Ser Ser Val
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Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
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His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
			50				55						60		

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Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp	65	70	75	80
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Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	130	135	140	
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Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175	
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Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205	
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220	
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235	240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270	
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285	
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300	
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315	320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350	
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365	
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380	
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395	400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430	
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445	
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460	

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Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475	480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510	
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525	
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	530	535	540	
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	545	550	555	560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	565	570	575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	580	585	590	
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	595	600	605	
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	610	615	620	
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	625	630	635	640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	645	650	655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	660	665	670	
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	675	680	685	
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	690	695	700	
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	705	710	715	720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	725	730	735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	740	745	750	
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	755	760	765	
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	770	775	780	
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	785	790	795	800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	805	810	815	
Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	820	825	830	
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg	Asp	835	840	845	
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	850	855	860	
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly				

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865	870	875	880
Thr Ile Thr Ser Gly Trp Thr Phe Gly Ala Gly Ala Ala Leu Gln Ile			
	885	890	895
Pro Phe Ala Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr			
	900	905	910
Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn			
	915	920	925
Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala			
	930	935	940
Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn			
	945	950	955
		960	
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val			
	965	970	975
Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln			
	980	985	990
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val			
	995	1000	1005
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn			
	1010	1015	1020
Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys			
	1025	1030	1035
Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro			
	1040	1045	1050
Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val			
	1055	1060	1065
Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His			
	1070	1075	1080
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn			
	1085	1090	1095
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln			
	1100	1105	1110
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val			
	1115	1120	1125
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro			
	1130	1135	1140
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn			
	1145	1150	1155
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn			
	1160	1165	1170
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu			
	1175	1180	1185
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu			
	1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu			
	1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met			
	1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys			
	1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro			
	1250	1255	1260

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Val Leu Lys Gly Val Lys Leu His Tyr Thr
1265 1270

<210> SEQ ID NO 3
 <211> LENGTH: 1273
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Variant 1 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 3

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1 5 10 15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20 25 30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35 40 45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50 55 60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65 70 75 80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85 90 95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100 105 110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115 120 125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130 135 140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145 150 155 160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165 170 175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180 185 190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195 200 205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210 215 220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225 230 235 240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245 250 255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260 265 270

Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
275 280 285

Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
290 295 300

Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305 310 315 320

Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
325 330 335

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Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	
			340					345					350			
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	
			355				360					365				
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	
	370					375					380					
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	
385					390					395					400	
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	
				405					410					415		
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	
			420					425					430			
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	
			435				440					445				
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	
	450					455					460					
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	
465					470					475					480	
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	
				485					490					495		
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	
			500					505					510			
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	
		515					520					525				
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	
	530					535					540					
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	
545					550					555					560	
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	
				565					570					575		
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	
			580					585					590			
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	
		595					600					605				
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	
	610					615					620					
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	
625				630						635					640	
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
				645					650					655		
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
			660					665					670			
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	
		675					680					685				
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	
	690					695					700					
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	
705					710					715					720	
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	
				725					730						735	

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Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740					745						750	
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
		755					760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
	770					775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
785					790					795					800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
			805						810					815	
Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg	Asp
		835					840					845			
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
	850					855					860				
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865					870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile
			885						890					895	
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900					905					910		
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
		915					920					925			
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala
	930					935						940			
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
945				950					955						960
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
			965						970					975	
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln
			980					985					990		
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
	995						1000					1005			
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
	1010					1015						1020			
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
	1025					1030					1035				
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
	1040					1045					1050				
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
	1055					1060					1065				
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
	1070					1075					1080				
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn	
	1085					1090					1095				
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln	
	1100					1105					1110				
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val	
	1115					1120					1125				
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro	

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1130	1135	1140
Glu Leu Asp Ser Phe Lys	Glu Glu Leu Asp Lys Tyr	Phe Lys Asn
1145	1150	1155
His Thr Ser Pro Asp Val Asp	Leu Gly Asp Ile Ser	Gly Ile Asn
1160	1165	1170
Ala Ser Val Val Asn Ile Gln	Lys Glu Ile Asp Arg	Leu Asn Glu
1175	1180	1185
Val Ala Lys Asn Leu Asn Glu	Ser Leu Ile Asp Leu	Gln Glu Leu
1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile	Lys Trp Pro Trp Tyr	Ile Trp Leu
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile	Ala Ile Val Met Val	Thr Ile Met
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys	Cys Ser Cys Leu Lys	Gly Cys Cys
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys	Phe Asp Glu Asp Asp	Ser Glu Pro
1250	1255	1260
Val Leu Lys Gly Val Lys Leu	His Tyr Thr	
1265	1270	

<210> SEQ ID NO 4

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 2 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 4

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val		
1	5	10 15
Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe		
20	25	30
Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu		
35	40	45
His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp		
50	55	60
Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp		
65	70	75 80
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu		
85	90	95
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser		
100	105	110
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile		
115	120	125
Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr		
130	135	140
Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr		
145	150	155 160
Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu		
165	170	175
Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe		
180	185	190
Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr		

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195	200	205
Pro Ile Asn Leu Val Arg Asp 210	Leu Pro Gln Gly Phe Ser Ala Leu Glu 215	
Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr 225	230	235 240
Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser 245	250	255
Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro 260	265	270
Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala 275	280	285
Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys 290	295	300
Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val 305	310	315 320
Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys 325	330	335
Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala 340	345	350
Trp Asn Arg Lys Arg Ile Ser Asn Cys Val Ala Asp Tyr Ser Val Leu 355	360	365
Tyr Asn Ser Ala Ser Phe Ser Thr Phe Lys Cys Tyr Gly Val Ser Pro 370	375	380
Thr Lys Leu Asn Asp Leu Cys Phe Thr Asn Val Tyr Ala Asp Ser Phe 385	390	395 400
Val Ile Arg Gly Asp Glu Val Arg Gln Ile Ala Pro Gly Gln Thr Gly 405	410	415
Lys Ile Ala Asp Tyr Asn Tyr Lys Leu Pro Asp Asp Phe Thr Gly Cys 420	425	430
Val Ile Ala Trp Asn Ser Asn Asn Leu Asp Ser Lys Val Gly Gly Asn 435	440	445
Tyr Asn Tyr Leu Tyr Arg Leu Phe Arg Lys Ser Asn Leu Lys Pro Phe 450	455	460
Glu Arg Asp Ile Ser Thr Glu Ile Tyr Gln Ala Gly Ser Thr Pro Cys 465	470	475 480
Asn Gly Val Glu Gly Phe Asn Cys Tyr Phe Pro Leu Gln Ser Tyr Gly 485	490	495
Phe Gln Pro Thr Asn Gly Val Gly Tyr Gln Pro Tyr Arg Val Val Val 500	505	510
Leu Ser Phe Glu Leu Leu His Ala Pro Ala Thr Val Cys Gly Pro Lys 515	520	525
Lys Ser Thr Asn Leu Val Lys Asn Lys Cys Val Asn Phe Asn Phe Asn 530	535	540
Gly Leu Thr Gly Thr Gly Val Leu Thr Glu Ser Asn Lys Lys Phe Leu 545	550	555 560
Pro Phe Gln Gln Phe Gly Arg Asp Ile Ala Asp Thr Thr Asp Ala Val 565	570	575
Arg Asp Pro Gln Thr Leu Glu Ile Leu Asp Ile Thr Pro Cys Ser Phe 580	585	590
Gly Gly Val Ser Val Ile Thr Pro Gly Thr Asn Thr Ser Asn Gln Val 595	600	605

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Ala Val Leu Tyr Gln Asp	Val Asn Cys Thr Glu	Val Pro Val Ala Ile
610	615	620
His Ala Asp Gln Leu Thr	Pro Thr Trp Arg Val Tyr Ser Thr Gly Ser	
625	630	635 640
Asn Val Phe Gln Thr Arg	Ala Gly Cys Leu Ile Gly Ala Glu His Val	
	645	650 655
Asn Asn Ser Tyr Glu Cys Asp	Ile Pro Ile Gly Ala Gly Ile Cys Ala	
	660	665 670
Ser Tyr Gln Thr Gln Thr	Asn Ser Pro Arg Arg Ala Arg Ser Val Ala	
	675	680 685
Ser Gln Ser Ile Ile Ala Tyr	Thr Met Ser Leu Gly Ala Glu Asn Ser	
	690	695 700
Val Ala Tyr Ser Asn Asn Ser	Ile Ala Ile Pro Thr Asn Phe Thr Ile	
	705	710 715 720
Ser Val Thr Thr Glu Ile	Leu Pro Val Ser Met Thr Lys Thr Ser Val	
	725	730 735
Asp Cys Thr Met Tyr Ile	Cys Gly Asp Ser Thr Glu Cys Ser Asn Leu	
	740	745 750
Leu Leu Gln Tyr Gly Ser	Phe Cys Thr Gln Leu Asn Arg Ala Leu Thr	
	755	760 765
Gly Ile Ala Val Glu Gln	Asp Lys Asn Thr Gln Glu Val Phe Ala Gln	
	770	775 780
Val Lys Gln Ile Tyr Lys	Thr Pro Pro Ile Lys Asp Phe Gly Gly Phe	
	785	790 795 800
Asn Phe Ser Gln Ile Leu	Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser	
	805	810 815
Phe Ala Glu Asp Ala Ala	Phe Asn Lys Val Thr Leu Ala Asp Ala Gly	
	820	825 830
Phe Ile Lys Gln Tyr Gly	Asp Cys Ala Gly Asp Ile Ala Ala Arg Asp	
	835	840 845
Leu Ile Cys Ala Gln Lys	Phe Asn Gly Leu Thr Val Ala Pro Pro Ala	
	850	855 860
Ala Thr Asp Glu Met Ile	Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly	
	865	870 875 880
Thr Ile Thr Ser Gly Trp	Thr Phe Gly Ala Gly Pro Ala Leu Gln Ile	
	885	890 895
Pro Phe Ala Met Gln Met	Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr	
	900	905 910
Gln Asn Val Leu Tyr Glu	Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn	
	915	920 925
Ser Ala Ile Gly Lys Ile	Gln Asp Ser Leu Ser Ser Thr Pro Ser Ala	
	930	935 940
Leu Gly Lys Leu Gln Asp	Val Val Asn Gln Asn Ala Gln Ala Leu Asn	
	945	950 955 960
Thr Leu Val Lys Gln Leu	Ser Ser Asn Phe Gly Ala Ile Ser Ser Val	
	965	970 975
Leu Asn Asp Ile Leu Ser	Arg Leu Asp Lys Val Glu Ala Glu Val Gln	
	980	985 990
Ile Asp Arg Leu Ile Thr	Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val	
	995	1000 1005

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Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn
1010						1015					1020			
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys
1025						1030					1035			
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro
1040						1045					1050			
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val
1055						1060					1065			
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His
1070						1075					1080			
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn
1085						1090					1095			
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln
1100						1105					1110			
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val
1115						1120					1125			
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro
1130						1135					1140			
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn
1145						1150					1155			
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn
1160						1165					1170			
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu
1175						1180					1185			
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu
1190						1195					1200			
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu
1205						1210					1215			
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met
1220						1225					1230			
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys
1235						1240					1245			
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro
1250						1255					1260			
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr					
1265						1270								

<210> SEQ ID NO 5

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 3 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 5

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5				10					15		

Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
			20					25					30		

Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40				45				

His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
		50				55					60				

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Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp	65	70	75	80
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu	85	90	95	
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser	100	105	110	
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile	115	120	125	
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	130	135	140	
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	145	150	155	160
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	180	185	190	
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205	
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220	
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235	240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270	
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285	
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300	
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315	320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350	
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365	
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380	
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395	400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430	
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445	
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460	
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys				

465				470				475				480			
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
485				490				495							
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
500				505				510							
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
515				520				525							
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
530				535				540							
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545				550				555				560			
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
565				570				575							
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
580				585				590							
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
595				600				605							
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
610				615				620							
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625				630				635				640			
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
645				650				655							
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
660				665				670							
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
675				680				685							
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
690				695				700							
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705				710				715				720			
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
725				730				735							
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
740				745				750							
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
755				760				765							
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
770				775				780							
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
785				790				795				800			
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
805				810				815							
Pro	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
820				825				830							
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg	Asp
835				840				845							
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
850				855				860							
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865				870				875				880			

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Thr Ile Thr Ser Gly Trp Thr Phe Gly Ala Gly Ala Ala Leu Gln Ile
 885 890 895
 Pro Phe Pro Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr
 900 905 910
 Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn
 915 920 925
 Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala
 930 935 940
 Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn
 945 950 955 960
 Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val
 965 970 975
 Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln
 980 985 990
 Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val
 995 1000 1005
 Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn
 1010 1015 1020
 Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys
 1025 1030 1035
 Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro
 1040 1045 1050
 Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val
 1055 1060 1065
 Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His
 1070 1075 1080
 Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn
 1085 1090 1095
 Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln
 1100 1105 1110
 Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val
 1115 1120 1125
 Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro
 1130 1135 1140
 Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
 1145 1150 1155
 His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
 1160 1165 1170
 Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
 1175 1180 1185
 Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
 1190 1195 1200
 Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
 1205 1210 1215
 Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
 1220 1225 1230
 Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
 1235 1240 1245
 Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
 1250 1255 1260

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Val Leu Lys Gly Val Lys Leu His Tyr Thr
1265 1270

<210> SEQ ID NO 6
 <211> LENGTH: 1273
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Variant 4 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 6

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1 5 10 15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20 25 30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35 40 45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50 55 60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65 70 75 80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85 90 95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100 105 110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115 120 125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130 135 140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145 150 155 160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165 170 175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180 185 190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195 200 205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210 215 220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225 230 235 240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245 250 255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260 265 270

Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
275 280 285

Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
290 295 300

Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305 310 315 320

Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
325 330 335

Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	
			340				345						350			
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	
			355				360						365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	
			370				375						380			
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	
			385				390						395	400		
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	
			405						410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	
			420						425						430	
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	
			435						440						445	
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	
			450						455						460	
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	
			465						470						475	
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	
			485						490						495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	
			500						505						510	
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	
			515						520						525	
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	
			530						535						540	
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	
			545						550						555	
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	
			565						570						575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	
			580						585						590	
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	
			595						600						605	
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	
			610						615						620	
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	
			625						630						635	
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
			645						650						655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
			660						665						670	
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	
			675						680						685	
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	
			690						695						700	
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	
			705						710						715	
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	
			725						730						735	
Asp	Cys	Thr	Met	Tyr												

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740						745						750					
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr		
755						760						765					
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln		
770						775						780					
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe		
785						790						795					
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser		
805						810						815					
Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly		
820						825						830					
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg	Asp		
835						840						845					
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala		
850						855						860					
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly		
865						870						875					
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile		
885						890						895					
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr		
900						905						910					
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn		
915						920						925					
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala		
930						935						940					
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn		
945						950						955					
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val		
965						970						975					
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln		
980						985						990					
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val		
995						1000						1005					
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn			
1010						1015						1020					
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys			
1025						1030						1035					
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro			
1040						1045						1050					
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val			
1055						1060						1065					
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His			
1070						1075						1080					
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn			
1085						1090						1095					
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln			
1100						1105						1110					
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val			
1115						1120						1125					
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro			
1130						1135						1140					

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Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
 1145 1150 1155
 His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
 1160 1165 1170
 Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
 1175 1180 1185
 Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
 1190 1195 1200
 Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
 1205 1210 1215
 Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
 1220 1225 1230
 Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
 1235 1240 1245
 Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
 1250 1255 1260
 Val Leu Lys Gly Val Lys Leu His Tyr Thr
 1265 1270

<210> SEQ ID NO 7
 <211> LENGTH: 1273
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Variant 5 of Mutated Sars CoV-2 Spike protein
 <400> SEQUENCE: 7

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
 1 5 10 15
 Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
 20 25 30
 Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
 35 40 45
 His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
 50 55 60
 Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
 65 70 75 80
 Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
 85 90 95
 Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
 100 105 110
 Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
 115 120 125
 Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
 130 135 140
 Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
 145 150 155 160
 Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
 165 170 175
 Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
 180 185 190
 Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
 195 200 205

Pro 210	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
Pro 225	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr 240
Leu	Leu	Ala	Leu	His 245	Arg	Ser	Tyr	Leu	Thr 250	Pro	Gly	Asp	Ser	Ser 255	Ser
Gly	Trp	Thr	Ala 260	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
Arg	Thr	Phe 275	Leu	Leu	Lys	Tyr	Asn 280	Glu	Asn	Gly	Thr	Ile 285	Thr	Asp	Ala
Val	Asp 290	Cys	Ala	Leu	Asp	Pro 295	Leu	Ser	Glu	Thr	Lys 300	Cys	Thr	Leu	Lys
Ser 305	Phe	Thr	Val	Glu	Lys 310	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val 320
Gln	Pro	Thr	Glu	Ser 325	Ile	Val	Arg	Phe	Pro 330	Asn	Ile	Thr	Asn	Leu	Cys
Pro	Phe	Gly	Glu 340	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser 350	Val	Tyr	Ala
Trp	Asn 355	Arg	Lys	Arg	Ile	Ser	Asn 360	Cys	Val	Ala	Asp	Tyr 365	Ser	Val	Leu
Tyr 370	Asn	Ser	Ala	Ser	Phe	Ser 375	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
Thr 385	Lys	Leu	Asn	Asp 390	Leu	Cys	Phe	Thr	Asn	Val 395	Tyr	Ala	Asp	Ser	Phe 400
Val	Ile	Arg	Gly 405	Asp	Glu	Val	Arg	Gln	Ile 410	Ala	Pro	Gly	Gln	Thr	Gly
Lys	Ile	Ala	Asp 420	Tyr	Asn	Tyr	Lys	Leu	Pro 425	Asp	Asp	Phe	Thr	Gly	Cys
Val	Ile	Ala 435	Trp	Asn	Ser	Asn 440	Asn	Leu	Asp	Ser	Lys	Val 445	Gly	Gly	Asn
Tyr 450	Asn	Tyr	Leu	Tyr	Arg	Leu 455	Phe	Arg	Lys	Ser	Asn 460	Leu	Lys	Pro	Phe
Glu 465	Arg	Asp	Ile	Ser	Thr 470	Glu	Ile	Tyr	Gln	Ala 475	Gly	Ser	Thr	Pro	Cys 480
Asn	Gly	Val	Glu 485	Gly	Phe	Asn	Cys	Tyr	Phe 490	Pro	Leu	Gln	Ser	Tyr	Gly 495
Phe	Gln	Pro	Thr 500	Asn	Gly	Val	Gly	Tyr 505	Gln	Pro	Tyr	Arg 510	Val	Val	Val
Leu	Ser 515	Phe	Glu	Leu	Leu	His	Ala 520	Pro	Ala	Thr	Val	Cys 525	Gly	Pro	Lys
Lys 530	Ser	Thr	Asn	Leu	Val	Lys 535	Asn	Lys	Cys	Val	Asn 540	Phe	Asn	Phe	Asn
Gly 545	Leu	Thr	Gly	Thr	Gly 550	Val	Leu	Thr	Glu	Ser	Asn 555	Lys	Lys	Phe	Leu 560
Pro	Phe	Gln	Gln 565	Phe	Gly	Arg	Asp	Ile	Ala 570	Asp	Thr	Thr	Asp	Ala	Val
Arg	Asp	Pro	Gln 580	Thr	Leu	Glu	Ile	Leu 585	Asp	Ile	Thr	Pro 590	Cys	Ser	Phe
Gly	Gly 595	Val	Ser	Val	Ile	Thr	Pro 600	Gly	Thr	Asn	Thr	Ser 605	Asn	Gln	Val

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Ala Val Leu Tyr Gln Asp	Val Asn Cys Thr Glu Val Pro Val Ala Ile
610	615 620
His Ala Asp Gln Leu Thr	Pro Thr Trp Arg Val Tyr Ser Thr Gly Ser
625	630 635 640
Asn Val Phe Gln Thr Arg Ala Gly Cys Leu Ile Gly Ala Glu His Val	
	645 650 655
Asn Asn Ser Tyr Glu Cys Asp Ile Pro Ile Gly Ala Gly Ile Cys Ala	
	660 665 670
Ser Tyr Gln Thr Gln Thr Asn Ser Pro Arg Arg Ala Arg Ser Val Ala	
	675 680 685
Ser Gln Ser Ile Ile Ala Tyr Thr Met Ser Leu Gly Ala Glu Asn Ser	
	690 695 700
Val Ala Tyr Ser Asn Asn Ser Ile Ala Ile Pro Thr Asn Phe Thr Ile	
	705 710 715 720
Ser Val Thr Thr Glu Ile Leu Pro Val Ser Met Thr Lys Thr Ser Val	
	725 730 735
Asp Cys Thr Met Tyr Ile Cys Gly Asp Ser Thr Glu Cys Ser Asn Leu	
	740 745 750
Leu Leu Gln Tyr Gly Ser Phe Cys Thr Gln Leu Asn Arg Ala Leu Thr	
	755 760 765
Gly Ile Ala Val Glu Gln Asp Lys Asn Thr Gln Glu Val Phe Ala Gln	
	770 775 780
Val Lys Gln Ile Tyr Lys Thr Pro Pro Ile Lys Asp Phe Gly Gly Phe	
	785 790 795 800
Asn Phe Ser Gln Ile Leu Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser	
	805 810 815
Pro Ala Glu Asp Ala Ala Phe Asn Lys Val Thr Leu Ala Asp Ala Gly	
	820 825 830
Phe Ile Lys Gln Tyr Gly Asp Cys Ala Gly Asp Ile Ala Ala Arg Asp	
	835 840 845
Leu Ile Cys Ala Gln Lys Phe Asn Gly Leu Thr Val Ala Pro Pro Ala	
	850 855 860
Ala Thr Asp Glu Met Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly	
	865 870 875 880
Thr Ile Thr Ser Gly Trp Thr Phe Gly Ala Gly Pro Ala Leu Gln Ile	
	885 890 895
Pro Phe Pro Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr	
	900 905 910
Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn	
	915 920 925
Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Pro Ser Ala	
	930 935 940
Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn	
	945 950 955 960
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val	
	965 970 975
Leu Asn Asp Ile Leu Ser Arg Leu Asp Pro Pro Glu Ala Glu Val Gln	
	980 985 990
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val	
	995 1000 1005
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn	

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1010	1015	1020
Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys		
1025	1030	1035
Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro		
1040	1045	1050
Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val		
1055	1060	1065
Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His		
1070	1075	1080
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn		
1085	1090	1095
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln		
1100	1105	1110
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val		
1115	1120	1125
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro		
1130	1135	1140
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn		
1145	1150	1155
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn		
1160	1165	1170
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu		
1175	1180	1185
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu		
1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu		
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met		
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys		
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro		
1250	1255	1260
Val Leu Lys Gly Val Lys Leu His Tyr Thr		
1265	1270	

<210> SEQ ID NO 8

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 6 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 8

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1 5 10 15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20 25 30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35 40 45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50 55 60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp

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65	70	75	80
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu	85	90	95
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser	100	105	110
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile	115	120	125
Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr	130	135	140
Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr	145	150	155
Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu	165	170	175
Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe	180	185	190
Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr	195	200	205
Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu	210	215	220
Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr	225	230	235
Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser	245	250	255
Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro	260	265	270
Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala	275	280	285
Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys	290	295	300
Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val	305	310	315
Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys	325	330	335
Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala	340	345	350
Trp Asn Arg Lys Arg Ile Ser Asn Cys Val Ala Asp Tyr Ser Val Leu	355	360	365
Tyr Asn Ser Ala Ser Phe Ser Thr Phe Lys Cys Tyr Gly Val Ser Pro	370	375	380
Thr Lys Leu Asn Asp Leu Cys Phe Thr Asn Val Tyr Ala Asp Ser Phe	385	390	395
Val Ile Arg Gly Asp Glu Val Arg Gln Ile Ala Pro Gly Gln Thr Gly	405	410	415
Lys Ile Ala Asp Tyr Asn Tyr Lys Leu Pro Asp Asp Phe Thr Gly Cys	420	425	430
Val Ile Ala Trp Asn Ser Asn Asn Leu Asp Ser Lys Val Gly Gly Asn	435	440	445
Tyr Asn Tyr Leu Tyr Arg Leu Phe Arg Lys Ser Asn Leu Lys Pro Phe	450	455	460
Glu Arg Asp Ile Ser Thr Glu Ile Tyr Gln Ala Gly Ser Thr Pro Cys	465	470	475
			480

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Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	530	535	540
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	545	550	555
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	565	570	575
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	580	585	590
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	595	600	605
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	610	615	620
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	625	630	635
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	645	650	655
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	660	665	670
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	675	680	685
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	690	695	700
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	705	710	715
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	725	730	735
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	740	745	750
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	755	760	765
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	770	775	780
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	785	790	795
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	805	810	815
Pro	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	820	825	830
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg	Asp	835	840	845
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	850	855	860
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	865	870	875

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Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile	885	890	895
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	900	905	910
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	915	920	925
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala	930	935	940
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn	945	950	955
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val	965	970	975
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln	980	985	990
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val	995	1000	1005
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn		1010	1015	1020
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys		1025	1030	1035
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro		1040	1045	1050
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val		1055	1060	1065
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His		1070	1075	1080
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn		1085	1090	1095
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln		1100	1105	1110
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val		1115	1120	1125
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro		1130	1135	1140
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn		1145	1150	1155
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn		1160	1165	1170
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu		1175	1180	1185
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu		1190	1195	1200
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu		1205	1210	1215
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met		1220	1225	1230
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys		1235	1240	1245
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro		1250	1255	1260
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr									

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1265                               1270

<210> SEQ ID NO 9
<211> LENGTH: 1273
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Variant 31 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 9

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
 1                               5                               10                               15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
 20                               25                               30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
 35                               40                               45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
 50                               55                               60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
 65                               70                               75                               80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
 85                               90                               95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100                               105                               110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115                               120                               125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130                               135                               140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145                               150                               155                               160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165                               170                               175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180                               185                               190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195                               200                               205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210                               215                               220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225                               230                               235                               240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245                               250                               255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260                               265                               270

Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
275                               280                               285

Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
290                               295                               300

Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305                               310                               315                               320

Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
325                               330                               335

Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala

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340							345					350				
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	
355							360				365					
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	
370							375				380					
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	
385	390									395	400					
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	
405							410				415					
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	
420							425				430					
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	
435							440				445					
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	
450							455				460					
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	
465	470									475	480					
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	
485							490				495					
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	
500							505				510					
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	
515							520				525					
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	
530							535				540					
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	
545	550									555	560					
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	
565							570				575					
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	
580							585				590					
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	
595							600				605					
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	
610							615				620					
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	
625	630									635	640					
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
645							650				655					
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
660							665				670					
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Gln	Gln	Ala	Gln	Ser	Val	Ala	
675							680				685					
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	
690							695				700					
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	
705	710									715	720					
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	
725							730				735					
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	
740							745				750					

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Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
	755						760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
	770					775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
	785				790					795					800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
			805						810					815	
Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg	Asp
	835						840				845				
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
	850					855					860				
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
	865				870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile
			885					890						895	
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900					905					910		
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
	915						920					925			
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala
	930					935					940				
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
	945				950					955					960
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
			965						970					975	
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln
			980					985					990		
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
	995						1000						1005		
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
	1010					1015						1020			
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
	1025					1030					1035				
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
	1040					1045					1050				
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
	1055					1060					1065				
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
	1070					1075					1080				
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn	
	1085					1090					1095				
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln	
	1100					1105					1110				
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val	
	1115					1120					1125				
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro	
	1130					1135					1140				

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Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn
1145						1150					1155			
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn
1160						1165					1170			
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu
1175						1180					1185			
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu
1190						1195					1200			
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu
1205						1210					1215			
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met
1220						1225					1230			
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys
1235						1240					1245			
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro
1250						1255					1260			
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr					
1265						1270								

<210> SEQ ID NO 10

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 32 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 10

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5					10				15		
Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
		20						25					30		
Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40				45				
His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
	50				55					60					
Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65				70					75					80	
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
		85						90					95		
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser
	100					105						110			
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile
	115					120					125				
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
	130				135					140					
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145				150					155					160	
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165					170					175		
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
	180					185						190			
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
	195					200					205				

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Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	530	535	540
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	545	550	555
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	565	570	575
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	580	585	590
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	595	600	605
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile			

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610	615	620
His Ala Asp Gln Leu Thr Pro Thr Trp Arg Val Tyr Ser Thr Gly Ser		
625	630	635 640
Asn Val Phe Gln Thr Arg Ala Gly Cys Leu Ile Gly Ala Glu His Val		
	645	650 655
Asn Asn Ser Tyr Glu Cys Asp Ile Pro Ile Gly Ala Gly Ile Cys Ala		
	660	665 670
Ser Tyr Gln Thr Gln Thr Asn Ser Pro Gln Gln Ala Gln Ser Val Ala		
	675	680 685
Ser Gln Ser Ile Ile Ala Tyr Thr Met Ser Leu Gly Ala Glu Asn Ser		
	690	695 700
Val Ala Tyr Ser Asn Asn Ser Ile Ala Ile Pro Thr Asn Phe Thr Ile		
	705	710 715 720
Ser Val Thr Thr Glu Ile Leu Pro Val Ser Met Thr Lys Thr Ser Val		
	725	730 735
Asp Cys Thr Met Tyr Ile Cys Gly Asp Ser Thr Glu Cys Ser Asn Leu		
	740	745 750
Leu Leu Gln Tyr Gly Ser Phe Cys Thr Gln Leu Asn Arg Ala Leu Thr		
	755	760 765
Gly Ile Ala Val Glu Gln Asp Lys Asn Thr Gln Glu Val Phe Ala Gln		
	770	775 780
Val Lys Gln Ile Tyr Lys Thr Pro Pro Ile Lys Asp Phe Gly Gly Phe		
	785	790 795 800
Asn Phe Ser Gln Ile Leu Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser		
	805	810 815
Phe Ala Glu Asp Ala Ala Phe Asn Lys Val Thr Leu Ala Asp Ala Gly		
	820	825 830
Phe Ile Lys Gln Tyr Gly Asp Cys Ala Gly Asp Ile Ala Ala Arg Asp		
	835	840 845
Leu Ile Cys Ala Gln Lys Phe Asn Gly Leu Thr Val Ala Pro Pro Ala		
	850	855 860
Ala Thr Asp Glu Met Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly		
	865	870 875 880
Thr Ile Thr Ser Gly Trp Thr Phe Gly Ala Gly Ala Ala Leu Gln Ile		
	885	890 895
Pro Phe Ala Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr		
	900	905 910
Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn		
	915	920 925
Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala		
	930	935 940
Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn		
	945	950 955 960
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val		
	965	970 975
Leu Asn Asp Ile Leu Ser Arg Leu Asp Pro Pro Glu Ala Glu Val Gln		
	980	985 990
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val		
	995	1000 1005
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn		
	1010	1015 1020

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Leu Ala  Ala Thr Lys Met  Ser  Glu Cys Val Leu Gly  Gln Ser Lys
1025                               1030           1035

Arg Val  Asp Phe Cys Gly  Lys  Gly Tyr His Leu Met  Ser Phe Pro
1040                               1045           1050

Gln Ser  Ala Pro His Gly  Val  Val Phe Leu His Val  Thr Tyr Val
1055                               1060           1065

Pro Ala  Gln Glu Lys Asn Phe  Thr Thr Ala Pro  Ala  Ile Cys His
1070                               1075           1080

Asp Gly  Lys Ala His Phe  Pro  Arg Glu Gly Val  Phe  Val Ser Asn
1085                               1090           1095

Gly Thr  His Trp Phe Val  Thr  Gln Arg Asn Phe  Tyr  Glu Pro Gln
1100                               1105           1110

Ile Ile  Thr Thr Asp Asn Thr  Phe Val Ser Gly  Asn  Cys Asp Val
1115                               1120           1125

Val Ile  Gly Ile Val Asn Asn  Thr Val Tyr Asp  Pro  Leu Gln Pro
1130                               1135           1140

Glu Leu  Asp Ser Phe Lys  Glu  Glu Leu Asp Lys  Tyr  Phe Lys Asn
1145                               1150           1155

His Thr  Ser Pro Asp Val  Asp  Leu Gly Asp Ile  Ser  Gly Ile Asn
1160                               1165           1170

Ala Ser  Val Val Asn Ile  Gln  Lys Glu Ile Asp  Arg  Leu Asn Glu
1175                               1180           1185

Val Ala  Lys Asn Leu Asn Glu  Ser Leu Ile Asp  Leu  Gln Glu Leu
1190                               1195           1200

Gly Lys  Tyr Glu Gln Tyr  Ile  Lys Trp Pro Trp  Tyr  Ile Trp Leu
1205                               1210           1215

Gly Phe  Ile Ala Gly Leu  Ile  Ala Ile Val Met  Val  Thr Ile Met
1220                               1225           1230

Leu Cys  Cys Met Thr Ser  Cys  Cys Ser Cys Leu  Lys  Gly Cys Cys
1235                               1240           1245

Ser Cys  Gly Ser Cys Cys  Lys  Phe Asp Glu Asp  Asp  Ser Glu Pro
1250                               1255           1260

Val Leu  Lys Gly Val Lys  Leu  His Tyr Thr
1265                               1270

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<210> SEQ ID NO 11

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 33 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 11

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Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1           5           10           15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20          25          30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35          40          45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50          55          60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65          70          75          80

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Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu	85	90	95
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser	100	105	110
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile	115	120	125
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	130	135	140
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	145	150	155
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	180	185	190
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475

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Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	530	535	540
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	545	550	555
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	565	570	575
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	580	585	590
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	595	600	605
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	610	615	620
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	625	630	635
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	645	650	655
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	660	665	670
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Gln	Gln	Ala	Gln	Ser	Val	Ala	675	680	685
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	690	695	700
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	705	710	715
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	725	730	735
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	740	745	750
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	755	760	765
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	770	775	780
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	785	790	795
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	805	810	815
Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	820	825	830
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg	Asp	835	840	845
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	850	855	860
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	865	870	875
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile			

885								890					895						
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr				
900								905					910						
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn				
915								920					925						
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala				
930								935					940						
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn				
945								950					955					960	
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val				
965								970					975						
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln				
980								985					990						
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val				
995								1000					1005						
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn					
1010								1015					1020						
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys					
1025								1030					1035						
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro					
1040								1045					1050						
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val					
1055								1060					1065						
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His					
1070								1075					1080						
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn					
1085								1090					1095						
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln					
1100								1105					1110						
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val					
1115								1120					1125						
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro					
1130								1135					1140						
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn					
1145								1150					1155						
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn					
1160								1165					1170						
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu					
1175								1180					1185						
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu					
1190								1195					1200						
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu					
1205								1210					1215						
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met					
1220								1225					1230						
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys					
1235								1240					1245						
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro					
1250								1255					1260						
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr										
1265								1270											

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<210> SEQ ID NO 12
 <211> LENGTH: 1273
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Variant 34 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 12

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Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1      5      10      15
Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20     25     30
Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35     40     45
His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50     55     60
Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65     70     75     80
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85     90     95
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100    105    110
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115    120    125
Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130    135    140
Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145    150    155    160
Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165    170    175
Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180    185    190
Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195    200    205
Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210    215    220
Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225    230    235    240
Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245    250    255
Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260    265    270
Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
275    280    285
Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
290    295    300
Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305    310    315    320
Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
325    330    335
Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala
340    345    350

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Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
	355						360					365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
	370					375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
			405					410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
			420					425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
	435						440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
	450					455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465					470					475					480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
			485						490					495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
			500					505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
	515						520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
	530					535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550					555					560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
			565						570					575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
			580					585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
	595						600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
	610					615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625					630					635					640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
			645						650					655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
		660						665					670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Gln	Gln	Ala	Gln	Ser	Val	Ala
	675						680					685			
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
	690					695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
			725						730				735		
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
	740							745					750		

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Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
	755						760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
	770					775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
	785				790					795					800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
			805						810					815	
Pro	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg	Asp
	835						840					845			
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
	850					855					860				
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
	865				870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile
			885					890						895	
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900					905					910		
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
		915					920					925			
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala
	930					935					940				
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
	945				950				955						960
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
			965					970						975	
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln
			980					985					990		
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
	995						1000					1005			
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
	1010					1015						1020			
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
	1025					1030					1035				
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
	1040					1045					1050				
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
	1055					1060					1065				
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
	1070					1075					1080				
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn	
	1085					1090					1095				
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln	
	1100					1105					1110				
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val	
	1115					1120					1125				
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro	
	1130					1135					1140				
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn	

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1145	1150	1155
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn		
1160	1165	1170
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu		
1175	1180	1185
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu		
1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu		
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met		
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys		
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro		
1250	1255	1260
Val Leu Lys Gly Val Lys Leu His Tyr Thr		
1265	1270	

<210> SEQ ID NO 13
 <211> LENGTH: 1273
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Variant 35 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 13

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val	
1 5 10 15	
Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe	
20 25 30	
Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu	
35 40 45	
His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp	
50 55 60	
Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp	
65 70 75 80	
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu	
85 90 95	
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser	
100 105 110	
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile	
115 120 125	
Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr	
130 135 140	
Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr	
145 150 155 160	
Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu	
165 170 175	
Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe	
180 185 190	
Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr	
195 200 205	
Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu	

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210	215	220
Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr		
225	230	235 240
Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser		
	245	250 255
Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro		
	260	265 270
Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala		
	275	280 285
Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys		
	290	295 300
Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val		
305	310	315 320
Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys		
	325	330 335
Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala		
	340	345 350
Trp Asn Arg Lys Arg Ile Ser Asn Cys Val Ala Asp Tyr Ser Val Leu		
	355	360 365
Tyr Asn Ser Ala Ser Phe Ser Thr Phe Lys Cys Tyr Gly Val Ser Pro		
	370	375 380
Thr Lys Leu Asn Asp Leu Cys Phe Thr Asn Val Tyr Ala Asp Ser Phe		
385	390	395 400
Val Ile Arg Gly Asp Glu Val Arg Gln Ile Ala Pro Gly Gln Thr Gly		
	405	410 415
Lys Ile Ala Asp Tyr Asn Tyr Lys Leu Pro Asp Asp Phe Thr Gly Cys		
	420	425 430
Val Ile Ala Trp Asn Ser Asn Asn Leu Asp Ser Lys Val Gly Gly Asn		
	435	440 445
Tyr Asn Tyr Leu Tyr Arg Leu Phe Arg Lys Ser Asn Leu Lys Pro Phe		
	450	455 460
Glu Arg Asp Ile Ser Thr Glu Ile Tyr Gln Ala Gly Ser Thr Pro Cys		
465	470	475 480
Asn Gly Val Glu Gly Phe Asn Cys Tyr Phe Pro Leu Gln Ser Tyr Gly		
	485	490 495
Phe Gln Pro Thr Asn Gly Val Gly Tyr Gln Pro Tyr Arg Val Val Val		
	500	505 510
Leu Ser Phe Glu Leu Leu His Ala Pro Ala Thr Val Cys Gly Pro Lys		
	515	520 525
Lys Ser Thr Asn Leu Val Lys Asn Lys Cys Val Asn Phe Asn Phe Asn		
	530	535 540
Gly Leu Thr Gly Thr Gly Val Leu Thr Glu Ser Asn Lys Lys Phe Leu		
545	550	555 560
Pro Phe Gln Gln Phe Gly Arg Asp Ile Ala Asp Thr Thr Asp Ala Val		
	565	570 575
Arg Asp Pro Gln Thr Leu Glu Ile Leu Asp Ile Thr Pro Cys Ser Phe		
	580	585 590
Gly Gly Val Ser Val Ile Thr Pro Gly Thr Asn Thr Ser Asn Gln Val		
	595	600 605
Ala Val Leu Tyr Gln Asp Val Asn Cys Thr Glu Val Pro Val Ala Ile		
610	615	620

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His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	
625					630					635					640	
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
			645						650					655		
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
			660					665						670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Gln	Gln	Ala	Gln	Ser	Val	Ala	
		675					680					685				
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	
	690					695					700					
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	
705					710					715					720	
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	
			725					730						735		
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	
		740					745						750			
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	
	755						760					765				
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	
	770					775					780					
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	
785					790					795					800	
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	
			805					810						815		
Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	
		820						825					830			
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg	Asp	
	835						840					845				
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	
	850					855					860					
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	
865					870					875					880	
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile	
			885					890						895		
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	
		900						905					910			
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	
	915						920					925				
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala	
	930					935					940					
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn	
945				950					955						960	
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val	
			965						970					975		
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln	
		980						985					990			
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val	
	995						1000						1005			
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn		
1010					1015								1020			

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Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys
1025						1030					1035			
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro
1040						1045					1050			
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val
1055						1060					1065			
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His
1070						1075					1080			
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn
1085						1090					1095			
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln
1100						1105					1110			
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val
1115						1120					1125			
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro
1130						1135					1140			
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn
1145						1150					1155			
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn
1160						1165					1170			
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu
1175						1180					1185			
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu
1190						1195					1200			
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu
1205						1210					1215			
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met
1220						1225					1230			
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys
1235						1240					1245			
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro
1250						1255					1260			
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr					
1265						1270								

<210> SEQ ID NO 14

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 36 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 14

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5					10					15	

Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
				20					25				30		

Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40				45				

His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
	50				55						60				

Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65					70					75				80	

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Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85						90					95	
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser
			100					105					110		
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile
		115					120					125			
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
	130					135					140				
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145					150					155					160
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165						170					175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
		180						185					190		
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
		195					200					205			
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
	210					215					220				
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr
225					230					235					240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser
			245						250					255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
			260					265					270		
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala
		275					280					285			
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys
	290					295					300				
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val
305					310					315					320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys
			325						330					335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala
			340					345					350		
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
		355					360					365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
	370					375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
			405					410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
			420					425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
		435					440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
	450					455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465					470					475					480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly

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485					490					495				
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val
			500					505				510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Lys
		515					520					525		
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe
	530					535					540			
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe
545					550				555					560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala
				565					570					575
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser
		580						585					590	Phe
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln
	595						600						605	Val
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala
610						615					620			Ile
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly
625					630					635				640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His
				645					650					655
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys
		660						665					670	Ala
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Gln	Gln	Ala	Gln	Ser	Val
	675						680					685		Ala
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn
	690					695					700			Ser
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr
705					710					715				720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser
				725					730					735
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn
		740						745					750	Leu
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu
		755					760					765		Thr
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala
	770					775					780			Gln
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly
785					790					795				800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg
				805					810					815
Pro	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala
			820						825				830	Gly
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ile	Ala	Ala	Arg
		835					840					845		Asp
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro
	850						855				860			Ala
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala
865						870				875				880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln
				885					890					895

Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900						905			910			
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
			915						920			925			
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala
			930						935			940			
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
			945						950			955			
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
			965						970			975			
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln
			980						985			990			
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
			995						1000			1005			
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
			1010						1015			1020			
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
			1025						1030			1035			
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
			1040						1045			1050			
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
			1055						1060			1065			
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
			1070						1075			1080			
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn	
			1085						1090			1095			
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln	
			1100						1105			1110			
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val	
			1115						1120			1125			
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro	
			1130						1135			1140			
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn	
			1145						1150			1155			
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn	
			1160						1165			1170			
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu	
			1175						1180			1185			
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu	
			1190						1195			1200			
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu	
			1205						1210			1215			
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met	
			1220						1225			1230			
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys	
			1235						1240			1245			
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro	
			1250						1255			1260			
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr						
			1265						1270						

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<210> SEQ ID NO 15
 <211> LENGTH: 1273
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Variant 37 of Mutated Sars CoV-2 spike protein

<400> SEQUENCE: 15

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
 1 5 10 15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
 20 25 30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
 35 40 45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
 50 55 60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
 65 70 75 80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
 85 90 95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
 100 105 110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
 115 120 125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
 130 135 140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
 145 150 155 160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
 165 170 175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
 180 185 190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
 195 200 205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
 210 215 220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
 225 230 235 240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
 245 250 255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
 260 265 270

Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
 275 280 285

Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
 290 295 300

Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
 305 310 315 320

Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
 325 330 335

Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala
 340 345 350

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Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
	355						360					365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
	370					375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
			405					410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
			420					425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
	435						440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
	450					455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465					470					475					480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
			485						490					495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
			500					505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515					520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
	530					535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550					555					560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
			565						570					575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
			580					585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
	610					615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625					630					635					640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
			645						650					655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
		660						665					670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Gln	Gln	Ala	Gln	Ser	Val	Ala
		675					680					685			
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
	690					695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
			725					730					735		
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
		740						745					750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr

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755	760	765
Gly Ile Ala Val Glu Gln Asp Lys Asn Thr Gln Glu Val Phe Ala Gln 770 775 780		
Val Lys Gln Ile Tyr Lys Thr Pro Pro Ile Lys Asp Phe Gly Gly Phe 785 790 795 800		
Asn Phe Ser Gln Ile Leu Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser 805 810 815		
Pro Ala Glu Asp Ala Ala Phe Asn Lys Val Thr Leu Ala Asp Ala Gly 820 825 830		
Phe Ile Lys Gln Tyr Gly Asp Cys Ala Gly Asp Ile Ala Ala Arg Asp 835 840 845		
Leu Ile Cys Ala Gln Lys Phe Asn Gly Leu Thr Val Ala Pro Pro Ala 850 855 860		
Ala Thr Asp Glu Met Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly 865 870 875 880		
Thr Ile Thr Ser Gly Trp Thr Phe Gly Ala Gly Pro Ala Leu Gln Ile 885 890 895		
Pro Phe Pro Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr 900 905 910		
Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn 915 920 925		
Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Pro Ser Ala 930 935 940		
Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn 945 950 955 960		
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val 965 970 975		
Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln 980 985 990		
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val 995 1000 1005		
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn 1010 1015 1020		
Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys 1025 1030 1035		
Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro 1040 1045 1050		
Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val 1055 1060 1065		
Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His 1070 1075 1080		
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn 1085 1090 1095		
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln 1100 1105 1110		
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val 1115 1120 1125		
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro 1130 1135 1140		
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn 1145 1150 1155		

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<210> SEQ ID NO 16
<211> LENGTH: 16
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Peptide 2

<400> SEQUENCE: 16

Ser Lys Arg Ser Phe Ile Glu Asp Leu Leu Phe Asn Lys Val Thr Leu
1             5             10            15

```

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<210> SEQ ID NO 17
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Peptide 5

<400> SEQUENCE: 17

Ile Lys Gln Tyr Gly Asp Cys Leu Gly Asp Ile Ala Ala Arg Asp Leu
1             5             10             15

Ile

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<210> SEQ ID NO 18
<211> LENGTH: 15
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Peptide 7

<400> SEQUENCE: 18

Asn Gly Leu Thr Val Leu Pro Pro Leu Leu Thr Asp Glu Met Ile
1             5             10             15

```

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<210> SEQ ID NO 19
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: SPIKELMX1 aa positions 812 to 828

<400> SEQUENCE: 19
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Pro	Ser	Lys	Arg	Ser	Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr
1				5					10					15	

Leu

<210> SEQ ID NO 20
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NCOA1LXD1 aa positions 627 to 643

<400> SEQUENCE: 20

Ser	Gln	Thr	Ser	His	Lys	Leu	Val	Gln	Leu	Leu	Thr	Thr	Thr	Ala	Glu
1			5						10					15	

Gln

<210> SEQ ID NO 21
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NCOA2LXD1 aa positions 635 to 651

<400> SEQUENCE: 21

Ser	Lys	Gly	Gln	Thr	Lys	Leu	Leu	Gln	Leu	Leu	Thr	Thr	Lys	Ser	Asp
1			5						10					15	

Gln

<210> SEQ ID NO 22
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NCOA1LXD2 aa positions 684 to 700

<400> SEQUENCE: 22

Thr	Glu	Arg	His	Lys	Ile	Leu	His	Arg	Leu	Leu	Gln	Glu	Gly	Ser	Pro
1			5						10					15	

Ser

<210> SEQ ID NO 23
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NCOA2LXD2 aa positions 684 to 700

<400> SEQUENCE: 23

Lys	Glu	Lys	His	Lys	Ile	Leu	His	Arg	Leu	Leu	Gln	Asp	Ser	Ser	Ser
1			5						10					15	

Pro

<210> SEQ ID NO 24
<211> LENGTH: 17
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: NCOA1LXD3 aa positions 743 to 759

<400> SEQUENCE: 24

Ser	Lys	Asp	His	Gln	Leu	Leu	Arg	Tyr	Leu	Leu	Asp	Lys	Asp	Glu	Lys
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

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1	5	10	15
---	---	----	----

Asp

<210> SEQ ID NO 25
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: NCOA2LXD3 aa positions 739 to 755

<400> SEQUENCE: 25

Lys	Lys	Glu	Asn	Ala	Leu	Leu	Arg	Tyr	Leu	Leu	Asp	Lys	Asp	Asp	Thr
1				5					10					15	

Lys

<210> SEQ ID NO 26
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: NCOA1LXD4 aa positions 907 to 923

<400> SEQUENCE: 26

Gln	Cys	Ile	Ser	Ser	Gln	Leu	Asp	Glu	Leu	Leu	Cys	Pro	Pro	Thr	Thr
1				5					10					15	

Val

<210> SEQ ID NO 27
 <211> LENGTH: 17
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: NCOA2LXD4 aa positions 1073 to 1089

<400> SEQUENCE: 27

Pro	Ser	Asp	Glu	Gly	Ala	Leu	Leu	Asp	Gln	Leu	Tyr	Leu	Ala	Leu	Arg
1				5					10					15	

Asn

<210> SEQ ID NO 28
 <211> LENGTH: 1273
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 28

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5					10					15	
Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
			20					25					30		
Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40				45				
His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
	50					55					60				
Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65					70					75				80	
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85						90					95	

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Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser	100	105	110
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile	115	120	125
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	130	135	140
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	145	150	155
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	180	185	190
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495

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Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
			500					505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515					520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
	530					535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550					555					560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
				565					570					575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
			580					585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
	610					615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625					630					635					640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
				645					650					655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
			660					665					670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
		675					680					685			
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
	690					695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
				725					730					735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740					745					750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
		755					760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
	770					775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
785					790					795					800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
				805					810					815	
Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp
		835					840					845			
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Leu	Pro	Pro	Leu
	850					855					860				
Leu	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865					870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile
				885					890					895	
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr

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900					905					910					
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
	915						920					925			
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala
	930					935					940				
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
	945					950					955				960
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
			965						970					975	
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln
		980						985					990		
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
		995					1000						1005		
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
	1010					1015					1020				
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
	1025					1030					1035				
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
	1040					1045					1050				
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
	1055					1060					1065				
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
	1070					1075					1080				
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn	
	1085					1090					1095				
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln	
	1100					1105					1110				
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val	
	1115					1120					1125				
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro	
	1130					1135					1140				
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn	
	1145					1150					1155				
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn	
	1160					1165					1170				
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu	
	1175					1180					1185				
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu	
	1190					1195					1200				
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu	
	1205					1210					1215				
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met	
	1220					1225					1230				
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys	
	1235					1240					1245				
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro	
	1250					1255					1260				
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr						
	1265					1270									

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<211> LENGTH: 1273
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 29

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1      5      10      15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20     25     30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35     40     45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50     55     60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65     70     75     80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85     90     95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100    105    110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115    120    125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130    135    140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145    150    155    160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165    170    175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180    185    190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195    200    205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210    215    220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225    230    235    240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245    250    255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260    265    270

Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
275    280    285

Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
290    295    300

Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305    310    315    320

Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
325    330    335

Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala
340    345    350

Trp Asn Arg Lys Arg Ile Ser Asn Cys Val Ala Asp Tyr Ser Val Leu
355    360    365

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Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
370						375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
			405					410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
			420					425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
		435					440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
450						455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465					470					475					480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
			485						490					495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
			500					505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515					520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
530						535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550					555					560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
			565						570					575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
			580					585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
610						615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625					630					635					640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
			645						650					655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
			660					665					670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
		675					680					685			
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
690						695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
				725					730					735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740					745					750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
		755					760						765		

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Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
770						775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
785					790					795					800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
			805						810					815	
Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Leu	Gly	Asp	Ile	Ala	Ala	Arg	Asp
		835					840					845			
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
	850					855					860				
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865					870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile
			885					890						895	
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900					905					910		
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
		915					920					925			
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala
	930					935					940				
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
945				950					955						960
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
			965					970						975	
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln
			980					985					990		
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
	995						1000					1005			
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
	1010					1015						1020			
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
	1025					1030					1035				
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
	1040					1045					1050				
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
	1055					1060					1065				
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
	1070					1075					1080				
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn	
	1085					1090					1095				
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln	
	1100					1105					1110				
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val	
	1115					1120					1125				
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro	
	1130					1135					1140				
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn	
	1145					1150					1155				
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn	

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1160	1165	1170
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu		
1175	1180	1185
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu		
1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu		
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met		
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys		
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro		
1250	1255	1260
Val Leu Lys Gly Val Lys Leu His Tyr Thr		
1265	1270	

<210> SEQ ID NO 30

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 30

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val		
1	5	10 15
Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe		
20	25	30
Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu		
35	40	45
His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp		
50	55	60
Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp		
65	70	75 80
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu		
85	90	95
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser		
100	105	110
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile		
115	120	125
Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr		
130	135	140
Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr		
145	150	155 160
Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu		
165	170	175
Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe		
180	185	190
Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr		
195	200	205
Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu		
210	215	220
Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr		

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225				230						235					240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser
				245					250					255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
			260					265					270		
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala
		275					280					285			
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys
	290					295					300				
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val
305					310					315					320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys
				325					330					335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala
			340					345					350		
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
	355					360						365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
	370					375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
			405					410					415		
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
		420						425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
	435					440						445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
	450					455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465				470					475						480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
			485					490					495		
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
		500						505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515				520						525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
	530					535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545				550					555						560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
			565					570					575		
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
		580						585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
	610					615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625				630						635					640

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Asn Val Phe Gln Thr Arg Ala Gly Cys Leu Ile Gly Ala Glu His Val	645	650	655
Asn Asn Ser Tyr Glu Cys Asp Ile Pro Ile Gly Ala Gly Ile Cys Ala	660	665	670
Ser Tyr Gln Thr Gln Thr Asn Ser Pro Arg Arg Ala Arg Ser Val Ala	675	680	685
Ser Gln Ser Ile Ile Ala Tyr Thr Met Ser Leu Gly Ala Glu Asn Ser	690	695	700
Val Ala Tyr Ser Asn Asn Ser Ile Ala Ile Pro Thr Asn Phe Thr Ile	705	710	715
Ser Val Thr Thr Glu Ile Leu Pro Val Ser Met Thr Lys Thr Ser Val	725	730	735
Asp Cys Thr Met Tyr Ile Cys Gly Asp Ser Thr Glu Cys Ser Asn Leu	740	745	750
Leu Leu Gln Tyr Gly Ser Phe Cys Thr Gln Leu Asn Arg Ala Leu Thr	755	760	765
Gly Ile Ala Val Glu Gln Asp Lys Asn Thr Gln Glu Val Phe Ala Gln	770	775	780
Val Lys Gln Ile Tyr Lys Thr Pro Pro Ile Lys Asp Phe Gly Gly Phe	785	790	795
Asn Phe Ser Gln Ile Leu Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser	805	810	815
Phe Ile Glu Asp Leu Leu Phe Asn Lys Val Thr Leu Ala Asp Ala Gly	820	825	830
Phe Ile Lys Gln Tyr Gly Asp Cys Ala Gly Asp Ala Ala Ala Arg Asp	835	840	845
Leu Ile Cys Ala Gln Lys Phe Asn Gly Leu Thr Val Ala Pro Pro Ala	850	855	860
Ala Thr Asp Glu Met Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly	865	870	875
Thr Ile Thr Ser Gly Trp Thr Phe Gly Ala Gly Ala Ala Leu Gln Ile	885	890	895
Pro Phe Ala Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr	900	905	910
Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn	915	920	925
Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala	930	935	940
Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn	945	950	955
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val	965	970	975
Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln	980	985	990
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val	995	1000	1005
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn	1010	1015	1020
Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys	1025	1030	1035

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Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro
1040						1045					1050			
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val
1055						1060					1065			
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His
1070						1075					1080			
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn
1085						1090					1095			
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln
1100						1105					1110			
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val
1115						1120					1125			
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro
1130						1135					1140			
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn
1145						1150					1155			
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn
1160						1165					1170			
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu
1175						1180					1185			
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu
1190						1195					1200			
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu
1205						1210					1215			
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met
1220						1225					1230			
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys
1235						1240					1245			
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro
1250						1255					1260			
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr					
1265						1270								

<210> SEQ ID NO 31

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 31

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5					10				15		
Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
			20					25					30		
Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40				45				
His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
	50				55					60					
Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65				70					75					80	
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85					90						95	

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Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser	100	105	110
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile	115	120	125
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	130	135	140
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	145	150	155
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	180	185	190
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val			

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500							505					510				
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	
		515					520					525				
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	
	530					535					540					
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	
545					550					555					560	
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	
				565					570					575		
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	
			580					585					590			
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	
	595						600					605				
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	
	610					615					620					
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	
625					630					635					640	
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
				645					650					655		
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
			660					665					670			
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	
	675					680						685				
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	
	690					695					700					
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	
705					710					715					720	
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	
				725					730					735		
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	
			740					745					750			
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	
	755						760					765				
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	
	770					775					780					
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	
785					790					795					800	
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	
			805						810					815		
Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	
			820					825					830			
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp	
	835						840					845				
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	
	850					855					860					
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	
865					870					875					880	
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile	
				885					890					895		
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	
		900						905					910			

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Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn
 915 920 925
 Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala
 930 935 940
 Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn
 945 950 955 960
 Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val
 965 970 975
 Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln
 980 985 990
 Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val
 995 1000 1005
 Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn
 1010 1015 1020
 Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys
 1025 1030 1035
 Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro
 1040 1045 1050
 Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val
 1055 1060 1065
 Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His
 1070 1075 1080
 Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn
 1085 1090 1095
 Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln
 1100 1105 1110
 Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val
 1115 1120 1125
 Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro
 1130 1135 1140
 Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
 1145 1150 1155
 His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
 1160 1165 1170
 Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
 1175 1180 1185
 Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
 1190 1195 1200
 Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
 1205 1210 1215
 Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
 1220 1225 1230
 Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
 1235 1240 1245
 Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
 1250 1255 1260
 Val Leu Lys Gly Val Lys Leu His Tyr Thr
 1265 1270

<210> SEQ ID NO 32

<211> LENGTH: 1273

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<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Variant 7 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 32
Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1           5           10          15
Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20          25          30
Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35          40          45
His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50          55          60
Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65          70          75          80
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85          90          95
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100         105        110
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115        120        125
Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130        135        140
Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145        150        155        160
Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165        170        175
Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180        185        190
Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195        200        205
Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210        215        220
Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225        230        235        240
Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245        250        255
Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260        265        270
Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
275        280        285
Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
290        295        300
Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305        310        315        320
Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
325        330        335
Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala
340        345        350
Trp Asn Arg Lys Arg Ile Ser Asn Cys Val Ala Asp Tyr Ser Val Leu
355        360        365

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Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
370						375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
				405					410					415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
			420					425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
		435					440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
450						455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465					470					475					480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
				485					490					495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
			500					505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515					520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
530						535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550					555					560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
				565					570					575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
			580					585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
610						615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625					630					635					640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
				645					650					655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
			660					665					670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
		675					680					685			
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
						695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
				725					730					735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740					745					750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
		755					760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln

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770	775	780
Val Lys Gln Ile Tyr Lys Thr Pro Pro Ile Lys Asp Phe Gly Gly Phe 785 790 795 800		
Asn Phe Ser Gln Ile Leu Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser 805 810 815		
Phe Ile Glu Asp Leu Leu Phe Asn Lys Val Thr Leu Ala Asp Ala Gly 820 825 830		
Phe Ile Lys Gln Tyr Gly Asp Cys Ala Gly Asp Ala Ala Arg Asp 835 840 845		
Leu Ile Cys Ala Gln Lys Phe Asn Gly Leu Thr Val Leu Pro Pro Leu 850 855 860		
Leu Thr Asp Glu Met Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly 865 870 875 880		
Thr Ile Thr Ser Gly Trp Thr Phe Gly Ala Gly Ala Ala Leu Gln Ile 885 890 895		
Pro Phe Ala Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr 900 905 910		
Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn 915 920 925		
Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala 930 935 940		
Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn 945 950 955 960		
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val 965 970 975		
Leu Asn Asp Ile Leu Ser Arg Leu Asp Pro Pro Glu Ala Glu Val Gln 980 985 990		
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val 995 1000 1005		
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn 1010 1015 1020		
Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys 1025 1030 1035		
Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro 1040 1045 1050		
Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val 1055 1060 1065		
Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His 1070 1075 1080		
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn 1085 1090 1095		
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln 1100 1105 1110		
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val 1115 1120 1125		
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro 1130 1135 1140		
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn 1145 1150 1155		
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn 1160 1165 1170		

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Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
1175 1180 1185

Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
1190 1195 1200

Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
1205 1210 1215

Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
1220 1225 1230

Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
1235 1240 1245

Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
1250 1255 1260

Val Leu Lys Gly Val Lys Leu His Tyr Thr
1265 1270

<210> SEQ ID NO 33
 <211> LENGTH: 1273
 <212> TYPE: PRT
 <213> ORGANISM: Artificial Sequence
 <220> FEATURE:
 <223> OTHER INFORMATION: Variant 8 of Mutated Sars-CoV-2 spike protein
 <400> SEQUENCE: 33

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1 5 10 15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20 25 30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35 40 45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50 55 60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65 70 75 80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85 90 95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100 105 110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115 120 125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130 135 140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145 150 155 160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165 170 175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180 185 190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195 200 205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210 215 220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225 230 235 240

Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser		
				245							250					255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro		
			260							265					270		
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala		
			275							280					285		
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys		
			290									300					
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val		
				305									315				320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys		
			325									330				335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala		
			340											350			
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu		
			355											365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro		
			370											380			
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe		
			385											395		400	
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly		
			405													415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys		
			420													430	
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn		
			435													445	
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe		
			450													460	
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys		
			465													475	480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly		
			485													495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val		
			500													510	
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys		
			515													525	
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn		
			530													540	
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu		
			545													555	560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val		
			565													575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe		
			580													585	590
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val		
			595													600	605
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile		
			610													615	620
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr</				

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Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
				645					650					655		
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
			660					665					670			
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	
		675						680				685				
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	
	690					695					700					
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	
	705				710					715					720	
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	
				725					730					735		
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	
			740					745					750			
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	
		755					760					765				
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	
	770					775					780					
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	
	785				790					795					800	
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	
			805					810						815		
Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	
			820					825					830			
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp	
		835					840					845				
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Leu	Pro	Pro	Leu	
	850					855					860					
Leu	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	
	865				870					875					880	
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile	
				885					890					895		
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	
			900					905					910			
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	
		915					920					925				
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala	
		930				935						940				
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn	
	945				950					955					960	
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val	
				965					970					975		
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln	
		980						985					990			
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val	
		995					1000					1005				
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn		
	1010					1015						1020				
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys		
	1025					1030					1035					
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro		

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1040	1045	1050
Gln Ser Ala Pro His Gly Val	Val Phe Leu His Val	Thr Tyr Val
1055	1060	1065
Pro Ala Gln Glu Lys Asn Phe	Thr Thr Ala Pro Ala	Ile Cys His
1070	1075	1080
Asp Gly Lys Ala His Phe Pro	Arg Glu Gly Val Phe	Val Ser Asn
1085	1090	1095
Gly Thr His Trp Phe Val Thr	Gln Arg Asn Phe Tyr	Glu Pro Gln
1100	1105	1110
Ile Ile Thr Thr Asp Asn Thr	Phe Val Ser Gly Asn	Cys Asp Val
1115	1120	1125
Val Ile Gly Ile Val Asn Asn	Thr Val Tyr Asp Pro	Leu Gln Pro
1130	1135	1140
Glu Leu Asp Ser Phe Lys Glu	Glu Leu Asp Lys Tyr	Phe Lys Asn
1145	1150	1155
His Thr Ser Pro Asp Val Asp	Leu Gly Asp Ile Ser	Gly Ile Asn
1160	1165	1170
Ala Ser Val Val Asn Ile Gln	Lys Glu Ile Asp Arg	Leu Asn Glu
1175	1180	1185
Val Ala Lys Asn Leu Asn Glu	Ser Leu Ile Asp Leu	Gln Glu Leu
1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile	Lys Trp Pro Trp Tyr	Ile Trp Leu
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile	Ala Ile Val Met Val	Thr Ile Met
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys	Cys Ser Cys Leu Lys	Gly Cys Cys
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys	Phe Asp Glu Asp Asp	Ser Glu Pro
1250	1255	1260
Val Leu Lys Gly Val Lys Leu	His Tyr Thr	
1265	1270	

<210> SEQ ID NO 34

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 9 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 34

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1 5 10 15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20 25 30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35 40 45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50 55 60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65 70 75 80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85 90 95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser

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100							105					110				
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile	
		115					120					125				
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	
	130					135					140					
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	
145					150					155					160	
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	
			165						170					175		
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	
		180						185					190			
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	
	195					200						205				
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	
	210					215					220					
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	
225					230					235					240	
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	
				245					250					255		
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	
			260					265					270			
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	
		275					280					285				
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	
	290					295					300					
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	
305					310					315					320	
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	
				325					330					335		
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	
			340					345					350			
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	
		355					360					365				
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	
	370					375					380					
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	
385					390					395					400	
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	
			405					410						415		
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	
			420					425					430			
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	
		435					440					445				
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	
	450					455					460					
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	
465					470					475					480	
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	
			485						490					495		
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	
			500					505					510			

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Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
	515						520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
	530					535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550				555						560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
				565					570					575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
			580					585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
	595						600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
	610					615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625					630					635					640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
				645				650						655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
			660					665					670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
		675				680					685				
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
	690					695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
				725					730					735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740					745					750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
		755					760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
	770					775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
785					790					795					800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
			805					810						815	
Pro	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp
		835						840					845		
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Leu	Pro	Pro	Leu
	850					855					860				
Leu	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865					870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile
				885					890					895	
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900					905						910	

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Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn
 915 920 925
 Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala
 930 935 940
 Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn
 945 950 955 960
 Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val
 965 970 975
 Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln
 980 985 990
 Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val
 995 1000 1005
 Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn
 1010 1015 1020
 Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys
 1025 1030 1035
 Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro
 1040 1045 1050
 Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val
 1055 1060 1065
 Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His
 1070 1075 1080
 Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn
 1085 1090 1095
 Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln
 1100 1105 1110
 Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val
 1115 1120 1125
 Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro
 1130 1135 1140
 Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
 1145 1150 1155
 His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
 1160 1165 1170
 Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
 1175 1180 1185
 Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
 1190 1195 1200
 Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
 1205 1210 1215
 Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
 1220 1225 1230
 Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
 1235 1240 1245
 Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
 1250 1255 1260
 Val Leu Lys Gly Val Lys Leu His Tyr Thr
 1265 1270

<210> SEQ ID NO 35

<211> LENGTH: 1273

<212> TYPE: PRT

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<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 10 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 35

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Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1           5           10           15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
          20           25           30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
          35           40           45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
          50           55           60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65           70           75           80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
          85           90           95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
          100          105          110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
          115          120          125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
          130          135          140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145           150           155           160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
          165           170           175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
          180          185          190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
          195          200          205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
          210          215          220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225           230           235           240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
          245           250           255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
          260          265          270

Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
          275          280          285

Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
          290          295          300

Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305           310           315           320

Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
          325          330          335

Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala
          340          345          350

Trp Asn Arg Lys Arg Ile Ser Asn Cys Val Ala Asp Tyr Ser Val Leu
          355          360          365

Tyr Asn Ser Ala Ser Phe Ser Thr Phe Lys Cys Tyr Gly Val Ser Pro

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370					375					380					
Thr 385	Lys	Leu	Asn	Asp	Leu 390	Cys	Phe	Thr	Asn	Val 395	Tyr	Ala	Asp	Ser	Phe 400
Val	Ile	Arg	Gly	Asp 405	Glu	Val	Arg	Gln	Ile 410	Ala	Pro	Gly	Gln	Thr	Gly 415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys 420
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn 435
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu 440	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe 450
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys 465
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly 485
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val 500
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys 515
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn 530
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu 545
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val 565
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe 580
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val 595
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile 610
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser 625
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val 645
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala 660
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala 675
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser 690
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile 705
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val 725
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu 740
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr 755
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln 770

Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	
Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp	
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Leu	Pro	Pro	Leu	
Leu	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile	
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala	
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn	
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val	
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln	
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val	
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn		
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys		
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro		
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val		
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His		
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn		
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln		
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val		
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro		
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn		
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn		

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Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu
1175						1180					1185			
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu
1190						1195					1200			
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu
1205						1210					1215			
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met
1220						1225					1230			
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys
1235						1240					1245			
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro
1250						1255					1260			
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr					
1265						1270								

<210> SEQ ID NO 36

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 11 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 36

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5					10				15		
Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
			20					25					30		
Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35				40					45				
His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
	50				55						60				
Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65				70					75					80	
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85					90					95		
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser
		100					105						110		
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile
	115					120					125				
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
	130				135					140					
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145				150					155					160	
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165					170					175		
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
		180					185					190			
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
	195					200					205				
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
	210					215				220					
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr
225				230					235					240	

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Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	530	535	540
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	545	550	555
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	565	570	575
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	580	585	590
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	595	600	605
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	610	615	620
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	625	630	635
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val			

645								650				655			
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
			660				665						670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
			675				680						685		
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
			690				695						700		
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
			705				710						715		
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
			725							730			735		
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740							745			750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
			755				760						765		
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
			770				775						780		
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
			785				790						795		
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
			805							810			815		
Pro	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820							825			830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp
			835				840						845		
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Leu	Pro	Pro	Leu
			850				855						860		
Leu	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
			865				870						875		
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile
			885							890			895		
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900							905			910		
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
			915				920						925		
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala
			930				935						940		
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
			945				950						955		
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
			965							970			975		
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln
			980							985			990		
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
			995				1000						1005		
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
			1010				1015						1020		
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
			1025				1030						1035		
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
			1040				1045						1050		

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Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val
 1055 1060 1065
 Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His
 1070 1075 1080
 Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn
 1085 1090 1095
 Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln
 1100 1105 1110
 Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val
 1115 1120 1125
 Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro
 1130 1135 1140
 Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
 1145 1150 1155
 His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
 1160 1165 1170
 Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
 1175 1180 1185
 Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
 1190 1195 1200
 Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
 1205 1210 1215
 Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
 1220 1225 1230
 Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
 1235 1240 1245
 Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
 1250 1255 1260
 Val Leu Lys Gly Val Lys Leu His Tyr Thr
 1265 1270

<210> SEQ ID NO 37

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 12 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 37

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
 1 5 10 15
 Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
 20 25 30
 Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
 35 40 45
 His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
 50 55 60
 Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
 65 70 75 80
 Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
 85 90 95
 Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
 100 105 110

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Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile	115	120	125
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	130	135	140
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	145	150	155
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	180	185	190
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510

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Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515					520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
	530					535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550					555					560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
				565					570					575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
			580					585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
	610					615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625					630					635					640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
				645					650					655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
			660					665					670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
		675					680					685			
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
	690					695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
				725					730					735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740					745					750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
		755					760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
	770					775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
785					790					795					800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
			805					810						815	
Pro	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp
		835					840					845			
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Leu	Pro	Pro	Leu
	850					855					860				
Leu	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865					870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile
			885					890						895	
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900					905					910		
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn

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915	920	925
Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Pro Ser Ala		
930	935	940
Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn		
945	950	955 960
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val		
965	970	975
Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln		
980	985	990
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val		
995	1000	1005
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn		
1010	1015	1020
Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys		
1025	1030	1035
Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro		
1040	1045	1050
Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val		
1055	1060	1065
Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His		
1070	1075	1080
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn		
1085	1090	1095
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln		
1100	1105	1110
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val		
1115	1120	1125
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro		
1130	1135	1140
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn		
1145	1150	1155
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn		
1160	1165	1170
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu		
1175	1180	1185
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu		
1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu		
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met		
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys		
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro		
1250	1255	1260
Val Leu Lys Gly Val Lys Leu His Tyr Thr		
1265	1270	

<210> SEQ ID NO 38

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

-continued

<220> FEATURE:

<223> OTHER INFORMATION: Variant 13 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 38

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Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1           5           10           15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20           25           30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35           40           45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50           55           60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65           70           75           80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85           90           95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100          105          110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115          120          125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130          135          140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145          150          155          160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165          170          175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180          185          190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195          200          205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210          215          220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225          230          235          240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245          250          255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260          265          270

Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
275          280          285

Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
290          295          300

Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305          310          315          320

Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
325          330          335

Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala
340          345          350

Trp Asn Arg Lys Arg Ile Ser Asn Cys Val Ala Asp Tyr Ser Val Leu
355          360          365

Tyr Asn Ser Ala Ser Phe Ser Thr Phe Lys Cys Tyr Gly Val Ser Pro
370          375          380

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Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395	400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430	
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445	
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460	
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475	480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510	
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525	
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	530	535	540	
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	545	550	555	560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	565	570	575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	580	585	590	
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	595	600	605	
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	610	615	620	
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	625	630	635	640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	645	650	655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	660	665	670	
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	675	680	685	
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	690	695	700	
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	705	710	715	720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	725	730	735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	740	745	750	
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	755	760	765	
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	770	775	780	

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Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	
785					790					795					800	
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	
			805						810					815		
Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	
		820						825					830			
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Leu	Gly	Asp	Ile	Ala	Ala	Arg	Asp	
	835						840					845				
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	
	850					855					860					
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	
865					870					875					880	
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile	
			885					890						895		
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	
			900					905					910			
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	
	915						920					925				
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala	
	930					935						940				
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn	
945				950					955						960	
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val	
			965					970						975		
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln	
		980						985					990			
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val	
	995						1000					1005				
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn		
	1010					1015						1020				
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys		
	1025					1030					1035					
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro		
	1040					1045					1050					
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val		
	1055					1060					1065					
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His		
	1070					1075					1080					
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn		
	1085					1090					1095					
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln		
	1100					1105					1110					
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val		
	1115					1120					1125					
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro		
	1130					1135					1140					
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn		
	1145					1150					1155					
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn		
	1160					1165					1170					
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu		

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1175	1180	1185
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu		
1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu		
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met		
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys		
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro		
1250	1255	1260
Val Leu Lys Gly Val Lys Leu His Tyr Thr		
1265	1270	

<210> SEQ ID NO 39

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 14 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 39

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val		
1	5	10
Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe		
20	25	30
Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu		
35	40	45
His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp		
50	55	60
Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp		
65	70	75
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu		
85	90	95
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser		
100	105	110
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile		
115	120	125
Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr		
130	135	140
Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr		
145	150	155
Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu		
165	170	175
Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe		
180	185	190
Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr		
195	200	205
Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu		
210	215	220
Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr		
225	230	235
Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser		

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245								250				255				
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	
			260					265					270			
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	
		275					280					285				
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	
	290					295					300					
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	
305					310					315					320	
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	
				325					330					335		
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	
			340					345					350			
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	
		355					360					365				
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	
	370					375					380					
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	
385					390					395					400	
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	
			405					410						415		
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	
		420						425					430			
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	
	435						440					445				
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	
	450					455					460					
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	
465					470					475					480	
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	
			485						490					495		
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	
			500					505					510			
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	
		515					520					525				
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	
	530					535					540					
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	
545					550					555					560	
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	
			565					570						575		
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	
			580					585					590			
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	
		595					600					605				
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	
	610					615					620					
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	
625					630					635					640	
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
			645					650						655		

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Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
			660					665					670			
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	
		675					680				685					
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	
	690					695					700					
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	
705				710						715					720	
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	
			725						730					735		
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	
		740					745						750			
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	
	755						760					765				
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	
	770					775					780					
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	
785					790					795					800	
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	
			805					810						815		
Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	
		820						825					830			
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Leu	Gly	Asp	Ile	Ala	Ala	Arg	Asp	
	835						840					845				
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	
	850					855					860					
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	
865					870					875					880	
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile	
		885						890						895		
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	
		900						905					910			
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	
	915						920					925				
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala	
	930					935					940					
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn	
945					950					955					960	
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val	
		965							970					975		
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln	
		980						985					990			
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val	
	995						1000						1005			
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn		
	1010					1015						1020				
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys		
	1025					1030					1035					
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro		
	1040					1045							1050			

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Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val
1055						1060					1065			
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His
1070						1075					1080			
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn
1085						1090					1095			
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln
1100						1105					1110			
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val
1115						1120					1125			
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro
1130						1135					1140			
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn
1145						1150					1155			
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn
1160						1165					1170			
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu
1175						1180					1185			
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu
1190						1195					1200			
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu
1205						1210					1215			
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met
1220						1225					1230			
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys
1235						1240					1245			
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro
1250						1255					1260			
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr					
1265						1270								

<210> SEQ ID NO 40

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 15 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 40

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5					10				15		

Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
		20						25					30		

Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40				45				

His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
		50				55					60				

Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65					70				75					80	

Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85					90						95	

Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser
			100				105						110		

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Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile
		115					120					125			
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
	130				135						140				
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145					150					155					160
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165						170					175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
		180						185					190		
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
	195					200						205			
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
	210					215					220				
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr
225					230					235					240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser
			245					250						255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
			260					265					270		
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala
	275						280					285			
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys
	290					295					300				
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val
305					310					315					320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys
			325						330					335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala
			340					345					350		
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
	355						360					365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
	370					375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
			405					410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
			420					425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
	435						440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
	450					455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465					470					475					480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
			485					490						495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
			500					505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys

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515						520						525					
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn		
530						535					540						
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu		
545					550					555					560		
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val		
				565					570					575			
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe		
			580					585					590				
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val		
	595						600					605					
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile		
	610					615					620						
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser		
625				630					635						640		
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val		
			645					650						655			
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala		
		660						665					670				
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala		
	675					680					685						
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser		
	690					695				700							
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile		
705				710						715					720		
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val		
			725					730						735			
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu		
		740						745					750				
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr		
	755						760					765					
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln		
	770				775						780						
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe		
785				790						795					800		
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser		
			805					810						815			
Pro	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly		
		820						825					830				
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Leu	Gly	Asp	Ile	Ala	Ala	Arg	Asp		
	835					840						845					
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala		
	850				855						860						
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly		
865					870					875					880		
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile		
			885					890						895			
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr		
		900						905					910				
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn		
	915						920						925				

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Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Ala Ser Ala
 930 935 940
 Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn
 945 950 955 960
 Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val
 965 970 975
 Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln
 980 985 990
 Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val
 995 1000 1005
 Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn
 1010 1015 1020
 Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys
 1025 1030 1035
 Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro
 1040 1045 1050
 Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val
 1055 1060 1065
 Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His
 1070 1075 1080
 Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn
 1085 1090 1095
 Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln
 1100 1105 1110
 Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val
 1115 1120 1125
 Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro
 1130 1135 1140
 Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
 1145 1150 1155
 His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
 1160 1165 1170
 Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
 1175 1180 1185
 Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
 1190 1195 1200
 Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
 1205 1210 1215
 Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
 1220 1225 1230
 Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
 1235 1240 1245
 Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
 1250 1255 1260
 Val Leu Lys Gly Val Lys Leu His Tyr Thr
 1265 1270

<210> SEQ ID NO 41

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

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<223> OTHER INFORMATION: Variant 16 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 41

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Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
 1           5           10           15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
 20           25           30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
 35           40           45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
 50           55           60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
 65           70           75           80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
 85           90           95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100           105           110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115           120           125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130           135           140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145           150           155           160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165           170           175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180           185           190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195           200           205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210           215           220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225           230           235           240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245           250           255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260           265           270

Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
275           280           285

Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
290           295           300

Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305           310           315           320

Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
325           330           335

Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala
340           345           350

Trp Asn Arg Lys Arg Ile Ser Asn Cys Val Ala Asp Tyr Ser Val Leu
355           360           365

Tyr Asn Ser Ala Ser Phe Ser Thr Phe Lys Cys Tyr Gly Val Ser Pro
370           375           380

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Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
			405					410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
			420					425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
		435						440				445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
	450					455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465					470					475					480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
			485						490					495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
			500					505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515					520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
	530					535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550					555					560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
			565						570					575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
			580					585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
	610					615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625					630					635					640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
			645						650					655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
			660					665					670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
		675					680					685			
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
	690					695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
			725						730					735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740					745					750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
		755					760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
	770					775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe

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785	790	795	800
Asn Phe Ser Gln Ile Leu Pro Asp Pro Ser Lys Pro Ser Lys Arg Ser	805	810	815
Phe Ile Glu Asp Leu Leu Phe Asn Lys Val Thr Leu Ala Asp Ala Gly	820	825	830
Phe Ile Lys Gln Tyr Gly Asp Cys Leu Gly Asp Ile Ala Ala Arg Asp	835	840	845
Leu Ile Cys Ala Gln Lys Phe Asn Gly Leu Thr Val Ala Pro Pro Ala	850	855	860
Ala Thr Asp Glu Met Ile Ala Gln Tyr Thr Ser Ala Leu Leu Ala Gly	865	870	875
Thr Ile Thr Ser Gly Trp Thr Phe Gly Ala Gly Pro Ala Leu Gln Ile	885	890	895
Pro Phe Ala Met Gln Met Ala Tyr Arg Phe Asn Gly Ile Gly Val Thr	900	905	910
Gln Asn Val Leu Tyr Glu Asn Gln Lys Leu Ile Ala Asn Gln Phe Asn	915	920	925
Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Pro Ser Ala	930	935	940
Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn	945	950	955
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val	965	970	975
Leu Asn Asp Ile Leu Ser Arg Leu Asp Pro Pro Glu Ala Glu Val Gln	980	985	990
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val	995	1000	1005
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn	1010	1015	1020
Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys	1025	1030	1035
Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro	1040	1045	1050
Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val	1055	1060	1065
Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His	1070	1075	1080
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn	1085	1090	1095
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln	1100	1105	1110
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val	1115	1120	1125
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro	1130	1135	1140
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn	1145	1150	1155
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn	1160	1165	1170
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu	1175	1180	1185

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Val Ala  Lys Asn Leu Asn Glu  Ser Leu Ile Asp Leu  Gln Glu Leu
  1190                      1195                1200

Gly Lys  Tyr Glu Gln Tyr  Ile  Lys Trp Pro Trp Tyr  Ile Trp Leu
  1205                      1210                1215

Gly Phe  Ile Ala Gly Leu  Ile  Ala Ile Val Met Val  Thr Ile Met
  1220                      1225                1230

Leu Cys  Cys Met Thr Ser Cys  Cys Ser Cys Leu Lys  Gly Cys Cys
  1235                      1240                1245

Ser Cys  Gly Ser Cys Cys Lys  Phe Asp Glu Asp Asp  Ser Glu Pro
  1250                      1255                1260

Val Leu  Lys Gly Val Lys Leu  His Tyr Thr
  1265                      1270

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<210> SEQ ID NO 42
<211> LENGTH: 1273
<212> TYPE: PRT
<213> ORGANISM: Artificial Sequence
<220> FEATURE:
<223> OTHER INFORMATION: Variant 17 of Mutated Sars-CoV-2 spike protein

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<400> SEQUENCE: 42

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Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
 1      5              10              15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
 20     25              30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
 35     40              45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
 50     55              60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
 65     70              75              80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
 85     90              95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100    105             110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115    120             125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130    135             140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145    150             155             160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165    170             175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180    185             190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195    200             205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210    215             220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225    230             235             240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245    250             255

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Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	530	535	540
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	545	550	555
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	565	570	575
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	580	585	590
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	595	600	605
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	610	615	620
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	625	630	635
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	645	650	655

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Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	660	665	670
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	675	680	685
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	690	695	700
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	705	710	715
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	725	730	735
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	740	745	750
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	755	760	765
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	770	775	780
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	785	790	795
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	805	810	815
Pro	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	820	825	830
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Leu	Gly	Asp	Ile	Ala	Ala	Arg	Asp	835	840	845
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	850	855	860
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	865	870	875
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile	885	890	895
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	900	905	910
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	915	920	925
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala	930	935	940
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn	945	950	955
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val	965	970	975
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln	980	985	990
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val	995	1000	1005
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn		1010	1015	1020
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys		1025	1030	1035
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro		1040	1045	1050
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val				

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1055	1060	1065
Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His		
1070	1075	1080
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn		
1085	1090	1095
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln		
1100	1105	1110
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val		
1115	1120	1125
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro		
1130	1135	1140
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn		
1145	1150	1155
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn		
1160	1165	1170
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu		
1175	1180	1185
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu		
1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu		
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met		
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys		
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro		
1250	1255	1260
Val Leu Lys Gly Val Lys Leu His Tyr Thr		
1265	1270	

<210> SEQ ID NO 43

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 18 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 43

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val		
1	5	10 15
Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe		
20	25	30
Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu		
35	40	45
His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp		
50	55	60
Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp		
65	70	75 80
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu		
85	90	95
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser		
100	105	110
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile		

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115					120					125					
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
130						135					140				
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145					150					155					160
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165						170					175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
		180						185					190		
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
	195					200						205			
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
	210					215					220				
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr
225					230					235					240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser
			245					250					255		
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
		260						265					270		
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala
		275					280					285			
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys
	290				295						300				
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val
305					310					315					320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys
			325						330					335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala
		340						345					350		
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
	355					360						365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
	370					375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
			405					410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
		420						425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
		435					440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
	450					455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465					470					475					480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
			485						490					495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
			500					505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515					520					525			

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Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
530						535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550				555						560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
				565					570					575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
		580						585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
	610					615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625					630					635					640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
				645					650					655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
		660						665					670		
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
		675					680				685				
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
	690					695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
				725					730					735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740					745					750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
		755					760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
		770				775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
785					790					795					800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
				805				810						815	
Pro	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Leu	Gly	Asp	Ile	Ala	Ala	Arg	Asp
		835					840					845			
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
		850				855					860				
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865					870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile
				885					890					895	
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900					905					910		
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
		915					920						925		

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Ser Ala Ile Gly Lys Ile Gln Asp Ser Leu Ser Ser Thr Pro Ser Ala
 930                      935                      940

Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn
 945                      950                      955                      960

Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val
                      965                      970                      975

Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln
                      980                      985                      990

Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val
                      995                      1000                      1005

Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn
 1010                      1015                      1020

Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys
 1025                      1030                      1035

Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro
 1040                      1045                      1050

Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val
 1055                      1060                      1065

Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His
 1070                      1075                      1080

Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn
 1085                      1090                      1095

Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln
 1100                      1105                      1110

Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val
 1115                      1120                      1125

Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro
 1130                      1135                      1140

Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
 1145                      1150                      1155

His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
 1160                      1165                      1170

Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
 1175                      1180                      1185

Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
 1190                      1195                      1200

Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
 1205                      1210                      1215

Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
 1220                      1225                      1230

Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
 1235                      1240                      1245

Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
 1250                      1255                      1260

Val Leu Lys Gly Val Lys Leu His Tyr Thr
 1265                      1270

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<210> SEQ ID NO 44

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 19 of Mutated Sars-CoV-2 spike protein

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<400> SEQUENCE: 44

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Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1           5           10          15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20          25          30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35          40          45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50          55          60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65          70          75          80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85          90          95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100         105         110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115         120         125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130         135         140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145         150         155         160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165         170         175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180         185         190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195         200         205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210         215         220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225         230         235         240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245         250         255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260         265         270

Arg Thr Phe Leu Leu Lys Tyr Asn Glu Asn Gly Thr Ile Thr Asp Ala
275         280         285

Val Asp Cys Ala Leu Asp Pro Leu Ser Glu Thr Lys Cys Thr Leu Lys
290         295         300

Ser Phe Thr Val Glu Lys Gly Ile Tyr Gln Thr Ser Asn Phe Arg Val
305         310         315         320

Gln Pro Thr Glu Ser Ile Val Arg Phe Pro Asn Ile Thr Asn Leu Cys
325         330         335

Pro Phe Gly Glu Val Phe Asn Ala Thr Arg Phe Ala Ser Val Tyr Ala
340         345         350

Trp Asn Arg Lys Arg Ile Ser Asn Cys Val Ala Asp Tyr Ser Val Leu
355         360         365

Tyr Asn Ser Ala Ser Phe Ser Thr Phe Lys Cys Tyr Gly Val Ser Pro
370         375         380

Thr Lys Leu Asn Asp Leu Cys Phe Thr Asn Val Tyr Ala Asp Ser Phe

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385				390					395					400			
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly		
				405					410					415			
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys		
				420					425					430			
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn		
				435					440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe		
				450					455					460			
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys		
				465					470					475			
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly		
				485					490					495			
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val		
				500					505					510			
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys		
				515					520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn		
				530					535					540			
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu		
				545					550					555			
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val		
				565					570					575			
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe		
				580					585					590			
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val		
				595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile		
				610					615					620			
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser		
				625					630					635			
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val		
				645					650					655			
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala		
				660					665					670			
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala		
				675					680					685			
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser		
				690					695					700			
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile		
				705					710					715			
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val		
				725					730					735			
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu		
				740					745					750			
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr		
				755					760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln		
				770					775					780			
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe		
				785					790					795			
												800					

Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
805															
Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
820															
825															
830															
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp
835															
840															
845															
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
850															
855															
860															
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865															
870															
875															
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile
885															
890															
895															
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
900															
905															
910															
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
915															
920															
925															
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala
930															
935															
940															
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
945															
950															
955															
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
965															
970															
975															
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln
980															
985															
990															
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
995															
1000															
1005															
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
1010															
1015															
1020															
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
1025															
1030															
1035															
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
1040															
1045															
1050															
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
1055															
1060															
1065															
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
1070															
1075															
1080															
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn	
1085															
1090															
1095															
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln	
1100															
1105															
1110															
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val	
1115															
1120															
1125															
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro	
1130															
1135															
1140															
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn	
1145															
1150															
1155															
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn	
1160															
1165															
1170															
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu	
1175															
1180															
1185															

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Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu
1190						1195					1200			
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu
1205						1210					1215			
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met
1220						1225					1230			
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys
1235						1240					1245			
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro
1250						1255					1260			
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr					
1265						1270								

<210> SEQ ID NO 45

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 20 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 45

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5					10					15	
Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
			20					25					30		
Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40				45				
His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
		50				55					60				
Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65					70					75				80	
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85					90					95		
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser
		100					105						110		
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile
		115					120					125			
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
	130					135					140				
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145					150					155				160	
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165						170					175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
		180						185					190		
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
		195					200					205			
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
	210					215					220				
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr
225					230					235				240	
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser
			245						250					255	

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Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
			260					265					270		
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala
		275					280					285			
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys
	290					295					300				
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val
305					310					315					320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys
				325					330					335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala
			340					345					350		
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
		355					360					365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
	370					375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385					390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
				405					410					415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
			420					425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
		435					440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
	450					455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
465					470					475					480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
				485					490					495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
			500					505					510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515					520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
	530					535					540				
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
545					550					555					560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
				565				570						575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
			580					585					590		
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595					600					605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
	610					615					620				
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
625				630					635						640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
				645				650						655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala

660						665						670							
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala				
675						680						685							
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser				
690						695						700							
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile				
705						710						715						720	
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val				
725						730						735							
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu				
740						745						750							
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr				
755						760						765							
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln				
770						775						780							
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe				
785						790						795						800	
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser				
805						810						815							
Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly				
820						825						830							
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp				
835						840						845							
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala				
850						855						860							
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly				
865						870						875						880	
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile				
885						890						895							
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr				
900						905						910							
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn				
915						920						925							
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala				
930						935						940							
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn				
945						950						955						960	
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val				
965						970						975							
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln				
980						985						990							
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val				
995						1000						1005							
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn					
1010						1015						1020							
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys					
1025						1030						1035							
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro					
1040						1045						1050							
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val					
1055						1060						1065							

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Pro Ala  Gln Glu Lys Asn Phe  Thr Thr Ala Pro  Ala  Ile Cys His
1070                                1075                                1080

Asp Gly  Lys Ala His Phe  Pro  Arg Glu Gly Val  Phe  Val Ser Asn
1085                                1090                                1095

Gly Thr  His Trp Phe Val  Thr  Gln Arg Asn Phe  Tyr  Glu Pro Gln
1100                                1105                                1110

Ile Ile  Thr Thr Asp Asn Thr  Phe Val Ser Gly Asn  Cys Asp Val
1115                                1120                                1125

Val Ile  Gly Ile Val Asn Asn  Thr Val Tyr Asp Pro  Leu Gln Pro
1130                                1135                                1140

Glu Leu  Asp Ser Phe Lys Glu  Glu Leu Asp Lys Tyr  Phe Lys Asn
1145                                1150                                1155

His Thr  Ser Pro Asp Val  Asp  Leu Gly Asp Ile  Ser  Gly Ile Asn
1160                                1165                                1170

Ala Ser  Val Val Asn Ile  Gln  Lys Glu Ile Asp Arg  Leu Asn Glu
1175                                1180                                1185

Val Ala  Lys Asn Leu Asn Glu  Ser Leu Ile Asp Leu  Gln Glu Leu
1190                                1195                                1200

Gly Lys  Tyr Glu Gln Tyr  Ile  Lys Trp Pro Trp Tyr  Ile Trp Leu
1205                                1210                                1215

Gly Phe  Ile Ala Gly Leu  Ile  Ala Ile Val Met Val  Thr Ile Met
1220                                1225                                1230

Leu Cys  Cys Met Thr Ser  Cys  Cys Ser Cys Leu Lys  Gly Cys Cys
1235                                1240                                1245

Ser Cys  Gly Ser Cys Cys Lys  Phe Asp Glu Asp Asp  Ser Glu Pro
1250                                1255                                1260

Val Leu  Lys Gly Val Lys Leu  His Tyr Thr
1265                                1270

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<210> SEQ ID NO 46

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 21 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 46

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Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1      5              10              15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20     25              30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35     40              45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50     55              60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65     70              75              80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85     90              95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100    105              110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115    120              125

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Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	130	135	140
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	145	150	155
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	180	185	190
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525

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Lys 530	Ser	Thr	Asn	Leu	Val	Lys 535	Asn	Lys	Cys	Val	Asn 540	Phe	Asn	Phe	Asn
Gly 545	Leu	Thr	Gly	Thr	Gly 550	Val	Leu	Thr	Glu	Ser	Asn 555	Lys	Lys	Phe	Leu 560
Pro	Phe	Gln	Gln	Phe 565	Gly	Arg	Asp	Ile	Ala 570	Asp	Thr	Thr	Asp	Ala 575	Val
Arg	Asp	Pro	Gln 580	Thr	Leu	Glu	Ile	Leu 585	Asp	Ile	Thr	Pro	Cys 590	Ser	Phe
Gly	Gly	Val 595	Ser	Val	Ile	Thr	Pro 600	Gly	Thr	Asn	Thr	Ser 605	Asn	Gln	Val
Ala 610	Val	Leu	Tyr	Gln	Asp	Val 615	Asn	Cys	Thr	Glu	Val 620	Pro	Val	Ala	Ile
His 625	Ala	Asp	Gln	Leu	Thr 630	Pro	Thr	Trp	Arg	Val 635	Tyr	Ser	Thr	Gly	Ser 640
Asn	Val	Phe	Gln	Thr 645	Arg	Ala	Gly	Cys	Leu 650	Ile	Gly	Ala	Glu	His 655	Val
Asn	Asn	Ser 660	Tyr	Glu	Cys	Asp	Ile 665	Pro	Ile	Gly	Ala	Gly	Ile 670	Cys	Ala
Ser	Tyr	Gln 675	Thr	Gln	Thr	Asn 680	Ser	Pro	Arg	Arg	Ala 685	Arg	Ser	Val	Ala
Ser 690	Gln	Ser	Ile	Ile	Ala 695	Tyr	Thr	Met	Ser	Leu	Gly 700	Ala	Glu	Asn	Ser
Val 705	Ala	Tyr	Ser	Asn	Asn 710	Ser	Ile	Ala	Ile	Pro 715	Thr	Asn	Phe	Thr	Ile 720
Ser	Val	Thr	Thr	Glu 725	Ile	Leu	Pro	Val	Ser	Met 730	Thr	Lys	Thr	Ser	Val 735
Asp	Cys	Thr 740	Met	Tyr	Ile	Cys	Gly 745	Asp	Ser	Thr	Glu	Cys	Ser 750	Asn	Leu
Leu	Leu	Gln 755	Tyr	Gly	Ser	Phe	Cys 760	Thr	Gln	Leu	Asn 765	Arg	Ala	Leu	Thr
Gly 770	Ile	Ala	Val	Glu	Gln 775	Asp	Lys	Asn	Thr	Gln	Glu 780	Val	Phe	Ala	Gln
Val 785	Lys	Gln	Ile	Tyr	Lys 790	Thr	Pro	Pro	Ile	Lys 795	Asp	Phe	Gly	Gly	Phe 800
Asn	Phe	Ser	Gln 805	Ile	Leu	Pro	Asp	Pro	Ser 810	Lys	Pro	Ser	Lys	Arg 815	Ser
Pro	Ile	Glu 820	Asp	Leu	Leu	Phe	Asn 825	Lys	Val	Thr	Leu	Ala	Asp 830	Ala	Gly
Phe	Ile	Lys 835	Gln	Tyr	Gly	Asp	Cys 840	Ala	Gly	Asp	Ala 845	Ala	Ala	Arg	Asp
Leu 850	Ile	Cys	Ala	Gln	Lys 855	Phe	Asn	Gly	Leu	Thr	Val 860	Ala	Pro	Pro	Ala
Ala 865	Thr	Asp	Glu	Met	Ile 870	Ala	Gln	Tyr	Thr	Ser 875	Ala	Leu	Leu	Ala	Gly 880
Thr	Ile	Thr	Ser	Gly 885	Trp	Thr	Phe	Gly	Ala 890	Gly	Ala	Ala	Leu	Gln	Ile 895
Pro	Phe	Pro 900	Met	Gln	Met	Ala	Tyr 905	Arg	Phe	Asn	Gly	Ile 910	Gly	Val	Thr
Gln 915	Asn	Val	Leu	Tyr	Glu	Asn 920	Gln	Lys	Leu	Ile	Ala 925	Asn	Gln	Phe	Asn
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala

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930	935	940
Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn 945 950 955 960		
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val 965 970 975		
Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln 980 985 990		
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val 995 1000 1005		
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn 1010 1015 1020		
Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys 1025 1030 1035		
Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro 1040 1045 1050		
Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val 1055 1060 1065		
Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His 1070 1075 1080		
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn 1085 1090 1095		
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln 1100 1105 1110		
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val 1115 1120 1125		
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro 1130 1135 1140		
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn 1145 1150 1155		
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn 1160 1165 1170		
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu 1175 1180 1185		
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu 1190 1195 1200		
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu 1205 1210 1215		
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met 1220 1225 1230		
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys 1235 1240 1245		
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro 1250 1255 1260		
Val Leu Lys Gly Val Lys Leu His Tyr Thr 1265 1270		

<210> SEQ ID NO 47

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 22 of Mutated Sars-CoV-2 spike protein

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<400> SEQUENCE: 47

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1			5						10					15	
Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
			20					25					30		
Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40					45			
His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
	50				55						60				
Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65				70					75					80	
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85						90					95	
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser
		100					105						110		
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile
	115					120						125			
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
	130					135					140				
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145					150				155					160	
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165						170					175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
		180					185						190		
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
		195				200						205			
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
	210					215					220				
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr
225				230					235					240	
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser
			245						250					255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
		260					265						270		
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala
		275					280					285			
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys
	290				295						300				
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val
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Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys
			325						330					335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala
		340						345					350		
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
	355					360					365				
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
	370					375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
385				390					395					400	

Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
				405					410						
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
				420					425						
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
				435					440						
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
				450					455						
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
				465					470						
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
				485					490						
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
				500					505						
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
				515					520						
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
				530					535						
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
				545					550						
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
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Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
				580					585						
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
				595					600						
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
				610					615						
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
				625					630						
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val
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Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala
				660					665						
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala
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Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
				690					695						
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
				705					710						
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
				725					730						
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
				740					745						
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
				755					760						
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
				770					775						
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
				785					790						

Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	
				805					810				815			
Phe	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	
				820					825				830			
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp	
				835					840				845			
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	
				850					855				860			
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	
				865					870				875			
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile	
				885					890				895			
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	
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Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	
				915					920				925			
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala	
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Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn	
				945					950				955			
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val	
				965					970				975			
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln	
				980					985				990			
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val	
				995					1000				1005			
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn		
				1010					1015				1020			
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys		
				1025					1030				1035			
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro		
				1040					1045				1050			
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val		
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Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His		
				1070					1075				1080			
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn		
				1085					1090				1095			
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln		
				1100					1105				1110			
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val		
				1115					1120				1125			
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro		
				1130					1135				1140			
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn		
				1145					1150				1155			
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn		
				1160					1165				1170			
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu		
				1175					1180				1185			
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu		

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1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu		
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met		
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys		
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro		
1250	1255	1260
Val Leu Lys Gly Val Lys Leu His Tyr Thr		
1265	1270	

<210> SEQ ID NO 48

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 23 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 48

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val		
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20	25	30
Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu		
35	40	45
His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp		
50	55	60
Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp		
65	70	75 80
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu		
85	90	95
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser		
100	105	110
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile		
115	120	125
Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr		
130	135	140
Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr		
145	150	155 160
Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu		
165	170	175
Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe		
180	185	190
Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr		
195	200	205
Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu		
210	215	220
Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr		
225	230	235 240
Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser		
245	250	255
Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro		

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260							265					270				
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	
		275					280					285				
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	
		290			295						300					
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	
305					310					315					320	
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	
				325					330					335		
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	
				340			345						350			
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	
		355				360						365				
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	
		370				375					380					
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	
385					390					395					400	
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	
				405					410					415		
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	
		420						425					430			
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	
		435				440						445				
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	
		450				455					460					
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	
465					470					475					480	
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	
				485					490					495		
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	
				500			505						510			
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	
		515				520						525				
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	
		530				535					540					
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	
545					550					555					560	
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	
				565					570					575		
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	
				580			585						590			
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	
		595				600						605				
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	
		610				615					620					
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	
625					630					635					640	
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
				645					650					655		
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
		660						665					670			

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Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala			
675																680		
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser			
690																695	700	
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile			
705																710	715	720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val			
725																730	735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu			
740																745	750	
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr			
755																760	765	
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln			
770																775	780	
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe			
785																790	795	800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser			
805																810	815	
Pro	Ile	Glu	Asp	Leu	Leu	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly			
820																825	830	
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp			
835																840	845	
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala			
850																855	860	
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly			
865																870	875	880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile			
885																890	895	
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr			
900																905	910	
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn			
915																920	925	
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala			
930																935	940	
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn			
945																950	955	960
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val			
965																970	975	
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln			
980																985	990	
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val			
995																1000	1005	
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn				
1010																1015	1020	
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys				
1025																1030	1035	
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro				
1040																1045	1050	
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val				
1055																1060	1065	

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Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His
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Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn
1085						1090					1095			
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln
1100						1105					1110			
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val
1115						1120					1125			
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro
1130						1135					1140			
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn
1145						1150					1155			
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn
1160						1165					1170			
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu
1175						1180					1185			
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu
1190						1195					1200			
Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu
1205						1210					1215			
Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met
1220						1225					1230			
Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys
1235						1240					1245			
Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro
1250						1255					1260			
Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr					
1265						1270								

<210> SEQ ID NO 49

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 24 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 49

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5					10				15		

Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
				20				25					30		

Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35				40					45				

His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
		50			55						60				

Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65				70					75					80	

Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85					90						95	

Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser
		100				105							110		

Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile
		115				120						125			

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Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
130						135					140				
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145					150					155					160
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165						170					175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
		180						185					190		
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
	195						200					205			
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
	210					215					220				
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr
	225				230					235					240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser
			245					250						255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
			260					265					270		
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala
		275					280					285			
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys
	290					295					300				
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val
	305				310					315					320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys
			325						330					335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala
		340						345					350		
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
		355					360					365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
	370					375					380				
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
	385				390					395					400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
			405					410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
		420						425					430		
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
		435					440					445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
	450					455					460				
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
	465				470				475						480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
			485					490						495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
		500					505						510		
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515					520					525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn

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530				535				540							
Gly 545	Leu	Thr	Gly	Thr	Gly 550	Val	Leu	Thr	Glu	Ser 555	Asn	Lys	Lys	Phe	Leu 560
Pro	Phe	Gln	Gln	Phe 565	Gly	Arg	Asp	Ile	Ala 570	Asp	Thr	Thr	Asp	Ala 575	Val
Arg	Asp	Pro	Gln	Thr 580	Leu	Glu	Ile	Leu 585	Asp	Ile	Thr	Pro	Cys 590	Ser	Phe
Gly	Gly	Val	Ser	Val 595	Ile	Thr	Pro 600	Gly	Thr	Asn	Thr	Ser 605	Asn	Gln	Val
Ala	Val 610	Leu	Tyr	Gln	Asp 615	Val	Asn	Cys	Thr	Glu	Val 620	Pro	Val	Ala	Ile
His 625	Ala	Asp	Gln	Leu 630	Thr	Pro	Thr	Trp	Arg 635	Val	Tyr	Ser	Thr	Gly	Ser 640
Asn	Val	Phe	Gln	Thr 645	Arg	Ala	Gly	Cys 650	Leu	Ile	Gly	Ala	Glu	His 655	Val
Asn	Asn	Ser	Tyr 660	Glu	Cys	Asp	Ile 665	Pro	Ile	Gly	Ala	Gly	Ile 670	Cys	Ala
Ser	Tyr 675	Gln	Thr	Gln	Thr	Asn	Ser 680	Pro	Arg	Arg	Ala	Arg 685	Ser	Val	Ala
Ser	Gln 690	Ser	Ile	Ile	Ala	Tyr 695	Thr	Met	Ser	Leu	Gly 700	Ala	Glu	Asn	Ser
Val 705	Ala	Tyr	Ser	Asn	Asn 710	Ser	Ile	Ala	Ile	Pro 715	Thr	Asn	Phe	Thr	Ile 720
Ser	Val	Thr	Thr	Glu 725	Ile	Leu	Pro	Val	Ser 730	Met	Thr	Lys	Thr	Ser 735	Val
Asp	Cys	Thr	Met 740	Tyr	Ile	Cys	Gly	Asp 745	Ser	Thr	Glu	Cys	Ser 750	Asn	Leu
Leu	Leu	Gln	Tyr 755	Gly	Ser	Phe	Cys 760	Thr	Gln	Leu	Asn	Arg 765	Ala	Leu	Thr
Gly 770	Ile	Ala	Val	Glu	Gln	Asp 775	Lys	Asn	Thr	Gln	Glu 780	Val	Phe	Ala	Gln
Val 785	Lys	Gln	Ile	Tyr	Lys 790	Thr	Pro	Pro	Ile	Lys 795	Asp	Phe	Gly	Gly	Phe 800
Asn	Phe	Ser	Gln	Ile 805	Leu	Pro	Asp	Pro	Ser 810	Lys	Pro	Ser	Lys	Arg 815	Ser
Pro	Ile	Glu	Asp 820	Leu	Leu	Phe	Asn	Lys 825	Val	Thr	Leu	Ala	Asp 830	Ala	Gly
Phe	Ile	Lys 835	Gln	Tyr	Gly	Asp	Cys 840	Ala	Gly	Asp	Ala	Ala	Ala 845	Arg	Asp
Leu 850	Ile	Cys	Ala	Gln	Lys	Phe 855	Asn	Gly	Leu	Thr	Val 860	Ala	Pro	Pro	Ala
Ala 865	Thr	Asp	Glu	Met	Ile 870	Ala	Gln	Tyr	Thr	Ser 875	Ala	Leu	Leu	Ala	Gly 880
Thr	Ile	Thr	Ser	Gly 885	Trp	Thr	Phe	Gly	Ala 890	Gly	Pro	Ala	Leu	Gln 895	Ile
Pro	Phe	Pro	Met 900	Gln	Met	Ala	Tyr	Arg 905	Phe	Asn	Gly	Ile	Gly 910	Val	Thr
Gln	Asn	Val 915	Leu	Tyr	Glu	Asn	Gln	Lys 920	Leu	Ile	Ala	Asn 925	Gln	Phe	Asn
Ser	Ala 930	Ile	Gly	Lys	Ile	Gln 935	Asp	Ser	Leu	Ser	Ser 940	Thr	Pro	Ser	Ala

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Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn
945                               950                               955                               960

Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val
                               965                               970                               975

Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln
                               980                               985                               990

Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val
                               995                               1000                               1005

Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn
1010                               1015                               1020

Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys
1025                               1030                               1035

Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro
1040                               1045                               1050

Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val
1055                               1060                               1065

Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His
1070                               1075                               1080

Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn
1085                               1090                               1095

Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln
1100                               1105                               1110

Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val
1115                               1120                               1125

Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro
1130                               1135                               1140

Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
1145                               1150                               1155

His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
1160                               1165                               1170

Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
1175                               1180                               1185

Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
1190                               1195                               1200

Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
1205                               1210                               1215

Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
1220                               1225                               1230

Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
1235                               1240                               1245

Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
1250                               1255                               1260

Val Leu Lys Gly Val Lys Leu His Tyr Thr
1265                               1270

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<210> SEQ ID NO 50

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 25 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 50

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Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val	1	5	10	15
Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe	20	25	30	
Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu	35	40	45	
His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp	50	55	60	
Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp	65	70	75	80
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu	85	90	95	
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser	100	105	110	
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile	115	120	125	
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	130	135	140	
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	145	150	155	160
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	180	185	190	
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205	
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220	
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235	240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270	
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285	
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300	
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315	320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350	
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365	
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380	
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395	400

Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	
				405					410							
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	
				420					425							
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	
				435					440							
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	
				450					455							
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	
				465					470							
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	
				485					490							
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	
				500					505							
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	
				515					520							
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	
				530					535							
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	
				545					550							
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	
				565					570							
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	
				580					585							
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	
				595					600							
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	
				610					615							
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	
				625					630							
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
				645					650							
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
				660					665							
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	
				675					680							
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	
				690					695							
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	
				705					710							
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	
				725					730							
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	
				740					745							
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	
				755					760							
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	
				770					775							
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	
				785					790							
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	
				800												

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805						810						815			
Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp
		835					840					845			
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
	850					855					860				
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865					870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile
			885					890						895	
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900					905					910		
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
	915						920					925			
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala
	930					935					940				
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
945				950						955					960
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
			965						970					975	
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln
			980					985					990		
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
	995					1000						1005			
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
	1010					1015					1020				
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
	1025					1030					1035				
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
	1040					1045					1050				
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
	1055					1060					1065				
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
	1070					1075					1080				
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn	
	1085					1090					1095				
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln	
	1100					1105					1110				
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val	
	1115					1120					1125				
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro	
	1130					1135					1140				
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn	
	1145					1150					1155				
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn	
	1160					1165					1170				
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu	
	1175					1180					1185				
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu	
	1190					1195					1200				

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Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
1205 1210 1215

Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
1220 1225 1230

Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
1235 1240 1245

Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
1250 1255 1260

Val Leu Lys Gly Val Lys Leu His Tyr Thr
1265 1270

<210> SEQ ID NO 51

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 26 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 51

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1 5 10 15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20 25 30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35 40 45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50 55 60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65 70 75 80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85 90 95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100 105 110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115 120 125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr
130 135 140

Tyr His Lys Asn Asn Lys Ser Trp Met Glu Ser Glu Phe Arg Val Tyr
145 150 155 160

Ser Ser Ala Asn Asn Cys Thr Phe Glu Tyr Val Ser Gln Pro Phe Leu
165 170 175

Met Asp Leu Glu Gly Lys Gln Gly Asn Phe Lys Asn Leu Arg Glu Phe
180 185 190

Val Phe Lys Asn Ile Asp Gly Tyr Phe Lys Ile Tyr Ser Lys His Thr
195 200 205

Pro Ile Asn Leu Val Arg Asp Leu Pro Gln Gly Phe Ser Ala Leu Glu
210 215 220

Pro Leu Val Asp Leu Pro Ile Gly Ile Asn Ile Thr Arg Phe Gln Thr
225 230 235 240

Leu Leu Ala Leu His Arg Ser Tyr Leu Thr Pro Gly Asp Ser Ser Ser
245 250 255

Gly Trp Thr Ala Gly Ala Ala Ala Tyr Tyr Val Gly Tyr Leu Gln Pro
260 265 270

Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala
		275					280					285			
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys
		290					295					300			
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val
		305					310					315			
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys
				325				330						335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala
				340				345				350			
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu
		355				360						365			
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro
		370				375						380			
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe
				390						395				400	
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly
				405				410						415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys
		420						425				430			
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn
		435				440						445			
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe
		450				455						460			
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys
				470						475				480	
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly
				485				490						495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val
		500						505				510			
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys
		515				520						525			
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn
		530				535						540			
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu
				550						555				560	
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val
				565				570						575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe
		580						585				590			
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val
		595				600						605			
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile
		610				615						620			
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser
				630						635				640	
Asn	Val	Phe	Gln	Thr	Arg	Ala									

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Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	675	680	685
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	690	695	700
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	705	710	715
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	725	730	735
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	740	745	750
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	755	760	765
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	770	775	780
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	785	790	795
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	805	810	815
Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	820	825	830
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp	835	840	845
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	850	855	860
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	865	870	875
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile	885	890	895
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	900	905	910
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	915	920	925
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala	930	935	940
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn	945	950	955
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val	965	970	975
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Lys	Val	Glu	Ala	Glu	Val	Gln	980	985	990
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val	995	1000	1005
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn		1010	1015	1020
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys		1025	1030	1035
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro		1040	1045	1050
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val		1055	1060	1065
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His				

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1070	1075	1080
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn		
1085	1090	1095
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln		
1100	1105	1110
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val		
1115	1120	1125
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro		
1130	1135	1140
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn		
1145	1150	1155
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn		
1160	1165	1170
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu		
1175	1180	1185
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu		
1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu		
1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met		
1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys		
1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro		
1250	1255	1260
Val Leu Lys Gly Val Lys Leu His Tyr Thr		
1265	1270	

<210> SEQ ID NO 52

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 27 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 52

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val
1 5 10 15

Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe
20 25 30

Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu
35 40 45

His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp
50 55 60

Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp
65 70 75 80

Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu
85 90 95

Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser
100 105 110

Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile
115 120 125

Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr

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130					135					140						
Tyr 145	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr 160	
					150						155					
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	
					165						170					
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	
					180						185					
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	
					195						200					
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	
					210						215					
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	
					225						230					
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	
					245						250					
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	
					260						265					
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	
					275						280					
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	
					290						295					
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	
					305						310					
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	
					325						330					
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	
					340						345					
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	
					355						360					
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	
					370						375					
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	
					385						390					
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	
					405						410					
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	
					420						425					
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	
					435						440					
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	
					450						455					
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	
					465						470					
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	
					485						490					
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	
					500						505					
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	
					515						520					
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	
					530						535					

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Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	545	550	555	560
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	565	570	575	
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	580	585	590	
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	595	600	605	
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	610	615	620	
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	625	630	635	640
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	645	650	655	
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	660	665	670	
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	675	680	685	
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	690	695	700	
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	705	710	715	720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	725	730	735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	740	745	750	
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	755	760	765	
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	770	775	780	
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	785	790	795	800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	805	810	815	
Pro	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly	820	825	830	
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp	835	840	845	
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala	850	855	860	
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly	865	870	875	880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Ala	Ala	Leu	Gln	Ile	885	890	895	
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr	900	905	910	
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn	915	920	925	
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Ala	Ser	Ala	930	935	940	

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Leu Gly Lys Leu Gln Asp Val Val Asn Gln Asn Ala Gln Ala Leu Asn
945                               950                               955                               960

Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val
                               965                               970                               975

Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln
                               980                               985                               990

Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val
                               995                               1000                               1005

Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn
1010                               1015                               1020

Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys
1025                               1030                               1035

Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro
1040                               1045                               1050

Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val
1055                               1060                               1065

Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His
1070                               1075                               1080

Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn
1085                               1090                               1095

Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln
1100                               1105                               1110

Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val
1115                               1120                               1125

Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro
1130                               1135                               1140

Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn
1145                               1150                               1155

His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn
1160                               1165                               1170

Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu
1175                               1180                               1185

Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu
1190                               1195                               1200

Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu
1205                               1210                               1215

Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met
1220                               1225                               1230

Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys
1235                               1240                               1245

Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro
1250                               1255                               1260

Val Leu Lys Gly Val Lys Leu His Tyr Thr
1265                               1270

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<210> SEQ ID NO 53

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 28 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 53

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Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val	1	5	10	15
Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe	20	25	30	
Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu	35	40	45	
His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp	50	55	60	
Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp	65	70	75	80
Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu	85	90	95	
Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser	100	105	110	
Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile	115	120	125	
Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr	130	135	140	
Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	145	150	155	160
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	180	185	190	
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205	
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220	
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235	240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270	
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285	
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300	
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315	320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350	
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365	
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380	
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395	400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly				

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405							410					415				
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	
			420					425					430			
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	
		435					440					445				
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	
	450					455					460					
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	
465					470					475					480	
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	
			485						490					495		
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	
			500					505					510			
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	
		515					520					525				
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	
	530					535					540					
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	
545					550					555					560	
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	
				565					570					575		
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	
			580					585					590			
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	
	595						600					605				
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	
	610					615					620					
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	
625				630						635					640	
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	
			645					650						655		
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	
		660					665						670			
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala	
	675					680						685				
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser	
	690					695					700					
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile	
705					710					715					720	
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val	
			725					730						735		
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu	
			740					745					750			
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr	
	755						760					765				
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln	
	770					775					780					
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe	
785					790					795					800	
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser	
			805					810						815		

Phe	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
820															
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp
835															
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
850															
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865															
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile
885															
Pro	Phe	Ala	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
900															
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
915															
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala
930															
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
945															
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
965															
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln
980															
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
995															
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
1010															
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
1025															
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
1040															
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
1055															
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
1070															
Asp	Gly	Lys	Ala	His	Phe	Pro	Arg	Glu	Gly	Val	Phe	Val	Ser	Asn	
1085															
Gly	Thr	His	Trp	Phe	Val	Thr	Gln	Arg	Asn	Phe	Tyr	Glu	Pro	Gln	
1100															
Ile	Ile	Thr	Thr	Asp	Asn	Thr	Phe	Val	Ser	Gly	Asn	Cys	Asp	Val	
1115															
Val	Ile	Gly	Ile	Val	Asn	Asn	Thr	Val	Tyr	Asp	Pro	Leu	Gln	Pro	
1130															
Glu	Leu	Asp	Ser	Phe	Lys	Glu	Glu	Leu	Asp	Lys	Tyr	Phe	Lys	Asn	
1145															
His	Thr	Ser	Pro	Asp	Val	Asp	Leu	Gly	Asp	Ile	Ser	Gly	Ile	Asn	
1160															
Ala	Ser	Val	Val	Asn	Ile	Gln	Lys	Glu	Ile	Asp	Arg	Leu	Asn	Glu	
1175															
Val	Ala	Lys	Asn	Leu	Asn	Glu	Ser	Leu	Ile	Asp	Leu	Gln	Glu	Leu	
1190															

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Gly	Lys	Tyr	Glu	Gln	Tyr	Ile	Lys	Trp	Pro	Trp	Tyr	Ile	Trp	Leu
1205						1210					1215			

Gly	Phe	Ile	Ala	Gly	Leu	Ile	Ala	Ile	Val	Met	Val	Thr	Ile	Met
1220					1225						1230			

Leu	Cys	Cys	Met	Thr	Ser	Cys	Cys	Ser	Cys	Leu	Lys	Gly	Cys	Cys
1235					1240						1245			

Ser	Cys	Gly	Ser	Cys	Cys	Lys	Phe	Asp	Glu	Asp	Asp	Ser	Glu	Pro
1250					1255						1260			

Val	Leu	Lys	Gly	Val	Lys	Leu	His	Tyr	Thr
1265					1270				

<210> SEQ ID NO 54

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 29 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 54

Met	Phe	Val	Phe	Leu	Val	Leu	Leu	Pro	Leu	Val	Ser	Ser	Gln	Cys	Val
1				5					10				15		

Asn	Leu	Thr	Thr	Arg	Thr	Gln	Leu	Pro	Pro	Ala	Tyr	Thr	Asn	Ser	Phe
			20					25					30		

Thr	Arg	Gly	Val	Tyr	Tyr	Pro	Asp	Lys	Val	Phe	Arg	Ser	Ser	Val	Leu
		35					40				45				

His	Ser	Thr	Gln	Asp	Leu	Phe	Leu	Pro	Phe	Phe	Ser	Asn	Val	Thr	Trp
	50					55					60				

Phe	His	Ala	Ile	His	Val	Ser	Gly	Thr	Asn	Gly	Thr	Lys	Arg	Phe	Asp
65					70					75				80	

Asn	Pro	Val	Leu	Pro	Phe	Asn	Asp	Gly	Val	Tyr	Phe	Ala	Ser	Thr	Glu
			85					90						95	

Lys	Ser	Asn	Ile	Ile	Arg	Gly	Trp	Ile	Phe	Gly	Thr	Thr	Leu	Asp	Ser
		100					105						110		

Lys	Thr	Gln	Ser	Leu	Leu	Ile	Val	Asn	Asn	Ala	Thr	Asn	Val	Val	Ile
		115				120						125			

Lys	Val	Cys	Glu	Phe	Gln	Phe	Cys	Asn	Asp	Pro	Phe	Leu	Gly	Val	Tyr
	130					135				140					

Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr
145					150					155				160	

Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu
			165						170					175	

Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe
		180						185					190		

Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr
		195					200					205			

Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu
	210					215					220				

Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr
225					230					235				240	

Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser
			245						250					255	

Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro
			260					265					270		

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Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	530	535	540
Gly	Leu	Thr	Gly	Thr	Gly	Val	Leu	Thr	Glu	Ser	Asn	Lys	Lys	Phe	Leu	545	550	555
Pro	Phe	Gln	Gln	Phe	Gly	Arg	Asp	Ile	Ala	Asp	Thr	Thr	Asp	Ala	Val	565	570	575
Arg	Asp	Pro	Gln	Thr	Leu	Glu	Ile	Leu	Asp	Ile	Thr	Pro	Cys	Ser	Phe	580	585	590
Gly	Gly	Val	Ser	Val	Ile	Thr	Pro	Gly	Thr	Asn	Thr	Ser	Asn	Gln	Val	595	600	605
Ala	Val	Leu	Tyr	Gln	Asp	Val	Asn	Cys	Thr	Glu	Val	Pro	Val	Ala	Ile	610	615	620
His	Ala	Asp	Gln	Leu	Thr	Pro	Thr	Trp	Arg	Val	Tyr	Ser	Thr	Gly	Ser	625	630	635
Asn	Val	Phe	Gln	Thr	Arg	Ala	Gly	Cys	Leu	Ile	Gly	Ala	Glu	His	Val	645	650	655
Asn	Asn	Ser	Tyr	Glu	Cys	Asp	Ile	Pro	Ile	Gly	Ala	Gly	Ile	Cys	Ala	660	665	670
Ser	Tyr	Gln	Thr	Gln	Thr	Asn	Ser	Pro	Arg	Arg	Ala	Arg	Ser	Val	Ala			

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675					680					685					
Ser	Gln	Ser	Ile	Ile	Ala	Tyr	Thr	Met	Ser	Leu	Gly	Ala	Glu	Asn	Ser
690						695					700				
Val	Ala	Tyr	Ser	Asn	Asn	Ser	Ile	Ala	Ile	Pro	Thr	Asn	Phe	Thr	Ile
705					710					715					720
Ser	Val	Thr	Thr	Glu	Ile	Leu	Pro	Val	Ser	Met	Thr	Lys	Thr	Ser	Val
				725					730					735	
Asp	Cys	Thr	Met	Tyr	Ile	Cys	Gly	Asp	Ser	Thr	Glu	Cys	Ser	Asn	Leu
			740					745					750		
Leu	Leu	Gln	Tyr	Gly	Ser	Phe	Cys	Thr	Gln	Leu	Asn	Arg	Ala	Leu	Thr
		755					760					765			
Gly	Ile	Ala	Val	Glu	Gln	Asp	Lys	Asn	Thr	Gln	Glu	Val	Phe	Ala	Gln
	770					775					780				
Val	Lys	Gln	Ile	Tyr	Lys	Thr	Pro	Pro	Ile	Lys	Asp	Phe	Gly	Gly	Phe
785					790					795					800
Asn	Phe	Ser	Gln	Ile	Leu	Pro	Asp	Pro	Ser	Lys	Pro	Ser	Lys	Arg	Ser
				805					810					815	
Pro	Ala	Glu	Asp	Ala	Ala	Phe	Asn	Lys	Val	Thr	Leu	Ala	Asp	Ala	Gly
			820					825					830		
Phe	Ile	Lys	Gln	Tyr	Gly	Asp	Cys	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp
		835					840					845			
Leu	Ile	Cys	Ala	Gln	Lys	Phe	Asn	Gly	Leu	Thr	Val	Ala	Pro	Pro	Ala
	850					855					860				
Ala	Thr	Asp	Glu	Met	Ile	Ala	Gln	Tyr	Thr	Ser	Ala	Leu	Leu	Ala	Gly
865					870					875					880
Thr	Ile	Thr	Ser	Gly	Trp	Thr	Phe	Gly	Ala	Gly	Pro	Ala	Leu	Gln	Ile
				885					890					895	
Pro	Phe	Pro	Met	Gln	Met	Ala	Tyr	Arg	Phe	Asn	Gly	Ile	Gly	Val	Thr
			900					905					910		
Gln	Asn	Val	Leu	Tyr	Glu	Asn	Gln	Lys	Leu	Ile	Ala	Asn	Gln	Phe	Asn
		915					920						925		
Ser	Ala	Ile	Gly	Lys	Ile	Gln	Asp	Ser	Leu	Ser	Ser	Thr	Pro	Ser	Ala
	930					935						940			
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn
945				950						955					960
Thr	Leu	Val	Lys	Gln	Leu	Ser	Ser	Asn	Phe	Gly	Ala	Ile	Ser	Ser	Val
				965					970					975	
Leu	Asn	Asp	Ile	Leu	Ser	Arg	Leu	Asp	Pro	Pro	Glu	Ala	Glu	Val	Gln
			980					985					990		
Ile	Asp	Arg	Leu	Ile	Thr	Gly	Arg	Leu	Gln	Ser	Leu	Gln	Thr	Tyr	Val
	995						1000						1005		
Thr	Gln	Gln	Leu	Ile	Arg	Ala	Ala	Glu	Ile	Arg	Ala	Ser	Ala	Asn	
	1010					1015						1020			
Leu	Ala	Ala	Thr	Lys	Met	Ser	Glu	Cys	Val	Leu	Gly	Gln	Ser	Lys	
	1025					1030						1035			
Arg	Val	Asp	Phe	Cys	Gly	Lys	Gly	Tyr	His	Leu	Met	Ser	Phe	Pro	
	1040					1045						1050			
Gln	Ser	Ala	Pro	His	Gly	Val	Val	Phe	Leu	His	Val	Thr	Tyr	Val	
	1055					1060						1065			
Pro	Ala	Gln	Glu	Lys	Asn	Phe	Thr	Thr	Ala	Pro	Ala	Ile	Cys	His	
	1070					1075						1080			

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Asp Gly	Lys Ala His Phe	Pro	Arg Glu Gly Val	Phe	Val Ser Asn
1085		1090		1095	
Gly Thr	His Trp Phe Val	Thr	Gln Arg Asn Phe Tyr	Glu Pro Gln	
1100		1105		1110	
Ile Ile	Thr Thr Asp Asn Thr	Phe Val Ser Gly Asn	Cys Asp Val		
1115		1120		1125	
Val Ile	Gly Ile Val Asn Asn	Thr Val Tyr Asp Pro	Leu Gln Pro		
1130		1135		1140	
Glu Leu	Asp Ser Phe Lys Glu	Glu Leu Asp Lys Tyr	Phe Lys Asn		
1145		1150		1155	
His Thr	Ser Pro Asp Val Asp	Leu Gly Asp Ile Ser	Gly Ile Asn		
1160		1165		1170	
Ala Ser	Val Val Asn Ile Gln	Lys Glu Ile Asp Arg	Leu Asn Glu		
1175		1180		1185	
Val Ala	Lys Asn Leu Asn Glu	Ser Leu Ile Asp Leu	Gln Glu Leu		
1190		1195		1200	
Gly Lys	Tyr Glu Gln Tyr Ile	Lys Trp Pro Trp Tyr	Ile Trp Leu		
1205		1210		1215	
Gly Phe	Ile Ala Gly Leu Ile	Ala Ile Val Met Val	Thr Ile Met		
1220		1225		1230	
Leu Cys	Cys Met Thr Ser Cys	Cys Ser Cys Leu Lys	Gly Cys Cys		
1235		1240		1245	
Ser Cys	Gly Ser Cys Cys Lys	Phe Asp Glu Asp Asp	Ser Glu Pro		
1250		1255		1260	
Val Leu	Lys Gly Val Lys Leu	His Tyr Thr			
1265		1270			

<210> SEQ ID NO 55

<211> LENGTH: 1273

<212> TYPE: PRT

<213> ORGANISM: Artificial Sequence

<220> FEATURE:

<223> OTHER INFORMATION: Variant 30 of Mutated Sars-CoV-2 spike protein

<400> SEQUENCE: 55

Met Phe Val Phe Leu Val Leu Leu Pro Leu Val Ser Ser Gln Cys Val					
1	5	10	15		
Asn Leu Thr Thr Arg Thr Gln Leu Pro Pro Ala Tyr Thr Asn Ser Phe					
20	25	30			
Thr Arg Gly Val Tyr Tyr Pro Asp Lys Val Phe Arg Ser Ser Val Leu					
35	40	45			
His Ser Thr Gln Asp Leu Phe Leu Pro Phe Phe Ser Asn Val Thr Trp					
50	55	60			
Phe His Ala Ile His Val Ser Gly Thr Asn Gly Thr Lys Arg Phe Asp					
65	70	75	80		
Asn Pro Val Leu Pro Phe Asn Asp Gly Val Tyr Phe Ala Ser Thr Glu					
85	90	95			
Lys Ser Asn Ile Ile Arg Gly Trp Ile Phe Gly Thr Thr Leu Asp Ser					
100	105	110			
Lys Thr Gln Ser Leu Leu Ile Val Asn Asn Ala Thr Asn Val Val Ile					
115	120	125			
Lys Val Cys Glu Phe Gln Phe Cys Asn Asp Pro Phe Leu Gly Val Tyr					
130	135	140			

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Tyr	His	Lys	Asn	Asn	Lys	Ser	Trp	Met	Glu	Ser	Glu	Phe	Arg	Val	Tyr	145	150	155	160
Ser	Ser	Ala	Asn	Asn	Cys	Thr	Phe	Glu	Tyr	Val	Ser	Gln	Pro	Phe	Leu	165	170	175	
Met	Asp	Leu	Glu	Gly	Lys	Gln	Gly	Asn	Phe	Lys	Asn	Leu	Arg	Glu	Phe	180	185	190	
Val	Phe	Lys	Asn	Ile	Asp	Gly	Tyr	Phe	Lys	Ile	Tyr	Ser	Lys	His	Thr	195	200	205	
Pro	Ile	Asn	Leu	Val	Arg	Asp	Leu	Pro	Gln	Gly	Phe	Ser	Ala	Leu	Glu	210	215	220	
Pro	Leu	Val	Asp	Leu	Pro	Ile	Gly	Ile	Asn	Ile	Thr	Arg	Phe	Gln	Thr	225	230	235	240
Leu	Leu	Ala	Leu	His	Arg	Ser	Tyr	Leu	Thr	Pro	Gly	Asp	Ser	Ser	Ser	245	250	255	
Gly	Trp	Thr	Ala	Gly	Ala	Ala	Ala	Tyr	Tyr	Val	Gly	Tyr	Leu	Gln	Pro	260	265	270	
Arg	Thr	Phe	Leu	Leu	Lys	Tyr	Asn	Glu	Asn	Gly	Thr	Ile	Thr	Asp	Ala	275	280	285	
Val	Asp	Cys	Ala	Leu	Asp	Pro	Leu	Ser	Glu	Thr	Lys	Cys	Thr	Leu	Lys	290	295	300	
Ser	Phe	Thr	Val	Glu	Lys	Gly	Ile	Tyr	Gln	Thr	Ser	Asn	Phe	Arg	Val	305	310	315	320
Gln	Pro	Thr	Glu	Ser	Ile	Val	Arg	Phe	Pro	Asn	Ile	Thr	Asn	Leu	Cys	325	330	335	
Pro	Phe	Gly	Glu	Val	Phe	Asn	Ala	Thr	Arg	Phe	Ala	Ser	Val	Tyr	Ala	340	345	350	
Trp	Asn	Arg	Lys	Arg	Ile	Ser	Asn	Cys	Val	Ala	Asp	Tyr	Ser	Val	Leu	355	360	365	
Tyr	Asn	Ser	Ala	Ser	Phe	Ser	Thr	Phe	Lys	Cys	Tyr	Gly	Val	Ser	Pro	370	375	380	
Thr	Lys	Leu	Asn	Asp	Leu	Cys	Phe	Thr	Asn	Val	Tyr	Ala	Asp	Ser	Phe	385	390	395	400
Val	Ile	Arg	Gly	Asp	Glu	Val	Arg	Gln	Ile	Ala	Pro	Gly	Gln	Thr	Gly	405	410	415	
Lys	Ile	Ala	Asp	Tyr	Asn	Tyr	Lys	Leu	Pro	Asp	Asp	Phe	Thr	Gly	Cys	420	425	430	
Val	Ile	Ala	Trp	Asn	Ser	Asn	Asn	Leu	Asp	Ser	Lys	Val	Gly	Gly	Asn	435	440	445	
Tyr	Asn	Tyr	Leu	Tyr	Arg	Leu	Phe	Arg	Lys	Ser	Asn	Leu	Lys	Pro	Phe	450	455	460	
Glu	Arg	Asp	Ile	Ser	Thr	Glu	Ile	Tyr	Gln	Ala	Gly	Ser	Thr	Pro	Cys	465	470	475	480
Asn	Gly	Val	Glu	Gly	Phe	Asn	Cys	Tyr	Phe	Pro	Leu	Gln	Ser	Tyr	Gly	485	490	495	
Phe	Gln	Pro	Thr	Asn	Gly	Val	Gly	Tyr	Gln	Pro	Tyr	Arg	Val	Val	Val	500	505	510	
Leu	Ser	Phe	Glu	Leu	Leu	His	Ala	Pro	Ala	Thr	Val	Cys	Gly	Pro	Lys	515	520	525	
Lys	Ser	Thr	Asn	Leu	Val	Lys	Asn	Lys	Cys	Val	Asn	Phe	Asn	Phe	Asn	530	535	540	

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Gly 545	Leu	Thr	Gly	Thr	Gly 550	Val	Leu	Thr	Glu	Ser 555	Asn	Lys	Lys	Phe	Leu 560
Pro	Phe	Gln	Gln	Phe 565	Gly	Arg	Asp	Ile	Ala 570	Asp	Thr	Thr	Asp	Ala 575	Val
Arg	Asp	Pro	Gln	Thr 580	Leu	Glu	Ile	Leu 585	Asp	Ile	Thr	Pro	Cys 590	Ser	Phe
Gly	Gly	Val 595	Ser	Val	Ile	Thr	Pro 600	Gly	Thr	Asn	Thr	Ser 605	Asn	Gln	Val
Ala	Val 610	Leu	Tyr	Gln	Asp	Val 615	Asn	Cys	Thr	Glu	Val 620	Pro	Val	Ala	Ile
His 625	Ala	Asp	Gln	Leu	Thr 630	Pro	Thr	Trp	Arg	Val 635	Tyr	Ser	Thr	Gly	Ser 640
Asn	Val	Phe	Gln	Thr 645	Arg	Ala	Gly	Cys	Leu 650	Ile	Gly	Ala	Glu	His 655	Val
Asn	Asn	Ser 660	Tyr	Glu	Cys	Asp	Ile 665	Pro	Ile	Gly	Ala	Gly	Ile 670	Cys	Ala
Ser	Tyr 675	Gln	Thr	Gln	Thr	Asn	Ser 680	Pro	Arg	Arg	Ala	Arg 685	Ser	Val	Ala
Ser 690	Gln	Ser	Ile	Ile	Ala	Tyr 695	Thr	Met	Ser	Leu	Gly 700	Ala	Glu	Asn	Ser
Val 705	Ala	Tyr	Ser	Asn	Asn	Ser 710	Ile	Ala	Ile	Pro 715	Thr	Asn	Phe	Thr	Ile 720
Ser	Val	Thr	Thr	Glu 725	Ile	Leu	Pro	Val	Ser 730	Met	Thr	Lys	Thr	Ser	Val 735
Asp	Cys	Thr 740	Met	Tyr	Ile	Cys	Gly 745	Asp	Ser	Thr	Glu	Cys 750	Ser	Asn	Leu
Leu	Leu	Gln 755	Tyr	Gly	Ser	Phe 760	Cys	Thr	Gln	Leu	Asn 765	Arg	Ala	Leu	Thr
Gly 770	Ile	Ala	Val	Glu	Gln	Asp 775	Lys	Asn	Thr	Gln	Glu 780	Val	Phe	Ala	Gln
Val 785	Lys	Gln	Ile	Tyr	Lys 790	Thr	Pro	Pro	Ile	Lys 795	Asp	Phe	Gly	Gly	Phe 800
Asn	Phe	Ser	Gln	Ile 805	Leu	Pro	Asp	Pro	Ser 810	Lys	Pro	Ser	Lys	Arg	Ser 815
Pro	Ala	Glu	Asp	Ala 820	Ala	Phe	Asn	Lys	Val 825	Thr	Leu	Ala	Asp	Ala	Gly
Phe	Ile	Lys 835	Gln	Tyr	Gly	Asp	Cys 840	Ala	Gly	Asp	Ala	Ala	Ala	Arg	Asp
Leu 850	Ile	Cys	Ala	Gln	Lys 855	Phe	Asn	Gly	Leu	Thr	Val 860	Ala	Pro	Pro	Ala
Ala 865	Thr	Asp	Glu	Met	Ile 870	Ala	Gln	Tyr	Thr	Ser 875	Ala	Leu	Leu	Ala	Gly 880
Thr	Ile	Thr	Ser	Gly 885	Trp	Thr	Phe	Gly	Ala 890	Gly	Pro	Ala	Leu	Gln	Ile 895
Pro	Phe	Pro 900	Met	Gln	Met	Ala	Tyr 905	Arg	Phe	Asn	Gly	Ile 910	Gly	Val	Thr
Gln	Asn	Val 915	Leu	Tyr	Glu	Asn 920	Gln	Lys	Leu	Ile	Ala 925	Asn	Gln	Phe	Asn
Ser 930	Ala	Ile	Gly	Lys	Ile 935	Gln	Asp	Ser	Leu	Ser	Ser 940	Thr	Pro	Ser	Ala
Leu	Gly	Lys	Leu	Gln	Asp	Val	Val	Asn	Gln	Asn	Ala	Gln	Ala	Leu	Asn

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945	950	955	960
Thr Leu Val Lys Gln Leu Ser Ser Asn Phe Gly Ala Ile Ser Ser Val			
	965	970	975
Leu Asn Asp Ile Leu Ser Arg Leu Asp Lys Val Glu Ala Glu Val Gln			
	980	985	990
Ile Asp Arg Leu Ile Thr Gly Arg Leu Gln Ser Leu Gln Thr Tyr Val			
	995	1000	1005
Thr Gln Gln Leu Ile Arg Ala Ala Glu Ile Arg Ala Ser Ala Asn			
	1010	1015	1020
Leu Ala Ala Thr Lys Met Ser Glu Cys Val Leu Gly Gln Ser Lys			
	1025	1030	1035
Arg Val Asp Phe Cys Gly Lys Gly Tyr His Leu Met Ser Phe Pro			
	1040	1045	1050
Gln Ser Ala Pro His Gly Val Val Phe Leu His Val Thr Tyr Val			
	1055	1060	1065
Pro Ala Gln Glu Lys Asn Phe Thr Thr Ala Pro Ala Ile Cys His			
	1070	1075	1080
Asp Gly Lys Ala His Phe Pro Arg Glu Gly Val Phe Val Ser Asn			
	1085	1090	1095
Gly Thr His Trp Phe Val Thr Gln Arg Asn Phe Tyr Glu Pro Gln			
	1100	1105	1110
Ile Ile Thr Thr Asp Asn Thr Phe Val Ser Gly Asn Cys Asp Val			
	1115	1120	1125
Val Ile Gly Ile Val Asn Asn Thr Val Tyr Asp Pro Leu Gln Pro			
	1130	1135	1140
Glu Leu Asp Ser Phe Lys Glu Glu Leu Asp Lys Tyr Phe Lys Asn			
	1145	1150	1155
His Thr Ser Pro Asp Val Asp Leu Gly Asp Ile Ser Gly Ile Asn			
	1160	1165	1170
Ala Ser Val Val Asn Ile Gln Lys Glu Ile Asp Arg Leu Asn Glu			
	1175	1180	1185
Val Ala Lys Asn Leu Asn Glu Ser Leu Ile Asp Leu Gln Glu Leu			
	1190	1195	1200
Gly Lys Tyr Glu Gln Tyr Ile Lys Trp Pro Trp Tyr Ile Trp Leu			
	1205	1210	1215
Gly Phe Ile Ala Gly Leu Ile Ala Ile Val Met Val Thr Ile Met			
	1220	1225	1230
Leu Cys Cys Met Thr Ser Cys Cys Ser Cys Leu Lys Gly Cys Cys			
	1235	1240	1245
Ser Cys Gly Ser Cys Cys Lys Phe Asp Glu Asp Asp Ser Glu Pro			
	1250	1255	1260
Val Leu Lys Gly Val Lys Leu His Tyr Thr			
	1265	1270	

1. A mutated SARS-COV-2 spike protein, wherein at least one of the aminoacids isoleucine or leucine at a position corresponding to aminoacid 818, 821, 822, 841, 844, 861, 864 and 865 of the wild type SARS-COV-2 protein sequence (SEQ ID NO: 1) is replaced by alanine.

2. A mutated SARS-COV-2 spike protein as claimed in claim 1 having the aminoacid sequence of SEQ ID NO: 2, 28, 29, 30 or 31.

3. A variant of a mutated SARS-COV-2 spike protein, wherein at least one of the aminoacids isoleucine or leucine at a position corresponding to aminoacid 818, 821, 822, 841, 844, 861, 864 and 865 of the wild type SARS-COV-2 protein sequence (SEQ ID NO: 1) are replaced by alanine, or a variant of a mutated SARS-COV-2 spike protein selected from SEQ ID NO: 2, 28, 29, 30, or 31.

4. A variant of the mutated SARS-COV-2 spike protein as claimed in claim 3, wherein said variant has the same aminoacid sequences of the corresponding mutated SARS-COV-2 spike protein in the following positions:

aminoacid sequence from a position corresponding to aminoacid 818 of SEQ ID NO: 1 to a position corresponding to aminoacid 822 of SEQ ID NO: 1,

aminoacid sequence from a position corresponding to aminoacid 841 of SEQ ID NO: 1 to a position corresponding to aminoacid 844 of SEQ ID NO: 1, and

aminoacid sequence from a position corresponding to aminoacid 861 of SEQ ID NO: 1 to position 865 of SEQ ID NO: 1

5. A variant as claimed in claim 3, wherein at least two of the following aminoacids of said mutated SARS-COV-2 spike protein are replaced by proline:

-alanine at positions corresponding to aminoacids 892, 899 and 942 of SEQ ID NO: 1,

phenylalanine at a position corresponding to aminoacid 817 of SEQ ID NO: 1,

lysine at a position corresponding to aminoacid 986 of SEQ ID NO: 1, and

valine at a position corresponding to aminoacid 987 of SEQ ID NO: 1.

6. A variant as claimed in claim 3, having sequence of SEQ ID NO: 3, SEQ ID NO: 4, SEQ ID NO: 5, SEQ ID NO: 6, SEQ ID NO: 7, SEQ ID NO: 8, SEQ ID NO: 32, SEQ ID NO: 33, SEQ ID NO: 34, SEQ ID NO: 35, SEQ ID NO: 36, SEQ ID NO: 37, SEQ ID NO: 38, SEQ ID NO: 39, SEQ ID NO: 40, SEQ ID NO: 41, SEQ ID NO: 42, SEQ ID NO: 43, SEQ ID NO: 44, SEQ ID NO: 45, SEQ ID NO: 46, SEQ ID NO: 47, SEQ ID NO: 48, SEQ ID NO: 49, SEQ ID NO: 50, SEQ ID NO: 51, SEQ ID NO: 52, SEQ ID NO: 53, SEQ ID NO: 54 or SEQ ID NO: 55.

7. A variant as claimed in claim 3, wherein the aminoacids arginine at a position corresponding to aminoacid 682, 683 and 685 of SEQ ID NO: 1 is replaced by glutamine.

8. A variant as claimed in claim 3, wherein the aminoacid arginine at a position corresponding to aminoacid 685 of SEQ ID NO: 1 is replaced by alanine.

9. A variant as claimed in claim 3, further containing, the amino acid modification at positions corresponding to the aminoacids of SEQ ID NO.1 listed below: T19R, EF156/157del, R158G, L452R, T478K, D614G, P681R and D950N, or A67V, H69del, V70del, T951, G142D, V143del, Y144del, Y145del, N211del, L212I, ins214EPE, G339D, S371L, S373P, S375F, K417N, N440K, G446S, S477N, T478K, E484A, Q493R, G496S, Q498R, N501Y, Y505H,

T547K, D614G, H655Y, N679K, P681H, N764K, D796Y, N856K, Q954H, N969K, L981F.

10. An immunogenic fragment of the mutated SARS-COV-2 spike protein as claimed in claim 1; a mutated SARS-COV-2 spike protein of SEQ ID NO: 2, 28, 29, 30, or 31; or of a variant thereof, comprising at least one sequence of said mutated SARS-COV-2 spike protein or variant selected from:

aminoacid sequence from a position corresponding to aminoacid 818 of SEQ ID NO: 1 to a position corresponding to aminoacid 822 of SEQ ID NO: 1,

aminoacid sequence from a position corresponding to aminoacid 841 of SEQ ID NO: 1 to a position corresponding to aminoacid 844 of SEQ ID NO: 1, and

aminoacid sequence from a position corresponding to aminoacid 861 of SEQ ID NO: 1 to position 865 of SEQ ID NO: 1.

11. An immunogenic fragment as claimed in claim 10, containing at least one sequence of said mutated SARS-COV-2 spike protein or variant selected from:

aminoacid sequence from position 818 to position 822 of SEQ ID NO: 2 or SEQ ID NO: 31,

aminoacid sequence from position 841 to position 844 of SEQ ID NO: 2, SEQ ID NO: 28, SEQ ID NO: 30 or SEQ ID NO: 31, and

aminoacid sequence from position 861 to position 865 of SEQ ID NO: 2, SEQ ID NO: 29, SEQ ID NO: 30 or SEQ ID NO: 31.

12. An immunogenic fragment as claimed in claim 10, further containing at least two, non-overlapping sequences selected from the following aminoacid sequences:

-aminoacid sequence from position 232 to position 246 of SEQ ID NO: 1,

aminoacid sequence from position 233 to position 247 of SEQ ID NO: 1,

aminoacid sequence from position 471 to position 503 of SEQ ID NO: 1,

aminoacid sequence from position 604 to position 625 of SEQ ID NO: 1,

aminoacid sequence from position 817 to position 833 of SEQ ID NO: 2 or SEQ ID NO: 31,

aminoacid sequence from position 891 to position 907 of SEQ ID NO: 1

aminoacid sequence from position 897 to position 913 of SEQ ID NO: 1,

aminoacid sequence from position 1164 to position 1191 of SEQ ID NO: 1, and

aminoacid sequence from position 1182 to position 1209 of SEQ ID NO: 1.

13. A protein comprising the sequence of a mutated SARS-COV-2 spike protein as claimed in claim 1; a mutated SARS-COV-2 spike protein of SEQ ID NO: 2, 28, 29, 30, or 31; a variant thereof; or an immunogenic fragment thereof; and one or more additional sequences.

14. An mRNA encoding for a mutated SARS-COV-2 spike protein as claimed in claim 1; a mutated SARS-COV-2 spike protein of SEQ ID NO: 2, 28, 29, 30, or 31; a variant thereof; an immunogenic fragment thereof; or a protein comprising said mutated SARS-COV-2 spike protein, variant, or immunogenic fragment.

15. An mRNA as claimed in claim 14, having a sequence wherein uridine residues are replaced with a modified nucleoside selected from pyridin-4-one ribonucleoside, 5-aza-uridine, 2-thio-5-azauridine, 2-thiouridine, 4-thio-

pseudouridine, 2-thio-pseudouridine, 5-hydroxyuridine, 3-methyluridine, 5-carboxymethyl-uridine, 1-carboxymethyl-pseudouridine, 5-propynyl-uridine, 1-propynyl-pseudouridine, 5-taurinomethyluridine, 1-taurinomethyl-pseudouridine, 5-taurinomethyl-2-thio-uridine, 1-taurinomethyl-4-thiouridine, 5-methyl-uridine, 1-methyl-pseudouridine, 4-thio-1-methyl-pseudouridine, 2-thio-1-methylpseudouridine, 1-methyl-1-deaza-pseudouridine, 2-thio-1-methyl-1-deaza-pseudouridine, dihydrouridine, dihydropseudouridine, 2-thio-dihydrouridine, 2-thio-dihydropseudouridine, 2-methoxyuridine, 2-methoxy-4-thio-uridine, 4-methoxy-pseudouridine, 4-methoxy-2-thio-pseudouridine, preferably with 1-methyl-pseudouridine.

16. A DNA encoding for a mutated SARS-COV-2 spike protein as claimed in claim 1; a mutated SARS-COV-2 spike protein of SEQ ID NO: 2, 28, 29, 30, or 31; a variant thereof; an immunogenic fragment thereof; or a protein comprising said mutated SARS-COV-2 spike protein, variant, or immunogenic fragment.

17. A plasmid containing the DNA as claimed in claim 16.

18. A viral vector, preferably selected from adeno or pox viruses, containing the DNA as claimed in claim 16.

19. A method for the prevention of COVID-19 in a subject, the method comprising administering to the subject a mutated SARS-COV-2 spike protein as claimed in claim 1; a mutated SARS-CoV-2 spike protein of SEQ ID NO: 2, 28, 29, 30, or 31; a variant thereof; an immunogenic fragment thereof; a protein comprising said mutated SARS-COV-2 spike protein, variant, or immunogenic fragment; a mRNA encoding said mutated SARS-COV-2 spike protein, variant, immunogenic fragment, or protein; a DNA encoding said mutated SARS-COV-2 spike protein, variant, immunogenic fragment, or protein; a plasmid containing said DNA; or a viral vector containing said DNA.

20. A vaccine composition containing a mutated SARS-COV-2 spike protein as claimed in claim 1; a mutated SARS-COV-2 spike protein of SEQ ID NO: 2, 28, 29, 30, or 31; a variant thereof; an immunogenic fragment thereof; or a protein comprising said mutated SARS-COV-2 spike protein, variant, or immunogenic fragment; a mRNA encoding said mutated SARS-COV-2 spike protein, variant, immunogenic fragment or protein; a DNA encoding said mutated SARS-COV-2 spike protein, variant, immunogenic fragment or protein; a plasmid containing said DNA; or a viral vector containing said DNA.

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